VASCULAR TECHNOLOGY

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This course is intended to provide the practicing vascular technologist/sonographer with an intensive review for certification. This syllabus adheres to the ARDMS study outline content, as well as to the CCI study outline; and should be used in conjunction with other pertinent references and resources. Although designed to assist the experienced practitioner in the successful completion of the certification examination, this course also provides those with less experience, a strong educational foundation upon which to build.

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Chapter 1 Arterial Gross Anatomy
(Central and Peripheral System)

A. AORTIC ARCH

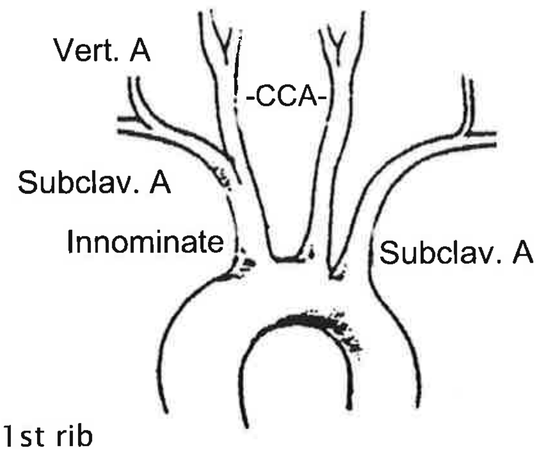
1. Proximal to Arch, the first branches of ascending aorta are the coronary A.

2. Innominate / Brachiocephalic

   - Divides into the right common carotid and subclavian arteries

3. Left common carotid artery (LCCA)

4. Left subclavian artery



B. UPPER EXTREMITY ARTERIES

1. Subclavian artery:

   - Runs laterally to outer border of 1st rib becoming axillary A.
   - Some branches: vertebral, thyrocervical, costocervical.

2. Axillary artery: after giving off several branches, becomes brachial A.

3. Brachial Artery: Branches into the radial and ulnar arteries at the inner aspect of the elbow, (also known as the antecubital fossa).

4. Radial artery: travels down lateral side of forearm into hand, branching to form:

   - Superficial palmar (volar) arch;
   - Terminates in deep palmar arch by joining deep branch of ulnar artery.

5. Ulnar artery: travels down medial side of forearm into hand, branching to form:

   - Deep palmar (volar) branch;
   - Terminates in superficial palmar arch.
   - Predominate source of blood flow to the hand

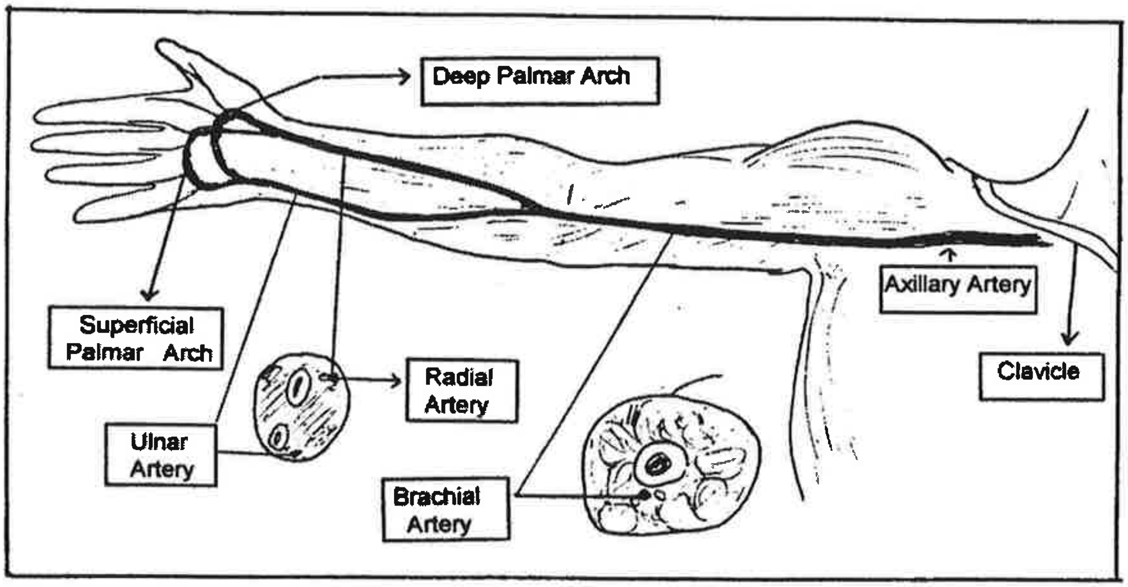
6. Superficial palmar (volar) arch includes:

   - Distal portion of the ulnar artery
   - Branch of the radial artery.

7. Deep palmar arch includes:

- Deep palmar branch of ulnar artery
- Distal portion of the radial artery.

8. Digital arteries arise from the palmar arches and extend into the fingers dividing into lateral and medial branches.



C. ABDOMINAL AORTA

1. Visceral branches

a. Celiac artery: (CA)

- Supplies stomach, liver, pancreas, duodenum, spleen.
- Branches into L. gastric, splenic, and common hepatic arteries.

b. Superior mesenteric artery: (SMA)

- Supplies the small intestine, cecum, parts of colon.
- Located about 1 cm behind the celiac A.
- Can be a common trunk of the celiac A. and SMA.

Find lt renal v
To find artery

c. Renal arteries:

- Supply blood to the kidneys, suprarenal glands, ureters.
- Multiple renal arteries not uncommon bilaterally.

- In transverse, a landmark for locating left renal A, is the left renal V which crosses the aorta anteriorly; the artery being just posterior.

d. Inferior mesenteric artery: (IMA)

- Supplies transverse, descending colon and part of rectum.
- Arises from abdominal aorta about 3-4 cm above bifurcation
- Can act as a collateral connection

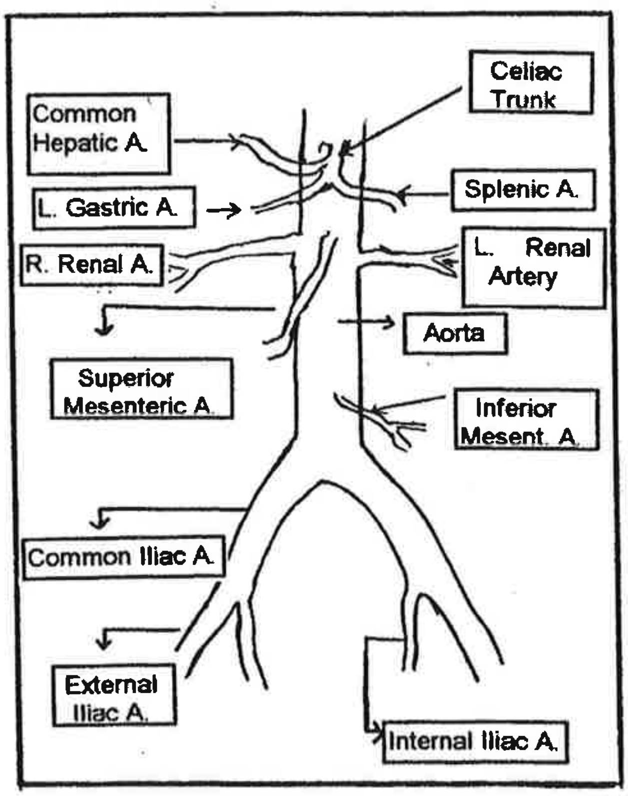
2. Terminal branches

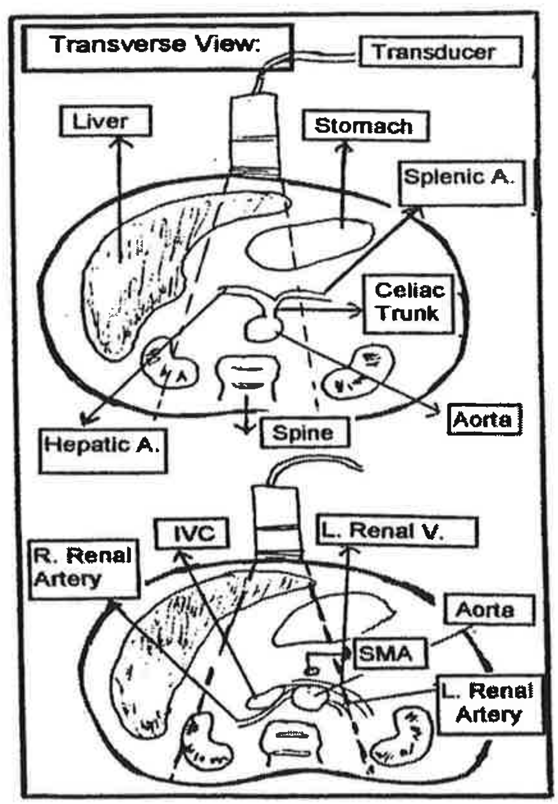
a. Both common iliac arteries (CIA) are the most distal branches of the aorta carrying blood to the pelvis, abdominal wall and lower limbs. The CIA divides into:

- Internal iliac artery (hypogastric).

- External iliac artery: travels along medial side of Psoas major muscle.

  Passing underneath inguinal ligament, becomes the CFA.





D. LOWER EXTREMITY ARTERIES

1. Common femoral artery (CFA) divides into:

After hunter's, CFA become SFA.

   a. Superficial femoral artery (SFA)

      - Runs the length of the thigh, passing through an opening In the tendon of the Adductor hiatus (adductor canal, or Hunter's canal),
      - Enters the popliteal fossa behind the knee.

   b. Deep femoral (profunda femoris) artery

      - A large branch; arises about 5 cm. from the inguinal ligament on the lateral side.
      - Can act as a collateral connection.

2. Popliteal artery

   a. Adductor hiatus: termination of the SFA and beginning of popliteal artery.

   b. Gives off number of genicular branches (can act as collaterals) to supply muscles, knee joint and skin.

   c. At interval between tibia and fibula, divides into anterior and posterior tibial arteries.

3. Anterior tibial (ATA)

   a. First branch off distal popliteal artery.

   b. At its lower end, becomes dorsalis pedis artery (DPA) and is directed across dorsum of foot towards base of great toe.

      Major branch of DPA: deep plantar artery; penetrating the sole of foot, it unites with lateral plantar artery to complete plantar arch

4. Posterior tibial (PTA)

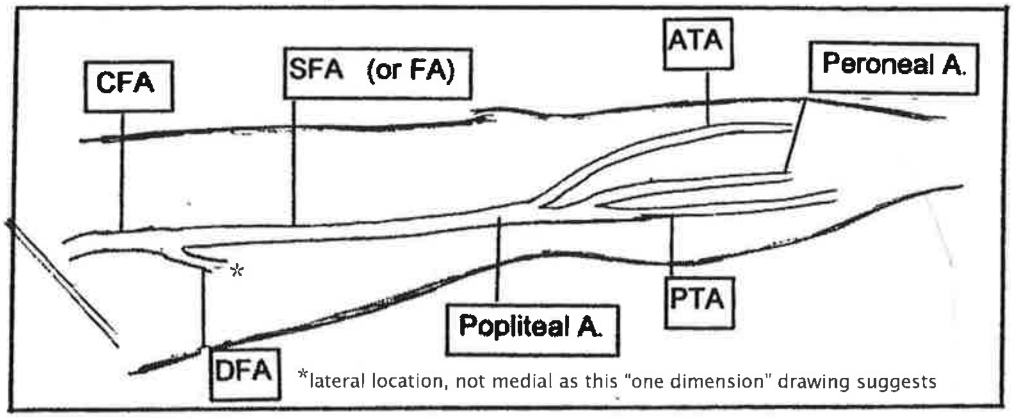
   a. Extends obliquely down posterior/medial side of the leg.

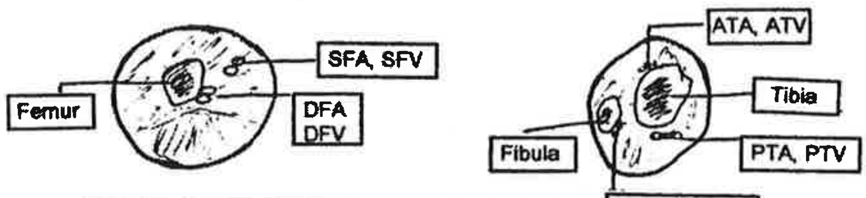
   b. Tibioperoneal trunk: short segment between ATA branch and branches of PTA and peroneal arteries.

   c. Major branches: lateral and medial plantar arteries, branching below medial malleolus to supply sole of foot.

5. Peroneal artery

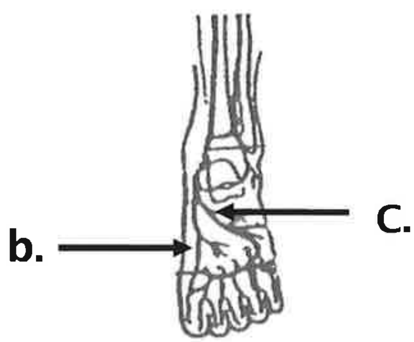
a. Passes toward fibula, traveling down medial side of that bone to supply structures of the lateral side of leg / foot.





Transverse Approach Examples

6. Digital arteries/Plantar arch



a. The plantar and dorsal metatarsals distribute blood to digits.

b. The plantar arch consists of deep plantar artery (branch of DPA) and

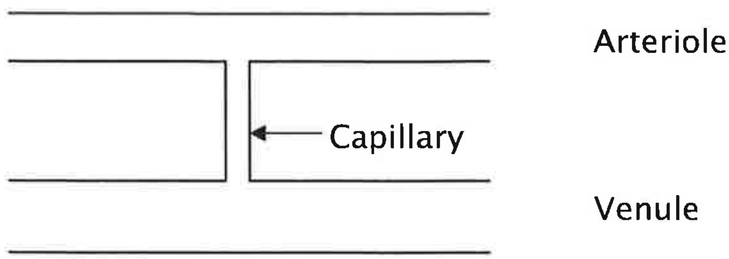
c. The lateral plantar artery (branch of PTA) which unites with the deep plantar artery.

7. Vessels of the microcirculation:

   a. Arteries transport gases, nutrients and other essential substances to the capillaries. Arteries progressively decrease in size from aorta (largest) to arterioles (smallest).

   b. Arterioles: considered resistance vessels; assist with regulating blood flow through contraction and relaxation.

   c. Capillaries: nutrients and waste products are exchanged between the tissue and blood.



E. MICROSCOPIC ANATOMY OF THE ARTERIAL WALL

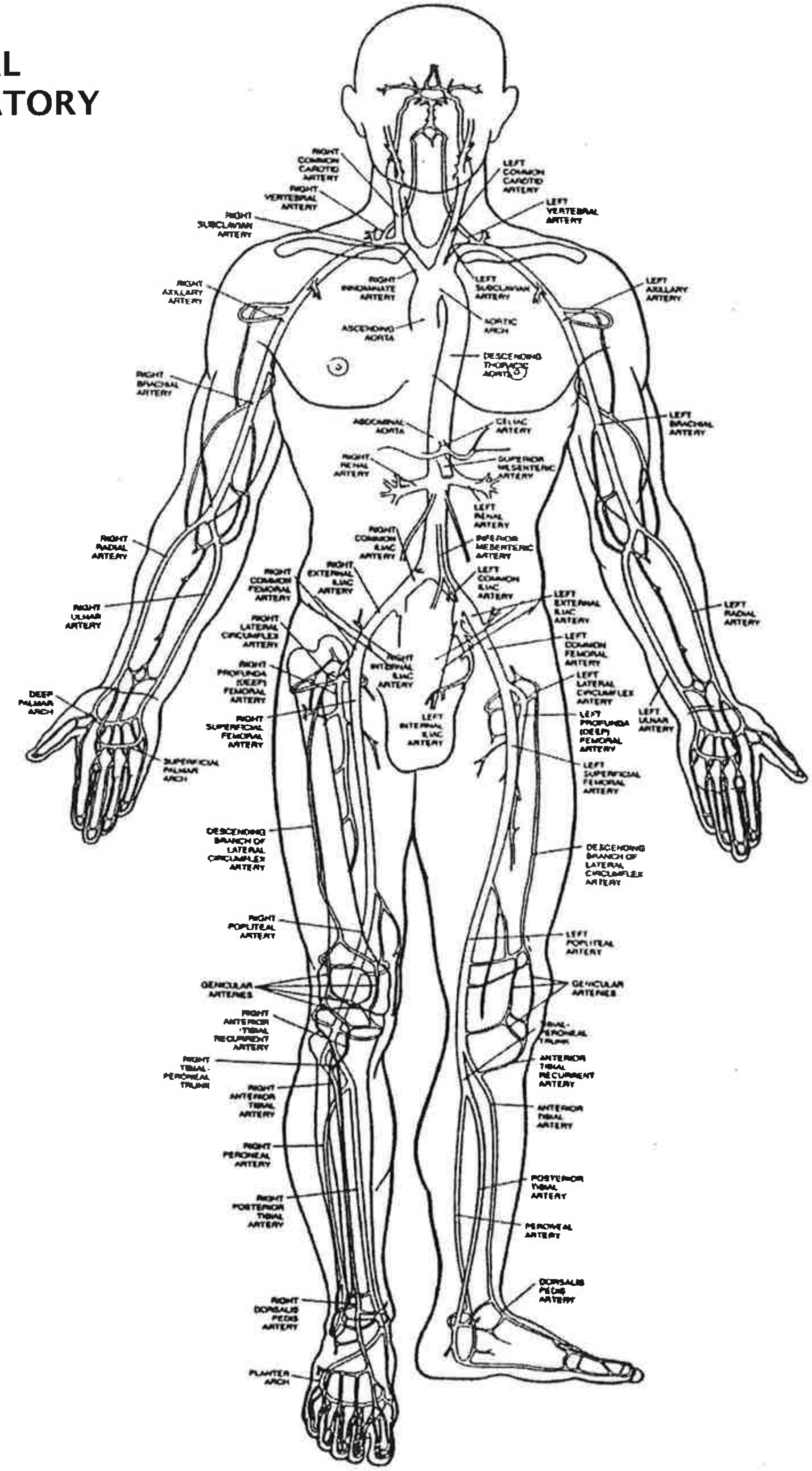
1. Tunica intima/inner layer is thin, consisting of a surface layer of smooth endothelium, base membrane and connective tissue.

2. Tunica media/intermediate layer is thicker, composed of smooth muscle and connective tissue, largely of the elastic type.

3. Tunica externa/outer layer (adventitia) is somewhat thinner than media, contains fibrous connective tissue; some muscle fibers.

4. As a rule, the adventitial layer contains the vasa vasorum, tiny vessels that carry blood to the walls of the larger arteries.

ARTERIAL CIRCULATORY SYSTEM



SOURCE:
Medasonics,
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Chapter 2 Physiology and Hemodynamics

A. THE ARTERIAL SYSTEM

1. Multi-branched elastic conduit set into oscillation by each beat of the heart.

2. Each beat pumps about 70 milliliters of blood into the aorta causing a blood pressure pulse.

3. Cardiac contraction begins:

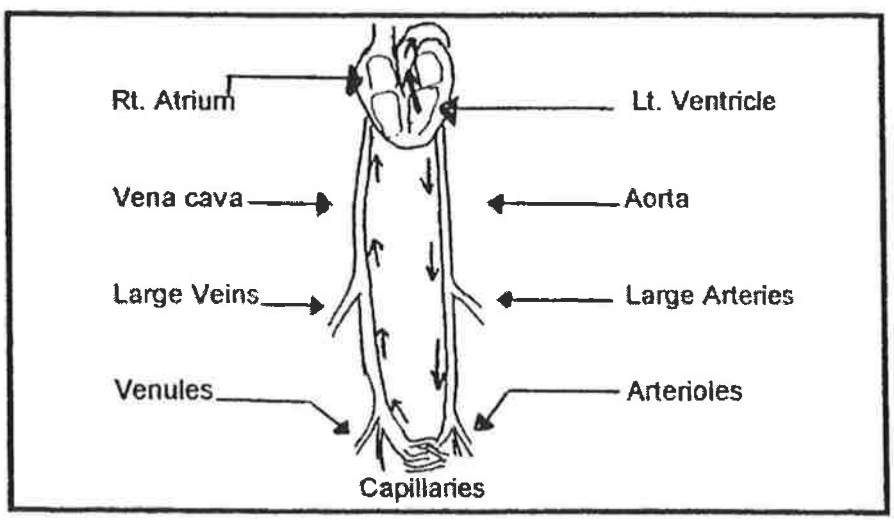
   - Pressure in the left ventricle rises rapidly
   - Left ventricle pressure exceeds that in the aorta
   - Aortic valve opens, blood is ejected, BP rises

4. Increased heart rate delivers an increased blood volume.

5. Patient's cardiac status plays an important role in the movement of blood throughout the vascular system.

6. The heart pump:

   - Generates the pressure to move the blood.
   - Results in a pressure wave (energy wave) that travels rapidly throughout the system, demonstrating a gradual transformation as it travels distally.



7. Pumping action of heart results in high volume of blood in arteries to maintain a high pressure gradient between the arteries and veins.

8. Cardiac output governs the amount of blood that enters the arterial system; arterial pressure and total **peripheral resistance**, determines the amount that leaves it.

9. Each cardiac contraction distends the arteries, which serve as reservoirs to store some blood volume and potential energy supplied to the system.

10. Pressure is greater at the heart, gradually decreasing as the blood moves further away. This pressure difference is necessary to maintain blood flow.

B. ENERGY (E)

1. Movement of any fluid medium between two points requires two things:

- A pathway along which the fluid can flow
- Difference in energy levels (pressure difference).

2. The amount of flow depends upon:

- Energy difference: includes losses resulting from fluid movement.
- Any resistance which tends to oppose such movement.

> **HINT**
>
> **Lower Resistance = Higher Flow rate**
> **Higher Resistance = Lower Flow rate**

3. The total energy contained in moving fluid is the sum of pressure (potential), kinetic and gravitational energies.

   a. Pressure (potential) energy is stored energy and is the major form of energy for circulation of blood; expressed in mmHg.

   b. Kinetic energy is small for circulating blood; expressed in terms of fluid density and its **velocity measurements**.

   c. Gravitational energy, **hydrostatic pressure (HP)**, is equivalent to the weight of the column of blood extending from the heart to level where pressure is measured.

Example: Averaged sized supine patient:

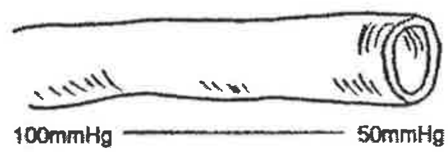
- Arteries and veins are nearly the same level as the heart.
- There is 0 mmHg (HP) against the arteries and veins at the ankle. (Ankle P = circulatory P plus about 0 mmHg)
- When standing, HP increases, adding about 100 mmHg against ankle vessels. (Ankle P = circulatory P plus 100mmHg)

4. An **energy gradient** is needed to move blood from one point to another. The greater the gradient; greater the flow.

    a. **Inertia:** relates to the tendency of a fluid to resist changes in its velocity (i.e. body at rest tends to stay at rest).

    b. As the blood moves farther out to the periphery, energy is dissipated largely in the form of heat.

    c. Energy continually restored by pumping action of heart.



C. FACTORS AFFECTING RESISTANCE TO FLOW

1. Movement of a fluid (blood) is dependent upon: physical properties of the fluid and what its moving through.

2.

$$R = 8\eta L / \pi r^4$$

R = Resistance η = viscosity L = vessel length
r = radius

- "R" is directly proportional to variables in numerator: (viscosity and length)
- "R" is inversely proportional to the variable in the denominator: (radius)

3. Although **viscosity** and vessel length have an effect on **resistance**, a change in vessel diameter has a more dramatic effect.

4. Internal **friction** within a fluid is measured by its **viscosity**.

    a Energy lost in form of heat (layers of rbc's rub against each other).

    b. Elevated hematocrit increases blood **viscosity** (thickness of a fluid); while severe anemia decreases blood **viscosity**. - decrease

resistance

c. Diminishing vessel size increases frictional forces and heat energy losses.

HINT:

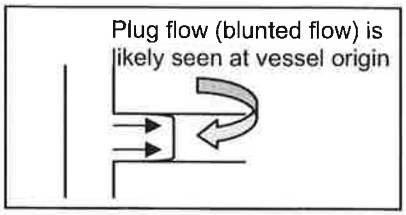
↑ Viscosity = ↓ Velocity

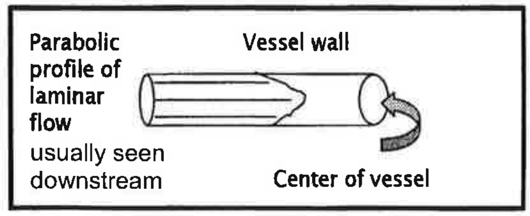
↓ Viscosity = ↑ Velocity

D. LAMINAR FLOW

1. Consists of layers of fluid particles moving against one another.

2. The fastest moving flow is in center; stationary layer remains at the wall. Laminar flow is considered stable flow.





E. ENERGY LOSSES

1. **Viscous** energy loss is due to increased friction between molecules and layers which ultimately causes energy loss.

2. **Inertial** losses occur with deviations from laminar flow, due to changes in direction and/or velocity.

   - The parabolic flow profile becomes flattened
   - Flow moves in a disorganized fashion.
   - This type of energy loss occurs at the **exit** of a stenosis.

F. POISEUILLE'S EQUATION

1. Defines relationship between: pressure, volume flow, resistance.

2. Helps answer question of how much fluid moves through vessel.

Q = P / R

Q = Volume Flow P= Pressure
R = Resistance

Remember this comparison:

Hemodynamics		**Electricity (Ohm's Law)**
Q = P / R		**I = V / R (I = E / R)**
Flow (Q)	=	Current (I)
Pressure (P)	=	Voltage (V) or (E)
Resistance (R)	=	Resistance (R)

3. **Poiseuille's equation** and the relationship between the contributors to flow and the flow itself is illustrated:

$$Q = \frac{(P1 - P2)\,\pi r^4}{8\,\eta\,L}$$

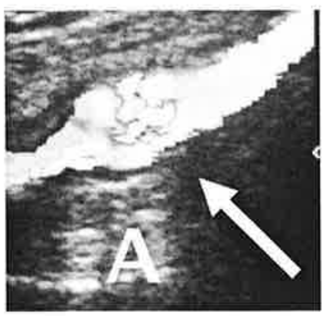
Q	=	Volume Flow
P1 - P2	=	Pressures at proximal/distal ends
r	=	Radius of the tube
L	=	Length of the tube
η	=	Viscosity of the fluid

a. Diameter change has most dramatic effect on resistance.

b. Radius (r^4) of vessel is <u>directly</u> proportional to volume flow.

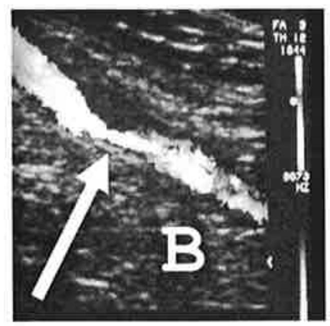
c. Small changes in radius may result in large changes in volume flow.

G. VELOCITY CHANGES : The Law of Conservation of Mass explains the relationship between velocity and area. (Q = A x V)



A = ↓ V

B = ↑ V



H. PRESSURE / VELOCITY RELATIONSHIPS—Bernoulli

1. Total energy contained in moving fluid is the sum of pressure, kinetic and gravitational energies.

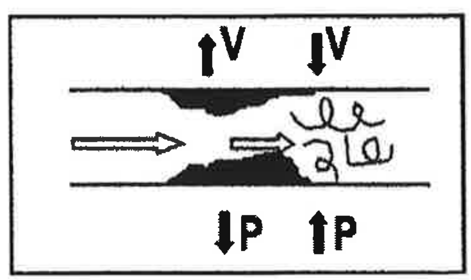
2. If one changes, the others make up for difference in order to maintain the original total fluid energy amount. (Performing exam on supine patient, there is NO change in hydrostatic pressure).

<u>HINT:</u>

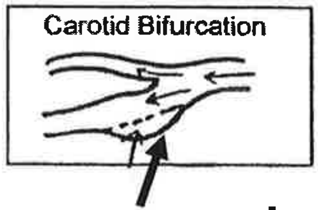
↑ Velocity = ↓ Pressure

↓ Velocity = ↑ Pressure

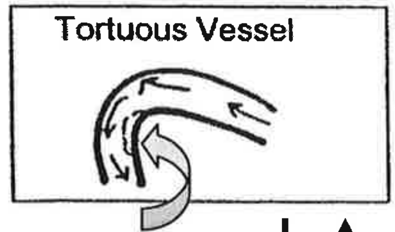
(Velocity E and Pressure E are inversely related)



3. Pressure gradients (flow separations) are set up. They occur because of a geometry change with or without intra-luminal disease; and because of curves.

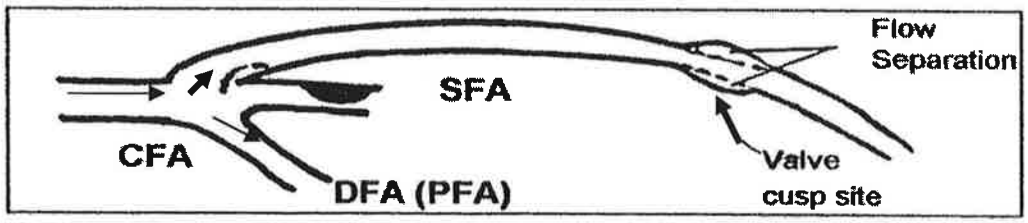


Flow separation: V↓ P↑



Flow Separation: V↓ P↑

4. Flow separations result in regions with stagnant or little movement. Example: bypass graft anastomosis site, or valve cusp site.



I. STEADY FLOW vs PULSATILE FLOW

1. **Steady flow** originates from a steady driving pressure.

   a. Easy to deal with because behavior is more predictable.
   b. In a rigid tube, energy losses are mainly **viscous**; can be described by **Poiseuille's equation**.

2. **Pulsatile flow** changes both the driving pressure conditions as well as the response of the system.

   a. Systole: forward flow throughout the periphery. (Fluid Acceleration)

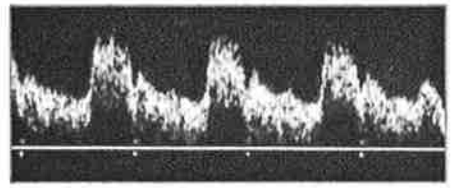
   b. Late systole/Early diastole: temporary flow reversal, due to a phase shifted negative pressure gradient and peripheral resistance, causing reflection of the wave proximally.

   c. Late diastole: flow is forward again, as reflective wave hits the proximal resistance of the next oncoming wave, and reverses.

J. PERIPHERAL RESISTANCE

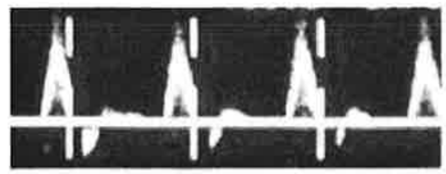
1. Low resistance flow: flow of a continuous (steady) nature feeding a dilated vascular bed.

   Example arteries: ICA, vertebral, renal, celiac, splenic, hepatic.



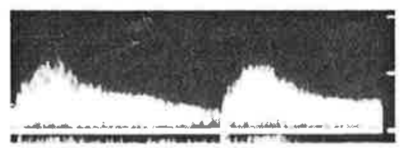
2. High resistance flow: flow of a pulsatile nature. Between incident pulses, hydraulic reflections travel back up the vessel from the periphery producing flow reversals in the vascular compartment.

   Example arteries: ECA, subclavian, aorta, iliac, extremity arteries, fasting SMA.



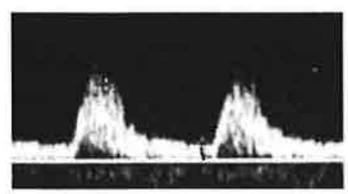
3. The reversal component of a high resistant signal may disappear distal to a stenosis because of decreased **peripheral resistance**, secondary to ischemia.

   Doppler flow distal to a significant stenosis is lower resistant. In addition it's more rounded in appearance and is weaker in strength.



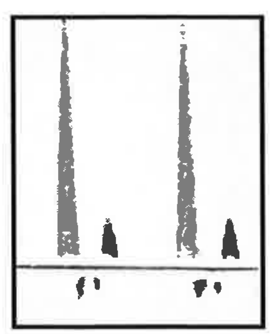
4. A normally high resistant (biphasic or triphasic) signal may become monophasic as it approaches the significant stenosis and/or arterial obstruction.

   Doppler flow proximal to a significant stenosis is higher resistant in quality (could have no diastole or minimal diastole (as seen in this example).



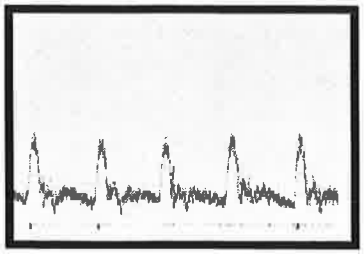
5. VASOCONSTRICTION

   Pulsatile changes in medium/small sized arteries of the limbs are increased. When this occurs, pulsatility changes are usually decreased in the minute arteries.



6. VASODILATATION

Pulsatile changes in medium/small sized arteries of the limbs are decreased (lower resistant). When this occurs, pulsatility changes are increased in minute arteries,



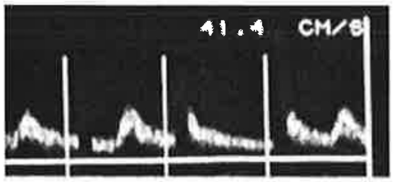
NOTE: As the inflow pressure falls as a result of stenosis; the natural response in periphery is to vasodilate to maintain flow.

K. COLLATERAL EFFECTS

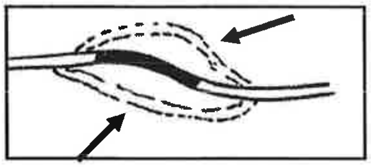
1. At rest, total blood flow may be fairly normal even in the presence of stenosis/ complete **occlusion** of main artery. Why? Development of a **collateral** network, and a compensatory decrease in **peripheral resistance.**

2. Arterial obstruction may alter flow in collateral channels nearby or further away from site of obstruction. Changes include:

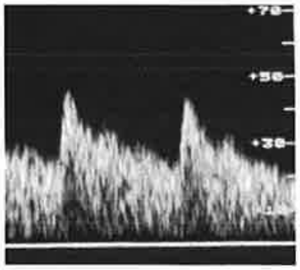
   - increased volume flow
   - reversed flow direction
   - increased velocity
   - waveform pulsatility changes



3. Location of collaterals helps provide tentative location of the obstruction.



L. EFFECTS OF EXERCISE



1. Exercise should induce peripheral vasodilatation which lowers the distal peripheral resistance, increasing blood flow.

2. Vasoconstriction and vasodilatation of blood vessels within skeletal muscles also influenced by sympathetic innervation fibers functioning primarily for regulation of body temperature.

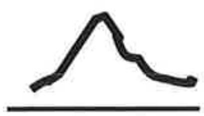
3. **Exercise** is probably best single vasodilator of resistance vessels within skeletal muscle.

4. <u>Autoregulation:</u> ability of most vascular beds to maintain constant level of blood flow over a wide range of perfusion pressures.

- Not present: perfusion pressure drops below a critical level.

BP rise:	= constriction of resistance vessels
BP fall:	= dilatation of resistance vessels

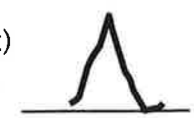
5. By decreasing resistance in the working muscle, **exercise** usually decreases "reflection" in exercising extremity.



- Example: a low resistant, monophasic flow signal is normally present in extremity arteries after vigorous exercise. The exercise causes reduced flow resistance (vasodilatation).

  Same monophasic pattern also seen pathologically. Peripheral dilatation occurs in response to proximal arterial obstruction.

6. Higher resistant signals may occur from normal (physiologic) vasoconstriction at the arteriolar level OR from distal arterial obstruction.



M. ADDITIONAL NOTES:

- Flow to a cool extremity (vasoconstriction) will have pulsatile signals.
- Flow to a warm extremity (vasodilatation) will have continuous, steady signals.

- Pulsatility changes do not differentiate well between occlusion and severe stenosis.
- Waveforms may not be altered with good collateralization.

Distal effects of obstructive disease may only be detectable following stress (i.e., exercise or hyperemic evaluation).

N. EFFECTS OF STENOSIS ON FLOW CHARACTERISTICS

1. Review: Laminar flow

- Even distribution of frequencies at systole: lower frequencies distributed at the walls, (boundary layer); faster moving flow in centerstream.

2. A hemodynamically significant stenosis causes a notable reduction in volume flow and pressure.

Cross sectional **area reduction** of 75% = **diameter reduction** of 50%.

3. Effects of flow abnormality produced by a stenosis depends on factors such as:

   a. Length, diameter, shape, degree of narrowing
   b. Multiple obstructions in the same vessel: resistance to flow is additive; it results in a higher resistance than in each individual narrowing.
   c. Obstructions in different vessels that are parallel: resistance to flow is less than the resistance in each individual narrowing because only part of the blood flow is going through each narrowing.
   d. Pressure gradient; peripheral resistance beyond stenosis

4. **STENOSIS PROFILE**

   a. Proximal to a stenosis: flow frequencies are usually dampened, with or without disturbance.

   b. Stenosis:

   Entrance into the stenosis produces an increase in Doppler shift frequencies (DSF), resulting in spectral broadening and **elevated velocities.**

   - **Flow disturbance** occurs due to interrupted flow stability with high velocities and eddy currents.

   - Abnormal "jet" (elevated velocities) may be isolated to area of stenosis, but also approaching and/or leaving it.

   c. Post-stenotic turbulence:

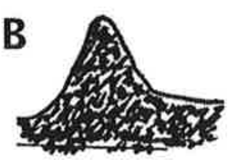
   - At stenosis **exit**, flow reversals, flow separations, vortices /eddy currents occur near edge of flow pattern.

   - Flow quality is comprised of multiple changes in **direction** and spectral broadening as displayed by the spectral pattern.

   - Energy expended as "heat" as eddys and vortices work against blood viscosity.

A

B



C

A Pre-Stenosis B Stenosis C Post-stenosis

Spectral broadening vertical

Chapter 3 Arterial Testing

(Signs, Symptoms, Disease Mechanisms)

Note: After patient referred for pertinent medical necessity, important to obtain complete history, provide limited physical exam, and clinical information necessary to relate to test findings.

Essential to maintain a warm environment for the patient in order to permit warming (and peripheral dilation) to occur. An adequate amount of acoustic coupling gel is applied to the skin for all types of Doppler exams.

A. SIGNS AND SYMPTOMS

1. Chronic occlusive disease

a. Claudication:

- Pain in muscles usually occurring during exercise; subsides with rest.
- Results from inadequate blood supply to muscle.
- Discomfort is predictable and subsides within minutes after exercise.
- Level of disease usually proximal to location of symptoms.
- Pseudo-claudication mimics vascular symptoms but is neurogenic or orthopedic in origin

b. Ischemic rest pain:

- A more severe symptom of diminished blood flow.
- Occurs when limb not dependent; BP decreased (such as when sleeping).

c. Tissue loss:

- Necrosis or death of tissue.
- Due to deficient or absent blood supply.

NOTE: A patient history may state: 4 block claudication. This means the patient c/o pain after walking 4 blocks.

2. **Acute arterial occlusion**

a. Symptoms include the 6 P's: pain, pallor, pulselessness, paresthesia, paralysis, polar.

b. May result from thrombus, embolism or trauma.

c. Emergency situation since the abrupt onset does not provide for the development of collateral channels.

3. **Vasospastic Disorders**

a. Raynaud's phenomenon: a condition that exists when symptoms of intermittent digital ischemia occur in response to cold exposure or emotional stress.

b. Changes in skin color may include pallor (whiteness), cyanosis (bluish color), or rubor (dark red color).

c. PRIMARY RAYNAUD'S

- Ischemia due to digital arterial spasm.
- Common in young women; may be hereditary, bilateral; history of symptoms for 2 years without progression/evidence of cause.
- Benign condition with excellent prognosis.

d. SECONDARY RAYNAUD'S

- Also known as obstructive Raynaud's syndrome.
- Normal vasoconstrictive responses of arterioles superimposed on a fixed artery obstruction. Ischemia constantly present.
- May be the first manifestation of Buerger's disease.

4. **Physical Exam**

Skin Changes

a. Color:

PALLOR: Result of deficient blood supply; skin pale.

RUBOR: Suggests dilated vessels or vessels dilated secondary to reactive hyperemia; skin is reddened.

CYANOSIS: A concentration of deoxygenated hemoglobin, causes bluish discoloration.

b. **Temperature**

- Touch patient's skin to determine warm/cold.
- Utilize skin thermometer to document precisely.

c. Lesions (Refer to Venous Section – Comparison chart)

- Ulcerations located: tibial area, foot, toes.
- Deep and more regular in shape.
- Quite painful as compared to venous ulcers.
- Duration of ulcer(s) is important, (i.e., days, weeks)
- Gangrene: death of tissue; usually due to deficient or absent blood supply.

d. Capillary filling

- An increase in the capillary refill time denotes decreased arterial perfusion.

e. Elevation/dependency changes

- Cadaveric pallor during elevation with ruborous red discoloration with dependency.

Palpation (pulses, aneurysms)

a. Rhythmic throbbing of artery in time with heartbeat signifies adequate circulatory status. Diminished/absent pulse suggests arterial insufficiency.

b. Grading pulses on scale of 0 (none) - 4+ (bounding) is fairly standard.

c. Aneurysms can be palpated and described as bounding.

d. Palpable "vibration" or "thrill" over pulse site may indicate a fistula, post-stenotic turbulence, or a patent dialysis access site.

e.

PALPABLE PULSES: aorta, femoral, popliteal, dorsalis pedis (DPA), posterior tibial (PTA) • The peroneal artery is not palpable

Auscultation (bruits)

- Bruit auscultation is more often done with carotid examination.

- Other than stethoscope use for carotid vessels and heart, additional sites include: aorta, femoral, and popliteal arteries.

- Additional detail about bruits is located in the Cerebrovascular Testing section, Chapter 17

B. RISK FACTORS

1. Diabetes

a. Atherosclerosis: more common; occurs at a younger age.
b. Higher incidence of disease: distal pop. and tibial arteries.
c. Medial calcification develops in LE arteries.
d. Poor sensation (neuropathy) may lead to increased injury
e. Higher incidence of gangrenous change, amputations.

2. Hypertension

a. Unclear whether high blood pressure is a causative factor or enhances the development of the atherosclerotic process.
b. Systemic hypertension is associated with greater incidence of coronary atherosclerosis. Increased BP taxes heart.

3. Hyperlipidemia

a. Elevated plasma lipids closely assoc. with development of atherosclerosis.
b. Frequent cause: Diet high in animal fat; metabolic problems associated with heredity.

4. Smoking

a. Studies suggest the chemicals in cigarettes irritate the endothelial lining of the vessels, causing vasoconstriction.

5. **Other** (not controllable): age, family history, male gender

C. MECHANISMS OF DISEASE (Applicable to all arterial systems)

1. Atherosclerosis (obliterans)

a. Most common arterial pathology: thickening, hardening, loss of elasticity of the artery walls.

b. Changes occur in intima and media layer of the vessel.

c. Major risk factors: smoking, hyperlipidemia, family history. Less important factors: hypertension, diabetes, sedentary lifestyle and arterial wall shear/stress.

d. Most common sites:

- Carotid bifurcation
- Vessel origins
- Infra-renal aorto-iliac system
- CFA bifurcation
- SFA at the adductor canal level
- Trifurcation region

e. Example: Leriche Syndrome—caused by obstruction of the aorta; usually occurs in males and is characterized by:

- Fatigue in hips, thighs, or calves with exercise
- Absence of femoral pulses
- Impotence
- Often times, pallor and coldness of LE

2. Embolism

a. Obstruction of vessel by foreign substance or blood clot.

b. Emboli may be solid, liquid or gaseous; may arise from the body or enter from without.

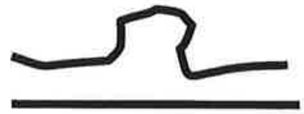
c. Most frequent cause: small plaque breaks loose (e.g. atherosclerotic lesion, arteritis, or angiographic procedure) and travels distally until it lodges in small vessel.

Example: Blue toe syndrome. Toe ischemia results; often improves, mainly from other smaller branches.

3. **Aneurysm**

a. True aneurysm is dilatation of all three arterial wall layers
Examples:

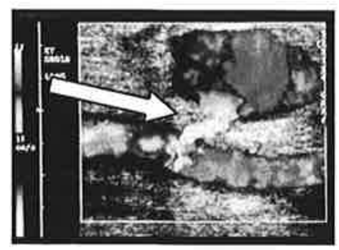
Fusiform: diffuse, circumferential dilatation.



Saccular : localized out-pouching

b. Dissecting aneurysm occurs when a small tear of the inner wall allows blood to form cavity between two wall layers. Often occurs in thoracic aorta.

c. Pseudoaneurysm results from a defect in main artery wall (e.g., post catheter insertion). Must be a channel communication from main artery to pulsatile structure outside vessel walls.



d. Most common location of a true aneurysm is infrarenal aorta. Other locations include: thoracic aorta, femoral, popliteal, renal.

- Patients with one aneurysm have higher incidence of 2.
- Cause unknown; ? poor nutrition, congenital defect, infection, or atherosclerosis.

Most frequent complication: Rupture of the aortic aneurysm; embolization of the peripheral aneurysms. (Note: Both types can accumulate thrombus inside)

4. **Non-atherosclerotic lesions**

a. **Arteritis**

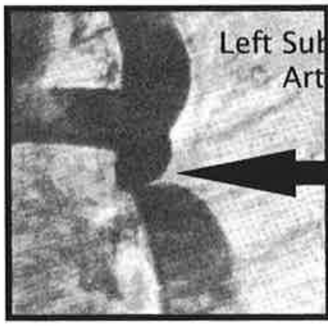
- Can affect tibial and peroneal arteries, as well as the smaller more distal arterioles and nutrient vessels.
- Inflammation of arterial wall leads to thrombosis of vessel;
- Type: Buerger's disease (thromboangiitis obliterans)
- Associated with heavy cigarette smoking.
- Occurs primarily in men <40 yr old.
- Patients present with occlusions of the distal arteries.
- Rest pain and ischemic ulceration present

b. Coarctation of the aorta

- One of several congenital anomalies of arterial system.
- Congenital narrowing or stricture of thoracic aorta ; but may affect abdominal aorta.
- Example: (see arrow)

CLINICAL FINDINGS:

- Hypertension due to decreased kidney perfusion.
- Manifestations of LE ischemia (e.g., decreased pulses and/or segmental pressures).



c. **Dissection**

- Can affect aorta and peripheral arteries
- Distinguishing ultrasound feature: thin membrane dividing the arterial lumen into 2 compartments
- Media is weakened; intima develops tear through which blood leaks into the media (false lumen)
- Flow velocities differ in each lumen
- Aortic dissections which can extend to iliacs, may occur consequent to hypertension or severe chest trauma
- Complications: stenosis, occlusion or thrombosis

d. **Vasospastic disorders:** Refer to Chapter 3

e. **Entrapment syndrome:** Refer to Chapter 14

Chapter 4 Doppler Waveform Analysis

1. **CAPABILITIES**

   a. Help confirm diagnosis /approximate location of arterial occlusive disease.

   b. Indicate severity of the occlusive process.

   c. Combined with Doppler segmental pressures

2. **LIMITATIONS**

   a. Patients with casts or extensive bandages.

   b. Waveforms may be affected by ambient temperature.

   c. Uncompensated congestive heart failure may result in dampened waveforms.

   d. Unable to discriminate stenosis from occlusion.

   e. Technically dependent test.

3. **PATIENT POSITIONING**

   a. Supine with the extremities at the same level as the heart. To decrease influence of hydrostatic pressure.

   b. The patient's hip is externally rotated, knee slightly bent.

   c. Other positions: Rt/Lt lateral decubitus position, or prone.

4. **PHYSICAL PRINCIPLES**

   a. <u>The Doppler effect</u>:

      - When a wave is reflected from a moving target, the frequency of the wave received is different (Doppler shift) from the transmitted wave.
      - This effect occurs with relative motion between the source and the receiver of the sound.
      - Blood is moving target; transducer is stationary source.

b. Continuous-wave (CW) Doppler is utilized.

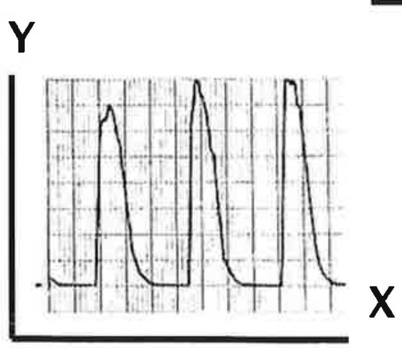
The reflected frequency is higher/lower than the transmitted frequency, depending on direction of flow.

c. Types of Doppler Velocimetry:

Analog: Employs a zero crossing frequency meter, to display the signals graphically on a strip chart recorder.

ZERO CROSSING FREQUENCY METER

- Circuitry counts each time the input signal crosses through zero (the baseline) within a time span.
- High frequency waves have many oscillations; low frequency waves have few.
- Direction of blood flow varies during the cardiac cycle.
- Machine estimates frequencies present in reflected signal and displays them.

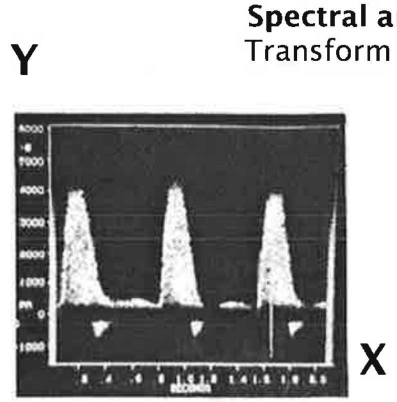


* Has acceptable accuracy

* Drawbacks include: noise, less sensitivity, high velocities are underestimated; low velocities are overestimated.

* Most equipment does a self calibration when the system is activated.

Spectral analysis: Individual frequencies displayed by Fast Fourier Transform (FFT) method.



* More commonly used during duplex evaluation

* Time displayed on the horizontal axis (X); frequency shifts displayed on the vertical axis (Y).

* Free of many of the analog recording drawbacks.

5. TECHNIQUE

a. Acoustic gel applied; 8 -10 MHz Doppler probe utilized.

b. Upper extremities: Doppler velocity wave forms recorded from the following arteries bilaterally:

- Subclavian
- Axillary (axilla)
- Brachial (at elbow-antecubital fossa)
- Radial (thumb side, at wrist)
- Ulnar (5th finger side, at wrist)

c. Lower extremities: Doppler velocity wave forms recorded from the following arteries bilaterally.

- Common femoral (CFA)
- Superficial femoral (SFA)
- Popliteal
- Posterior tibial (PTA) (medial malleolus)
- Dorsalis pedis (DPA) (top of foot)
- Peroneal (if necessary) location: lateral malleolus

d. The audible AND wave form qualities are observed, documented, and combined with Doppler segmental pressures.

e. POTENTIAL SOURCES OF TECHNICAL ERROR

- Improper probe position
- Inadvertent probe motion
- Incorrect angle of incidence
- Inadequate amount of gel
- Excessive pressure on the probe tip
- Insufficient period of rest before testing

6. INTERPRETATION—Qualitative

a. Normal signals:

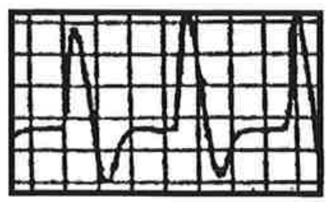
b. Abnormal signals:

TRIPHASIC

- Rapid upslope
- Sharp peak
- Rapid down-stroke
- Flow reversal
- Resumption of forward flow
- Example: UE, LE arteries

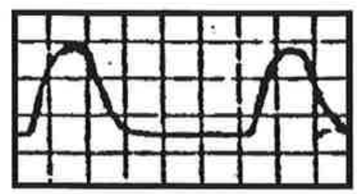
BIPHASIC

- Rapid upslope
- Sharp peak
- Fairly rapid downstroke
- Flow reversal
- No resumption of forward flow
- Also considered normal in some patients



MONOPHASIC

- Slow upslope
- Rounded peak
- Slow down-stroke
- No reversal



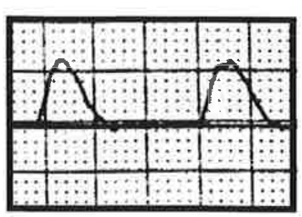
c. Essential to observe for deterioration in the signals from one level to the next.

d. A monophasic/dampened (pulsatile) signal is often obtained proximal to an obstruction. (In this example there is no diastolic flow.)

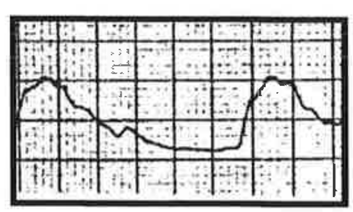
e. Well collateralized occlusions can appear similar to flow distal to a stenosis.

> Vasodilatation of the distal vessels often occur with proximal obstruction, reducing the pulsatility; causing the signals to have lower resistant (steady) flow quality.

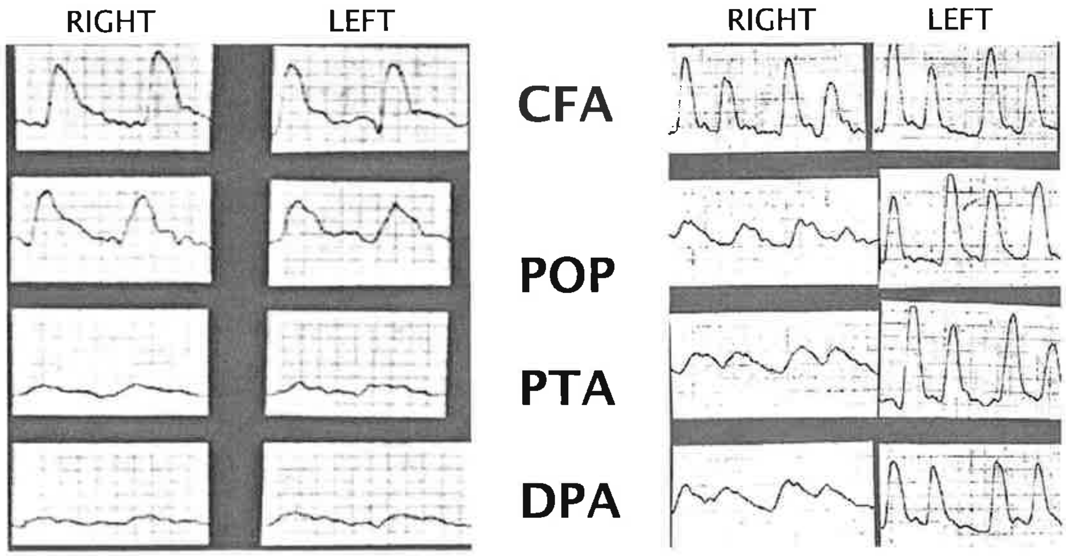
Miscellaneous Examples:



d. Proximal to Obstruction (monophasic, pulsatile)



e. Distal to Obstruction (monophasic, more steady)



Waveforms only: suggestive of bilateral AI, multi-level occlusive disease

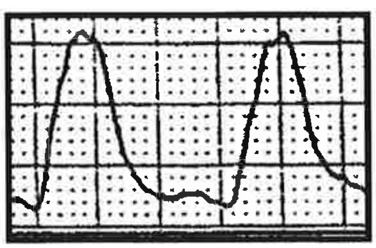
Waveforms only: suggestive of R. fem-pop occlusive disease and some element of aorto-iliac disease

f. Upper extremity signals:

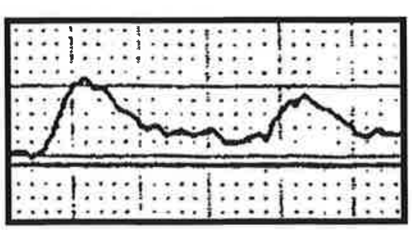
- Subclavian A: high resistant, multiphasic flow. Proximal occlusion or critical stenosis will make the signal more monophasic.

- Strandness and Sumner describe arteriovenous shunts in the skin of fingertips that cause flow patterns in hand to be tremendously variable; e.g., continuous signals can be found in the brachial, radial, and ulnar arteries in the relaxed and warm patient.

g. Doppler waveforms post-exercise: (If obtained)



- NORMAL: pre-exercise wave form qualities are maintained and/or augmented. No reverse component. Usually just pressures obtained post-exercise.

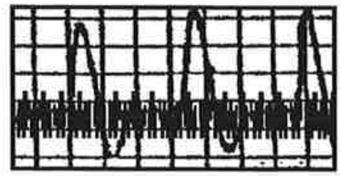


- ABNORMAL: Slow upstroke with more rounded peak, slow downstroke, no reverse component.

h. Absent Doppler signals may suggest occlusion or a pre-occlusive vessel (string sign.) Analog Doppler not capable of portraying velocities of less than 6 cm/sec.

i. <u>Troubleshooting:</u>

- *Recorder stylus not recording?* Check that proper test selection and/or probe type has been made.



- *"60 cycle" noise on tracing?* Decrease gain; turn system off/on; increase filter; try another plug receptacle.

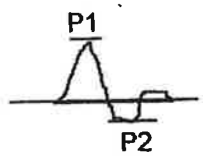
- *Recording stylus "stuck" at lower or upper portion of tracing?* Activate "re-set" control to re-center.

7. **INTERPRETATION—Quantitative** (using spectral analysis)

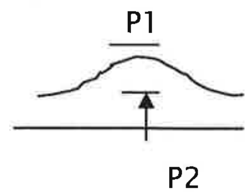
a. **<u>Pulsatility Index (PI)</u>**: Calculated by dividing the peak-to-peak frequency difference (P1 - P2) by the mean (average) frequency.

$\frac{P_1 - P_2}{mean}$

NORMAL



ABNORMAL

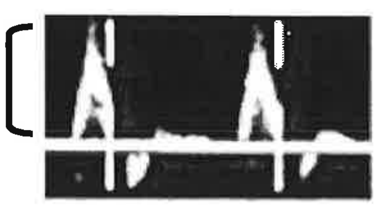


- Differentiate inflow from outflow disease. (Example: aortoiliac from femoral disease). What is going on proximally?

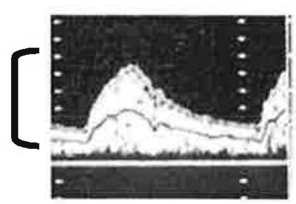
b. **<u>Acceleration Time:</u>** helps to differentiate inflow (aorto-iliac) disease from outflow (S-femoral) disease.

- Based on principle that proximal arterial obstruction results in a slowing of the time interval between the onset of systole to the point of maximum peak.

- Example criteria: An acceleration time of >133 msec suggests presence of proximal iliac disease.



Normal (CFA)



Abnormal (CFA)

Chapter 5

Doppler Segmental Pressures (LE)

1. **CAPABILITIES**

   a. Assess presence/severity of arterial disease.

   b. Combined with Doppler velocity or volume pulse waveforms.

2. **LIMITATIONS**

   a. Cannot discriminate between stenosis and occlusion, precisely localize area of obstruction, nor discriminate between CFA and external iliac disease

   b. Calcified vessels (medial calcinosis) render falsely elevated Doppler pressures; e.g., diabetics, end stage renal disease.

   c. Uncompensated CHF may result in decreased ankle/brachial indices.

   d. Artifactually elevated high thigh pressures when narrow cuff used on thigh.

   e. Difficult to interpret in presence of multi-level disease.

3. **PATIENT POSITIONING**

   a. Patient should rest 20 minutes prior to the exam, especially when vascular disease present.

   b. Supine, with legs about same level as heart, so that hydrostatic pressure (which affects a sitting patient) cannot affect the BP measurements.

4. **PHYSICAL PRINCIPLES**

   Refer to "Doppler Velocity Waveforms" evaluation.

REMINDER about cuff artifact:

- If the cuff is too large for a limb segment, BP is falsely lower
- If the cuff is too narrow for a limb segment, BP is falsely higher

5. TECHNIQUE

a. Appropriately sized blood pressure cuffs applied to both extremities.

- Place cuff "straight" on the extremity site, not encircling any bony prominence.
- Should fit snugly.
- Ideally the cuff bladder should be placed over the artery.
- Appropriate technique ensures the bladder inflation transmits pressure quickly into the tissue to compress the artery.
- Width of cuff should be about 20% greater than diameter of llmb

b.

Four Cuff Method: (bilateral)

- Brachial (upper arm)
- High Thigh
- Above the knee (AK) (low thigh)
- Below the knee (BK) (calf)
- Ankle
- Size: 12 x 40 cm with longer cuff bladders for thighs

c.

Three Cuff Method: (bilateral)

- Brachial (upper arm)
- One thigh cuff (19 x 40 cm)
- Below the knee (calf)
- Ankle
- Size: 12 x 40 cm; with exception of thigh cuff

d. <u>Difference between the 4 cuff and 3 cuff method is:</u>

- Two thigh cuffs (provide proximal and distal pressure measurements) but artifactually elevated BP's are obtained.

- The 3 cuff method utilizes one large thigh cuff (high on the thigh), providing a more accurate pressure reading.

e. <u>Optimization of the Doppler signal</u>:

- Utilize an 8-10 MHz Doppler frequency probe.
- Angle the CW Doppler probe 45-60 degrees to the skin.
- Due to vessel angulation, probe angle behind the knee may be closer to 90 degrees to the skin.
- Angle the probe so blood flow moves antegrade (towards the probe).

f. Using a hand held sphygmomanometer, or an automatic cuff inflator, segmental Doppler pressures are obtained in the following order bilaterally:

ORDER OF SEGMENTAL PRESSURES

- Brachial (upper arm) using brachial artery
- Ankle (Use PTA and DPA; peroneal A. only if necessary).
- Calf (BK) (Use the site - PTA or DPA - that had the highest pressure, for obtaining the pressure for this level)
- Above the knee (AK) (Same as calf; may need to use popliteal A. if difficult to obtain)
- High thigh (HT) (Same as above knee)
- If one large thigh cuff used, protocol same as for the calf.

know order + why

NOTE: You must start at the ankle and move proximally to eliminate the possibility of underestimating the systolic pressure measurement.

For example, the high thigh cuff is inflated to above systole, deflated slowly to determine the BP; cuff is fully deflated; then the next distal cuff (AK) is inflated. Because this is done quickly, there isn't enough time for the arterial blood to completely flow normally into the leg, so the obtained BP would likely be falsely lower than it should be.

Note: Usually, the higher of the two pedal Doppler pressures is used to obtain the remainder of the segmental pressures in the leg beginning with the calf level.

g. Toe pressures are discussed later in this section.

h. Hints regarding segmental blood pressures:

REMEMBER!

- Complete cessation of blood flow is required; cuffs inflated 20-30 mmHg beyond last audible Doppler arterial signal; OR
- Inflate the cuff 20-30 mmHg higher than the highest brachial pressure.
- If pressure measurements must be repeated, the cuff should be fully deflated for about a minute prior to repeat inflation.
- The systolic pressure is recorded as that pressure at which the first audible Doppler arterial signal returns.

6. INTERPRETATION

a. Ankle/brachial index (ABI) is calculated by dividing the ankle pressure by the higher of the two brachial pressures.

b. Another term for this is ankle/arm pressure index (API).

c. The following criteria may vary slightly from Vascular Lab to Vascular Lab.

ANKLE/BRACHIAL INDEX

> 1.0	=	Normal
>0.9 - 1.0	=	Asymptomatic disease or mild arterial disease.
0.5 - 0.9	=	Claudication (moderate disease)
< 0.5	=	Rest pain (severe arterial disease)

Note: Incompressible vessels have falsely elevated and inaccurate pressures. An ABI of >1.3—1.5 is considered incompressible.

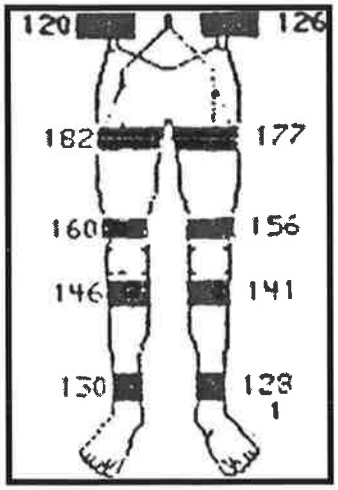
d. Additional observations:

- Some authors feel an absolute ankle pressure (< 50 mmHg), rather than the ABI of .5 has stronger weight for predicting symptoms at rest.
- Strandness' work suggests that an ABI of ≥.5 represents single segment disease; <.5 suggests multiple lesions.

e.
- Segmental pressure drops of >30 mmHg (some authors suggest 20-30 mmHg) between two consecutive levels, suggests significant obstruction.
- A horizontal difference of ≥ 20-30 mmHg suggests obstructive disease at or above the level in the leg with the lower pressure.

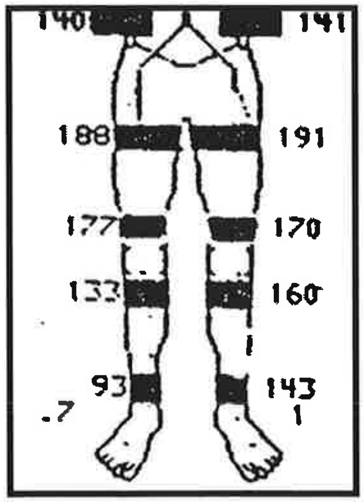
NOTE: The following examples are for self-assessment and are different from examples shown during slide presentation.

<u>EXAMPLE #1</u>



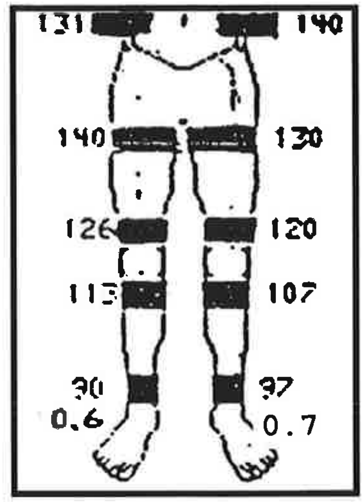
NORMAL

<u>EXAMPLE #2</u>



ABNORMAL

<u>EXAMPLE #3</u>



ABNORMAL

(Source of drawings is unknown)

f. <u>Answers to the previous examples:</u>

#1 Normal

#2 Femoral-popliteal and tibial vessel occlusive disease: >30mmHg gradient between the AK to calf and calf to ankle levels on the right; and 20-30 mmHg difference from right to left at calf and ankle levels.

#3 Aorto-iliac (A-I) occlusive disease: low pressures at both high thigh levels only.

g.

<u>(4) CUFF TECHNIQUE</u>

- The high thigh pressure is normally ≥ 30 mmHg than highest brachial pressure.
- The AK and BK pressures should be at least the same as the highest brachial.

<u>(3) CUFF TECHNIQUE</u>

- Thigh pressure is similar to the highest brachial pressure.
- Does not allow for differentiation of HT to AK pressures.

h. Pressure levels for foot ulcer healing:

- Toe pressures of $\leq$30 mm Hg are evident in foot and toe ulcers (lesions) that failed to heal.

- Ankle pressures cannot always be relied upon; (e.g., diabetic patients) toe pressures may be more reliable.

7. EXERCISE

a. Compare resting values to those obtained after exercise.

b. Helps differentiate between true and pseudo-claudication.

c. Determine presence/absence of collaterals.

d. Contraindications include: shortness of breath, hypertension, cardiac problems, stroke, walking problems.

e. **TECHNIQUE** (After resting exam)

- Patient walks on a constant load treadmill at a $\leq$12% elevation.

- Speed: 1.5 MPH for a maximum of 5 minutes or until symptoms increase to such severity that patient must stop.

- Document: duration of walking, MPH, onset, location and progression of symptoms.

- Post-exercise Doppler pressures obtained:
  both ankles (abnormal first), then higher brachial.

- Post-exercise ABI's are obtained :

  ◊ Immediately (Normal: ABI increases)
  ◊ Abnormally, ABI decreases minimally or to a severe amount.
  ◊ With drop after exercise, pressures obtained every two minutes until pre-exercise pressures are attained.

f. **INTERPRETATION**

INCORPORATES:

- Duration of exercise
- Length of time to recover.
- Pressure changes from pre to post exercise.

Single level Disease:

- Takes 2 to 6 minutes for the ABI's to increase back to resting levels after they dropped to low or unrecordable levels after exercise.

Multi-level Disease:

- Takes from 6 - 12 minutes for the ABI's to increase back to resting levels after they remained low or at unrecordable levels after exercise.

g. **REACTIVE HYPEREMIA**

- An alternate method for stressing the peripheral circulation. Used when patients: cannot walk long enough, use a cane or walker, have pulmonary problems; poor cardiac status, or other situations.

- Bilateral thigh cuffs (19 x 40) inflated to suprasystolic pressure levels (usually 20-30 mm Hg above the higher brachial BP) maintaining the pressure 3 to 5 minutes.

Produces ischemia and vasodilatation distal to the occluding cuffs.

- Upon release of cuff occlusion, ABI's are obtained.

- Normal limbs may show a transient drop of 17-34%.

Single level Disease

- $\leq$ 50% drop in ankle pressure with reactive hyperemia.

Multi-level Disease

- >50% ankle pressure drop is seen

Treadmill testing is the preferable test because it produces a physiologic stress that reproduces a patient's ischemic symptoms.

Chapter 6 Doppler Segmental Pressures (UE)

1. **CAPABILITIES** See previous "Lower Extremities"

2. **LIMITATIONS** See previous "Lower Extremities"

3. **PATIENT POSITIONING**

   - Supine with arms relaxed at the patient's sides.
   - Cuffs should be on snug, but not tight.

4. **PHYSICIAL PRINCIPLES**

   (Refer to previous "Doppler Velocity Waveforms" Evaluation)

5. **TECHNIQUE**

   a. 12 x 40 cm cuff placed snugly on the upper arm; 10 x 40 cm cuff on the forearm bilaterally.

   b. Brachial artery used to obtain upper arm BP; radial and ulnar arteries used to obtain the forearm pressure.

   c. Pressures are combined with Doppler velocity wave forms from the sites previously described.

   d. **Digits** are discussed in Chapter 9

6. **ALLEN TEST** to evaluate patency of the palmar arch.

   Technique:

   a. With manual compression of radial A by the technologist, patient clenches ipsilateral fist (<1 min.) inducing pallor.

   b. With manual compression of the radial artery continuing, patient is asked to relax the hand.

c. Interpretation:

NORMAL	ABNORMAL
• Reappearance of the normal color to indicate the ulnar artery is providing flow to the palmar arch.	• Color does not reappear to indicate: an ulnar artery occlusion or palmar arch obstruction.

d. Limitations:

- Excessive dorsiflexion of wrist may compress radial and ulnar arteries leading to a false positive test.

- If hand is opened, fingers forcibly extended; the skin over palm can be stretched, (could lead to pallor due to compression of small vessels).

e. Documentation:

- Use a PPG on the index finger to document arterial pulsations before and after the "clenched fist."

- Although more difficult to compress, the ulnar artery can be evaluated similarly to assess radial artery flow.

7. INTERPRETATION

a. A 15-20 mm Hg difference from one brachial pressure to the other, suggests a >50% stenosis of subclavian artery and/or the vessel under the cuff. (Refer to Chapter 21)

b. A >15-20 mmHg drop from upper arm to forearm suggests:

- A brachial artery obstruction distal to the upper cuff
- Obstruction in both radial and ulnar arteries.
- Obstruction in single forearm artery which has decreased pressure.

c. A difference of $\geq$20 mm Hg between radial and ulnar pressures suggests obstruction in vessel with lower P.

d. See Chapter 9 (Digits) for additional information.

Chapter 7 Penile Non-imaging and Imaging

1. CAPABILITIES

Helps determine whether impotence is related to peripheral vascular insufficiency.

2. LIMITATIONS

a. High level of anxiety or antagonism.
b. Duplex needed for velocities and anatomic conditions.
c. Sensitivity to injectable medication; and/or anticoagulation therapy may prevent use of the injection component.

3. PATIENT POSITIONING

a. Supine, with appropriate draping done to maintain privacy.

4. PHYSICAL PRINCIPLES

a. Doppler previously described.
b. Plethysmographic techniques described in next section.
c. Duplex scanning principles.

5. TECHNIQUE / INTERPRETATION: Non-imaging

Technique:

a. Doppler velocity signals obtained bilat. (CFA, PTA, DPA).

b. Appropriate BP cuffs placed on upper arms, and ankles (refer to LE arterial):

- Calculate ABI (previously described, Chapter 5)
- Poor arterial inflow to LE due to proximal obstruction may affect penile arterial inflow.

c. Penile pressures obtained with Doppler or PPG as the "end point detector." Probe placement: lateral or ventral. Cuff size: 2.5 cm X 12.5 cm

d. Penile plethysmographic waveforms obtained as needed.

Interpretation:

a. Evaluation of the Doppler waveform quality: CFA, PTA and DPA as described previously.

b.

PENILE / BRACHIAL INDEX		
• Normal	=	$\geq$ 0.75
• Marginal	=	0.65 - 0.74
• Abnormal*	=	< 0.65

* Consistent with vasculogenic impotence

c. Reduced pressure highly suggestive of more proximal arterial disease (e.g., aorto-iliac: internal iliac arteries.

d. Abnormal plethysmographic tracings show marked reduction in amplitude.

6. TECHNIQUE / INTERPRETATION: Imaging

Technique: (Use a 7 or 10 MHz transducer)

a. Obtain informed consent

b. Prior to injection of medication, the cavernous (cavernosal) arteries: are measured (A/P diameter) in transverse approach; PSV, EDV obtained.

c. Specific medication is injected into lateral aspect, proximal shaft of penis to induce erection.

d. After 1-2 minutes post-injection, the same measurements are obtained as before injection (see b. above)

e. Dorsal veins flow velocity measured.

f. Important: If rigid erection is maintained for up to 3 hours, patient must be instructed to contact his Urologist immediately to reverse the priapism.

Interpretation: (Varies in the literature)

- Normal: Diameter of cavernous arteries should increase post-injection.
- PSV measurements should increase.
  Normal range: PSV approx. 30 cm/sec or higher. Anything less often considered abnormal.
- Dorsal vein velocities should not increase; an increase could suggest venous leak. Normal: <3 cm/sec; Abnormal: >20 cm/sec.

Chapter 8 Plethysmography (UE, LE)

1. CAPABILITIES

a. In combination with Doppler segmental pressures helps differentiate true claudication from non-vascular sources.

b. Detect presence/absence of arterial disease while defining its functional aspects.

c. Help localize the level of obstruction.

- Assessment of follow-up treatment.
- PPG mainly used for evaluation of digits and penile vessels.

2. LIMITATIONS

a. Cannot be specific to one vessel. Tracing reflects all arterial flow beneath the cuff.

b. Cannot discriminate between major arteries and collateral branches. (Venous influence not completely eliminated).

c. Difficult to perform volume plethysmography on obese patients.

3. PATIENT POSITIONING

Most exams can be done with the limbs in a resting position (patient supine). The patient can be sitting for evaluation of upper limb digits.

4. PHYSICAL PRINCIPLES

<u>Volume (air) Plethysmography</u>
(Measurement of volume change)

a. Pneumatic cuffs are placed around specific levels of the extremities or digits.

b. A measured amount of air is sequentially inflated into a cuff to pressure ranging from 10 to 65 mm Hg, depending on cuff size.

c. As arterial flow moves under the cuff, momentary volume changes in the limb segment occur.

d. These changes beneath cuff are converted to pulsatile pressure changes within the air-filled cuff bladder.

e. A pressure transducer converts the pressure changes into analog waveforms for display on strip-chart recorder.

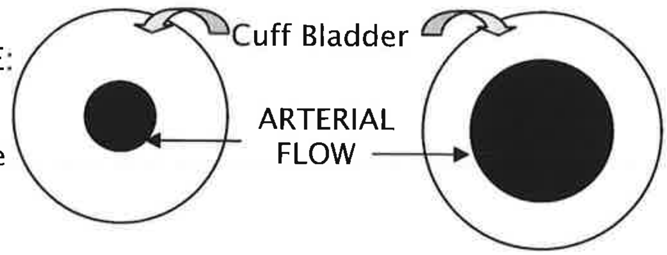


Photo-plethysmography (PPG)

a. Consists of transducer, amplifier, strip-chart recorder.

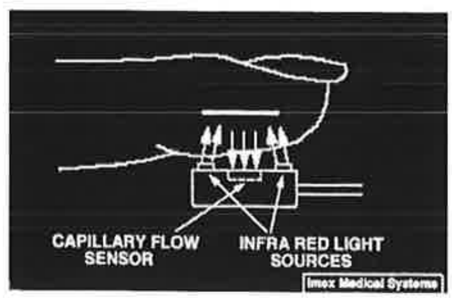
b. Detects cutaneous blood flow, rather than truly measuring volume change

c. Photocell consists of light emitting diode and photo-sensor.

d. Diode transmits infrared light into subcutaneous tissue with backscattered light reflected back to the adjacent photo-sensor.

e. The cutaneous blood flow determines the reflection.

f. Blood attenuates light in proportion to its content in tissue. Increased blood flow results in decreased reflection. However, that's displayed as an increase/positive deflection on the waveform.



5. **TECHNIQUE**

Volume Plethysmography

a. Patient supine, with heels slightly elevated on cushion.

b. Using a 3 or 4 cuff method, appropriately sized pneumatic cuffs applied snugly to: thigh, calf and ankle bilaterally.

c. Machine performs a self-calibration when activated.

d. Bilateral brachial pressures obtained during the Doppler segmental pressure exam.

e. Appropriate amount of air used to bring cuff pressure to predetermined levels. Begin with upper part of extremity, moving distally. Record at least 3 pulse cycles.

f. Artifacts not uncommon due to improper cuff application.

g. Similar "Gain" setting is maintained throughout the study. If a different setting is used, note on the recording paper.

h. Volume plethysmography and pressures are complementary

<u>Photo-plethysmography (PPG)</u> See DIGITS, Chapter 9

6. INTERPRETATION

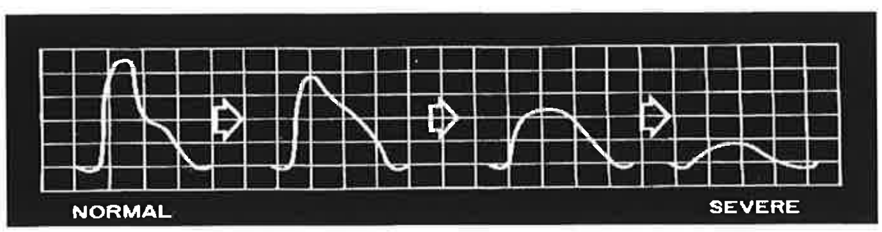
Both "volume" and "photo" plethysmography are evaluated using qualitative criteria as described by DE Strandness, MD and J Raines, PhD. Major descriptions follow:

a. Normal: fairly rapid upslope, sharp systolic peak with reflected wave.

b Mildly abnormal: Sharp peak, absent reflected wave, downslope is bowed away from baseline.

c. Moderately abnormal: Flattened systolic peak, upslope and downslope more delayed; reflected wave (notch) absent.

d. Severely abnormal: Low amplitude,or may be absent.



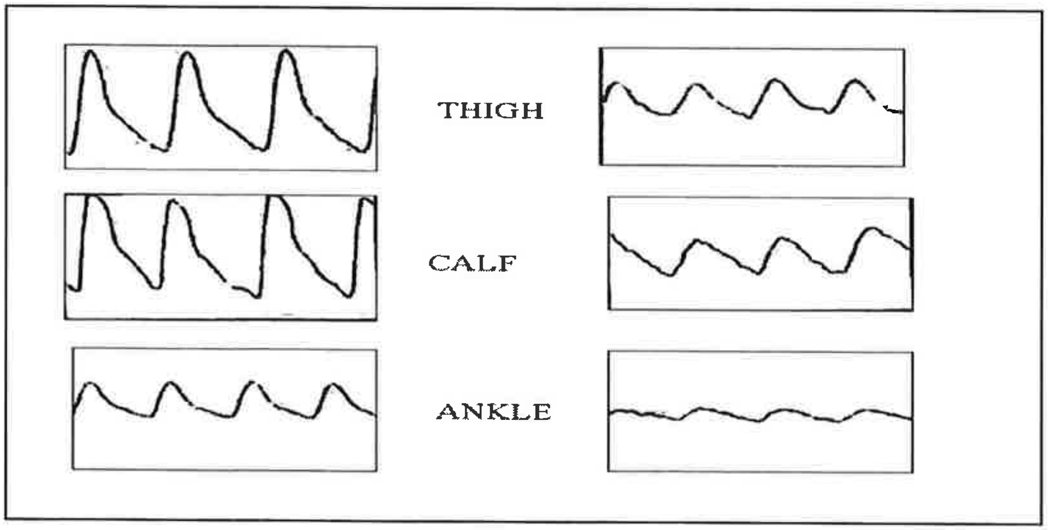
ADDITIONAL NOTES:

e. Abnormal waveforms always reflect hemodynamically significant disease proximal to level of tracing. Severity generally underestimated.

f. Reduced amplitude with no changes in the contour is likely to reflect insignificant disease, unless it is unilateral.

g. EXAMPLE STUDY:

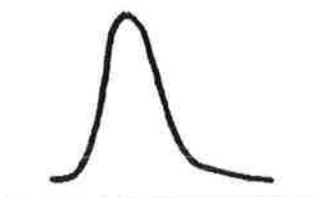
RIGHT LEFT



Abnormal plethysmographic waveform quality suggests poor arterial flow at all levels on the left . There is normal waveform quality seen on the right at all levels.

h. A fair waveform quality may accompany abnormal segmental pressures because collaterals can underestimate significance of obstruction based upon plethysmography.

Normal



Based on collaterals

Note: Amplitude similar; waveform qualities different.

i. Additional qualitative criteria found in the next chapter.

j.

TROUBLESHOOTING

- *Can't center the stylus?* Check the "mode." AC mode for arterial; DC mode for venous.
- *Stylus wandering on paper?* Activate the "re-set" control; and be sure correct exam function is selected.
- *Unable to obtain clean waveform?* Reapply the PPG. Patient tremors make an exam nearly impossible.
- *No tracing?* Check the exam mode, paper, and connection points.

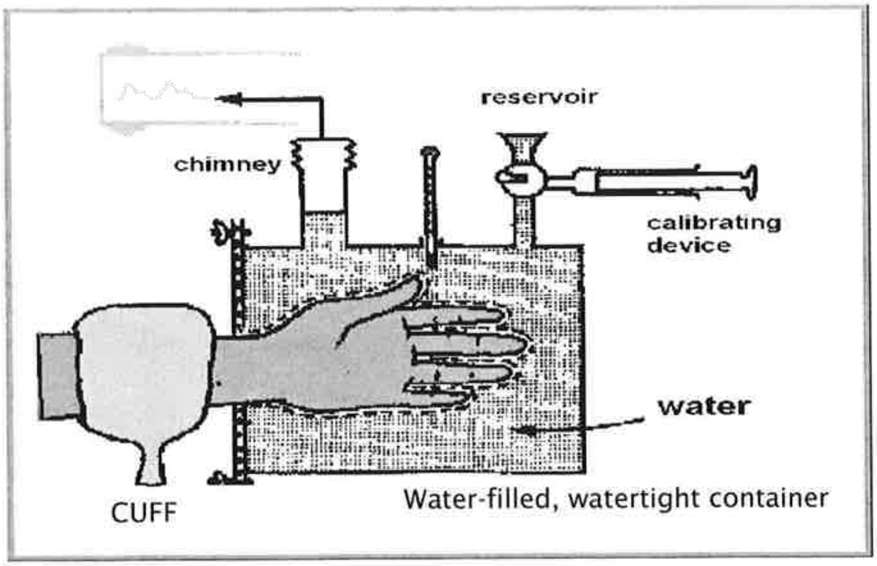
7. DISPLACEMENT PLETHYSMOGRAPHY:

a. Any change in volume of the enclosed part will displace an equal amount of water. Displacement is measured by the height of the water in the chimney.

b. Accurate for detecting changes in limb volume.

c. Pulse plethysmography refers to transient changes in (limb) volume related to the pulse by pulse activity of the left ventricle; the (body) part expands when arterial inflow exceeds venous outflow.

In the example below, hand is inside a surgical rubber glove (as depicted by the dotted/dashed line). The water temperature is constant.



From: Noninvasive Diagnostic Techniques in Vascular Disease Eugene F. Bernstein, M.D., 3rd edition, C.V. Mosby 1985

Chapter 9 Digit Plethysmography and Pressures

1. CAPABILITIES

a. Help detect presence of arterial disease.

b. Differentiate fixed arterial obstruction from vasospasm.

c. Assess effects of treatment.

2. LIMITATIONS

a. Quality of tracings greatly affected by the vasoconstricted state of arteries before testing begins. (Has patient: come in from cold weather? been smoking? nervous?)

b. With **volume plethysmography**, cuffs applied too tightly, can obliterate or diminish the pulse wave forms.

c. With **photo plethysmography** improper contact with skin surface will cause poor results.

d. Bandages that can't be removed; ulcerations and/or gangrene may prevent placement of cuff or photocell.

e. Patient with extremity tremor.

3. PATIENT POSITIONING

a. Toe evaluation: supine with some elevation of head.

b. Finger evaluation: sitting with arms resting on pillow placed on patient's lap.

4. PHYSICAL PRINCIPLES

The principles for volume and photo plethysmography have been discussed previously.

5. TECHNIQUE—TOES

a. Patient should be kept as warm as possible.

b. The study is done in combination with a complete LE arterial exam, or a limited version such as an ABI.

c. An appropriately sized cuff, the width at least 1.2 times that of the toe, (2.5 - 3 cm) is applied to base of great toe.

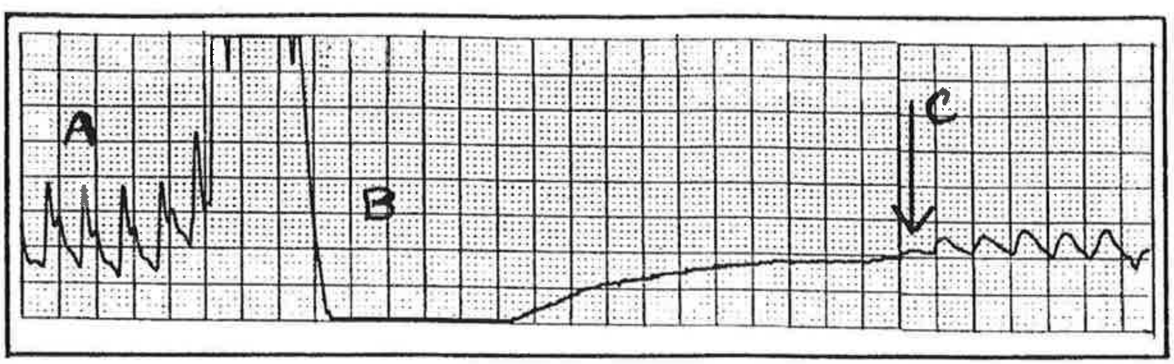
d. Equipment does a self calibration when activated.

e. Plethysmographic waveforms are recorded for all toes as previously described. (If poor waveforms, warm the toes).

f. Usual method for obtaining great toe pressures is PPG: (photocell as end point detector). METHOD FOLLOWS:

- Digit cuff placed at base of great toe.
- Photocell securely attached to plantar side of toe using double-stick tape or velcro strap.
- Pulses recorded; paper speed=5mm/sec. **(A)**
- While pulsations are recorded, cuff is inflated to 20-30mmHg above highest brachial P.
- No pulsations should be seen. **(B)**
- Cuff is slowly deflated, watching for return of first pulse to define the pressure level. **(C)**
- Can also be with volume plethysmography.

EXAMPLE:



The increase in toe volume (rise from A to B) secondary to obstruction of venous outflow. (Plethysmography measures all volume changes).

6. **TECHNIQUE: Fingers without cold stress**

a. Upper extremity arterial study is completed initially.

b. Evaluate Doppler signals (triphasic, biphasic, monophasic) and obtain pressures.

c. Doppler exam of the palmar arch, to verify patency.

d. Apply finger cuffs (sizes: 2 - 2.5 cm)

e. Pressures and waveforms obtained (using similar method described for toes).

7. **TECHNIQUE: Fingers with cold stress**

Performed in cases of symptoms occurring due to **cold sensitivity**, (Toes can also be evaluated).

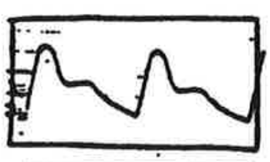
a. After resting study is performed, hands are immersed in ice cold water for 3 minutes if possible.

b. Following cold stress, waveforms and pressures are obtained: immediately after and five minutes after.

c. Document patient symptoms, skin color observations and other pertinent findings on the report form.

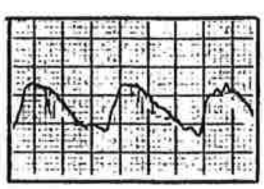
8. **INTERPRETATION**

a. Normal waveform qualities:



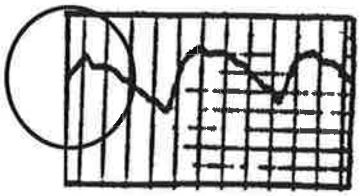
- Sharp upstroke during peak systole
- Prolonged down-stroke with notch (reflected wave) approximately half way down.
- Amplitude is usually greater than toe tracings

b. Abnormal **Obstructive** waveform qualities:



- Occlusion located anywhere proximal to the tip of the finger causes the pulse to assume an "obstructive" form.
- Slow upslope to rounded peak.
- Downslope that bows away from baseline.

c. Abnormal **Peaked** waveform qualities:



- Upslope is slower than normal.
- Sharp, (anacrotic) notch is present
- Reflected wave located high on the downslope.
- Has characteristics of both normal and the obstructive waveforms.

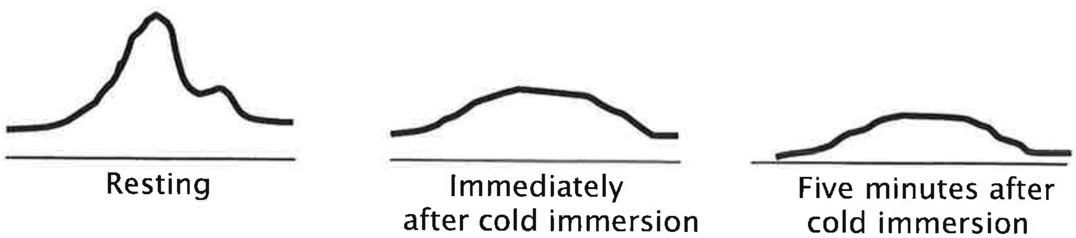
d. Other observations:

- Organic (fixed) obstructive disease has abnormal Doppler arterial signals, systolic pressures and PPG tracings.

- Functional (intermittent) obstructive disease has normal Doppler arterial signals, systolic pressures, and/or PPG tracings; but abnormal findings after cold stimulation.

- Sumner and Strandness described the "peaked" pulse characteristically seen in the digit pulse contours of patients with Raynaud's phenomena.

Following cold immersion, abnormal cold sensitivity is likely if the amplitude fails to return to baseline levels within 5 minutes.



e.

DIGIT PRESSURE MEASUREMENTS

- UE digits: Finger/brachial indices 0.8-0.9
- LE digits: Toe/brachial Index (TBI)
  - ◊ Normal toe P's vary from 60-80% of brachial P.
  - ◊ The presence of artifactually high ankle pressures from arterial calcinosis usually negates a toe/ankle P index.
  - ◊ Digital artery occlusion: severely reduced pressure.

Chapter 10 Transcutaneous Oximetry TcPO_2

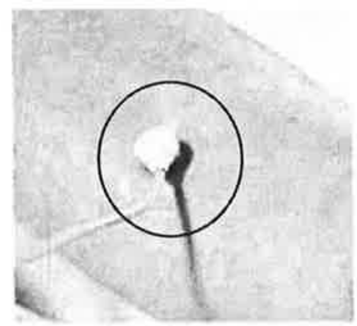
1. **CAPABILITIES**

- Wound healing and amputation level determination.
- Reflects tissue oxygen tension and relies on a balance between oxygen supply and consumption.

2. **LIMITATIONS**

- Inability to keep electrode fairly flat on the skin surface.
- Electrode placement must be on intact skin.

3. **PATIENT POSITIONING**



Patient supine, head slightly elevated; room is warm

4. **PHYSICAL PRINCIPLES**

a. Electrode heats skin to 45° C.

- Blood flow increases
- Lipid layer in fatty tissue melts
- More "O_2" escapes through skin, measured by sensor in electrode

b. As "O_2" escapes, a chemical reaction occurs in electrolyte solution, located between skin and electrode surface.

c. Electrode converts chemical reaction to a "current" reading which is converted to a pO_2 reading in mmHg.

5. **TECHNIQUE**

a. Skin cleansed with alcohol wipe; air dried.

b. Airtight self-adhesive fixation ring placed on the skin

c. Few drops of electrolyte solution put inside ring.

d. Electrode/sensor is affixed to fixation ring.

e. Reference reading (upper lat. chest) obtained first; followed by specific sites (e.g., near toes, BK, AK)

f. After manual calibration, it takes approximately 15-20 minutes/site to obtain "pO_2" reading.

6. INTERPRETATION

a. Normal values vary in literature and with manufacturer. Healing should occur with a pO_2 reading of 70-80 mmHg;

b. Borderline healing: 30-40 mmHg.

c. Non-healing (poor values) = 10-15 mmHg. This number also varies.

Chapter 11 Duplex/Color Flow Imaging (UE)

1. **CAPABILITIES**

   a. Localize stenosis / occlusion; evaluate degree of stenosis.

   b. Determine the presence/absence of aneurysm.

   c. Post-op study: hemodialysis access or arterial bypass graft.

   d. Detect AVF's or other unusual abnormality.

2. **LIMITATIONS**

   a. Limited access to extremity (e.g., dressings, skin staples or sutures, open wounds, IV site).

   b. Pertaining to hemodialysis access grafts:

      —Graft angulation

      —Difficult to adequately evaluate the outflow vein in an obese patient.

3. **PATIENT POSITIONING**

   a. Patient is supine with small pillow under head.

   b. Extremity close to the examiner.

   c. Arm is at a 45 degree angle from the body, and externally rotated. ("pledge position")

4. **PHYSICAL PRINCIPLES**

   a. Duplex scanning: combination of real-time B-mode imaging (gray scale evaluation) and Doppler spectral analysis.

   b. Doppler Color Flow Imaging: Doppler information is displayed on image after evaluated for phase (direction toward or away from transducer) and its frequency content (hue or shade of the color).

c. Sample size for acquiring pulsed Doppler information is usually 1-1.5mm. Size is increased incrementally if needed.

d. In-depth coverage of this process is covered in the morning "Physics Section."

5. TECHNIQUE: Native Arteries

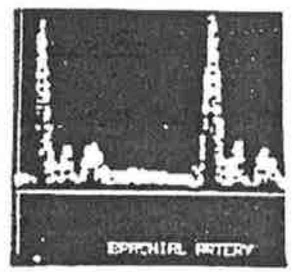
a. Utilize a 7 or 5MHz linear array transducer.

b. Neck vessels identified, with attention given to innominate artery on the R. The LCCA branches off the Arch.

c. Color/duplex scanning is also used to evaluate the following:

Subclavian
Axillary
Brachial
Radial
Ulnar
Palmar arch (if needed)

Example UE arterial signal →



d. Uncommon for arteries in the upper extremities to become stenotic. Main use is evaluation of dialysis access grafts.

6. TECHNIQUE: Hemodialysis Access

a. Auscultate the access for bruit and/or palpate for a "thrill" (vibration). A patent dialysis access, as well as a stenotic one can produce a "thrill."

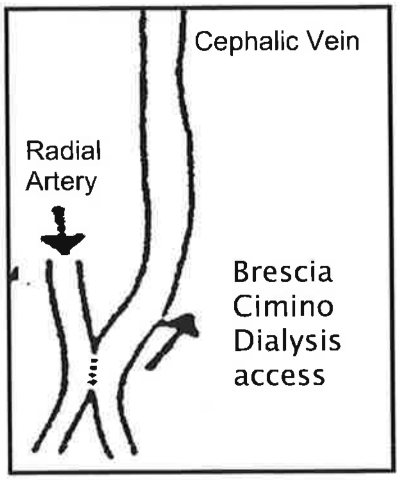
b. Utilize a 7 or 5 MHz linear array transducer.

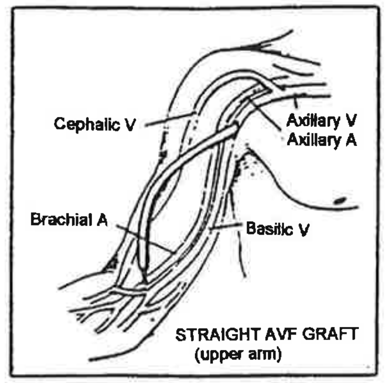
c. Evaluate dialysis access grafts as follows:

- Inflow artery
- Arterial anastomosis
- Continue through the body of the graft
- Observe for aneurysm, puncture sites, peri-graft fluid.
- If color available, observe the image for flow changes, turbulence, flow channel changes.
- Venous anastomosis
- Outflow vein

Note: Dialysis access (e.g.Brescia-cimino fistula) assessment sites include: inflow artery, anastomosis, outflow vein.

d. Dialysis access examples include: Brescia-Cimino fistula; straight, looped synthetic grafts.





7. <u>**Technique** - General Observations:</u>

a. Longitudinal/transverse approaches used to evaluate gray scale for thrombus, stenosis etc.

b. Doppler peak systolic velocities (PSV) obtained at appropriate sites and as needed. Volume flow may be evaluated.

c. Documentation consists of video tape or prints.

8. INTERPRETATION: Native Arteries

a. <u>Stenosis</u>:

- Currently, no criteria for classifying disease as there is for LE.
- Normal peak systolic velocities vary widely with skin temperature changes. Doppler signal quality is usually triphasic.
- If a >50% diameter reduction is present, observe for characteristics of the "stenosis profile."

b. <u>Occlusion</u>:

- Observe for lack of Doppler signals (image and/or waveform) and the proverbial "thump" which is obtained proximal to occlusion.

c. <u>Aneurysm:</u>

- Dilation of the vessel from degeneration and/or weakening of the wall.

- Ulnar artery aneurysms can form in response to using the palm as a hammer.

- Subclavian aneurysms often are associated with embolization to the digits.

9. INTERPRETATION: Hemodialysis Access

a. Identify/document if present (location, extent, and type) any aneurysmal changes, puncture sites, peri-graft fluid, thrombus.

b. Peak systolic velocities (PSV) and end diastolic velocities (EDV) vary as to the type of access; normally both are elevated. Low PSV obtained in access graft could indicate arterial inflow problems.

c. No standardized velocity criteria at this time. Follow-up studies provide specific comparison to previous studies on same patient.

d. Venous anastomosis and outflow vein are most common sites for stenosis (likely caused by increased arterial pressure introduced to the vein and/or intimal hyperplasia).

e. Other hemodynamic complications include:

- Large blood volumes shunted from artery to lower resistant venous circulation, can increase venous return resulting in congestive heart failure.

- A "steal syndrome:" the distal arterial blood flow is reversed into the lower resistant venous circulation and can cause pain on exertion, pallor and coolness of the skin distal to the shunt.

Chapter 12 Duplex / Color Flow Imaging (LE)

1. **CAPABILITIES**

   a. Determine presence/absence of >50% diameter reduction or occlusions.

   b. Determine presence/absence of aneurysms.

   c. Follow up of bypass grafts.

   d. To localize the stenotic lesion prior to balloon angioplasty.

2. **LIMITATIONS**

   a. Presence of dressings, skin staples, sutures; open wounds.

   b. Incisional tenderness, hematomas.

   c. Obesity - may be difficult to image vessels

   d. Calcific shadowing from diabetes and/or atherosclerosis.

3. **PATIENT POSITIONING**

   a. Patient supine with head on pillow.

   b. Extremity positioned close to examiner.

   c. Pt's hip minimally rotated externally, with knee flexed.

   d. Prone or lateral decubitus position may be required for assessment of popliteal artery.

4. **PHYSICAL PRINCIPLES**

   Detailed presentation in the morning "Physics" session. The Doppler equation may be written as:

$$\boxed{Df = \frac{2\ Fo\ V\ Cos\ Ø}{c}}$$

- Df = Doppler shift frequency.
- Fo = Carrier frequency
- c = Speed of ultrasound through soft tissue (1540m/sec)
- V = Velocity of the moving reflectors
- Cos ø = Angle

a.

> The number 2 represents <u>two</u> Doppler shifts:
>
> - Red blood cell is first an observer of a stationary ultrasound field.
> - Then acts as a wave source when the waves scatter from its surface

<u>CALCULATING FOR VELOCITY:</u>

$$\boxed{V = \frac{c \quad Df}{2\,Fo \quad Cos\,Ø}}$$

- Df = Known value, <u>measured</u> by the duplex. Proportional to velocity of the source.
- V = Calculated value which represents velocity of flow (moving reflectors). To calculate, the frequency (Df) must be known.
- Cos Ø = Angle (Ø) determined from the image; cosine theta is calculated. (e.g., cos Ø of 90 = 0).

b. Source of error (solving for velocity) is the Doppler angle theta (Ø) which <u>increases</u> its nonlinear influence as the angle becomes closer to 90 degrees.

c.

> **Ideal Doppler Angle for Vascular exams is 60 degrees**, obtained centerstream, parallel to walls. Vessel tortuosity and other conditions, will cause the Doppler angle to be < 60 degrees, which is usable.
>
> **Not reliable: Doppler angle of > 60 degrees.**

d. The effect on the Doppler shift frequency display when the carrier frequency is changed: Decreased transducer frequency = decreased Doppler waveform size (e.g., 8 MHz changed to a 4 MHz)

5. TECHNIQUE: Native Arteries

a. Scan with a 7 or 5 MHz linear array transducer for the following:

- Distal external iliac artery
- Common femoral artery (CFA)
- CFA bifurcation
  —Superficial femoral artery (SFA)
  —Profunda femoris

- S-femoral artery (proximal - distal)
- Popliteal artery (proximal - distal)
- Trifurcation
  - Anterior tibial artery (ATA)
  - Posterior tibial artery (PTA)
  - Peroneal artery

b. Observe gray scale for plaque or other abnormalities.

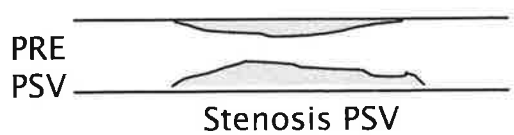
c. Observe color flow patterns.

d. Obtain peak systolic velocities (PSV) from each major vessel (proximal, mid and distal if vessel is long) and more if disease suspected.

- Characterize the Doppler signals: triphasic (normal), biphasic, or monophasic (abnormal).
- Observe wave form for any spectral broadening.

e. If a >50% diameter reduction is suspected, obtain:

- Pre-stenotic PSV (approaching the stenosis).
- PSV (highest) in stenosis.



f. Post-stenotic turbulence signals.

- Document as previously described for UE.

6. TECHNIQUE: Bypass grafts

a. Know type, location and age:

- Synthetic (e.g., Gortex)

Reversed Saphenous Vein Graft (RSVG)

- Small end is now proximal
- Large end is distal
- Vein valves stay open due to arterial flow pressure
- Branches are ligated.

In-situ Vein Graft

- GSV stays in place
- Small end is distal
- Large end is proximal
- Prior to surgery, valves broken up with special instrument; branches ligated

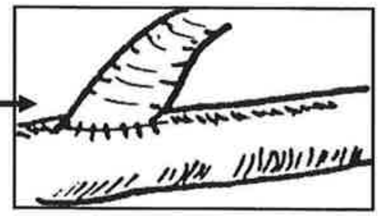
b. Protocol usually combined with ABI's.

c. Vein bypass graft evaluation includes: observation of gray scale, color flow and PSV's of the following:

- Inflow artery
- Proximal anastomosis
- Entire length of the vein bypass graft
- Distal anastomosis
- Outflow artery
- Also check for branches (that could form AV fistulas), valves, and/or other abnormalities.

d. Synthetic bypass graft evaluation include observations of gray scale, color image, PSV's of the following:

- Inflow artery
- Proximal anastomosis
- Mid graft
- Distal anastomosis
- Outflow artery



7. **INTERPRETATION: Native Arteries**

a. NORMAL:

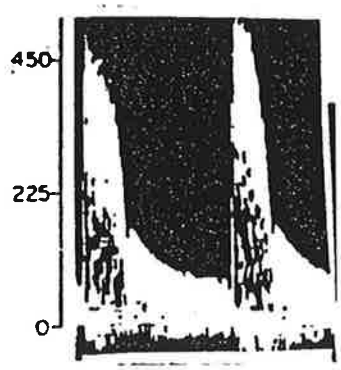
- Doppler signals are triphasic; a change to biphasic can be significant; however some patients normally have biphasic flow without evidence of disease.

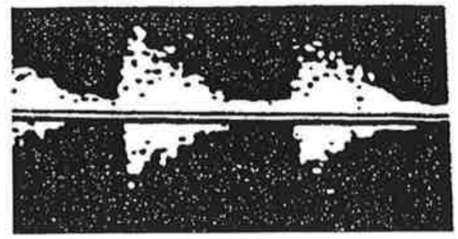
b. ABNORMAL FINDINGS:

Compare stenotic PSV to pre-stenotic PSV

- 2:1 ratio = ≥50% diameter reduction
- 4:1 ratio = ≥75% diameter reduction
- >400 cm/sec PSV = ≥75% diameter reduction
- Never accept the value of "numbers" alone; post-stenotic turbulence should be present.

- Observe for occlusions; aneurysms.





Pre-stenotic Doppler spectra (monophasic and dampened)

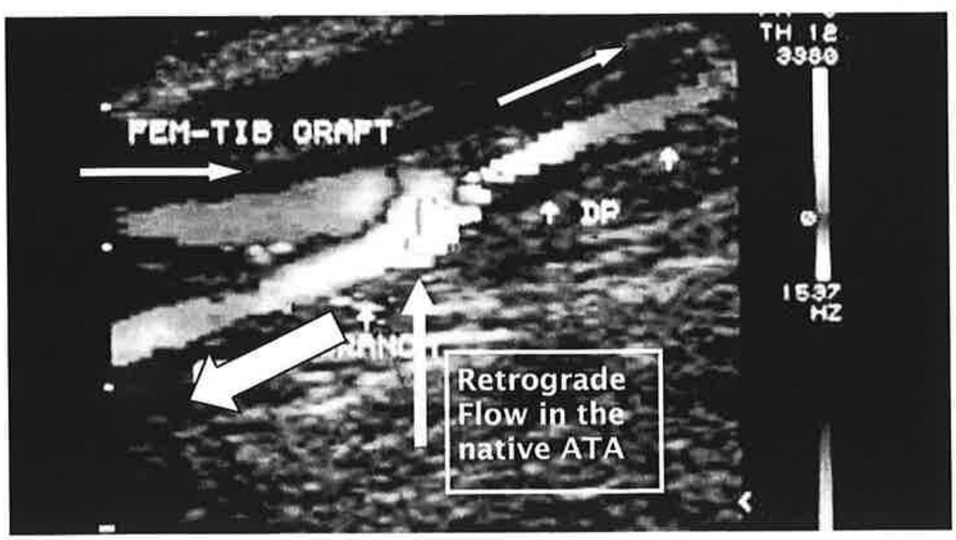
Doppler spectra obtained AT the stenosis (highest PSV's documented)

Post-stenotic turbulence (Evident distal to a > 50% stenosis)

8. **INTERPRETATION: Bypass Grafts**

a. NORMAL:

- As stated under "Native Arteries."
- Retrograde flow in the native artery may be evident at the distal nastomosis of a RSVG, which provides an additional source of collateral flow. (Retrograde flow results from a pressure gradient)



b. ABNORMAL FINDINGS:

Compare results with previous studies and observe for changes such as:

- Decrease of 30 cm/sec in any graft segment.
- Reduced PSV's in smallest graft diameter that were greater previously.
- Change from triphasic to biphasic signals.
- Decrease in ABI of > 0.15.
- Observe for post-complications such as: AV fistula (can siphon off graft flow); valve cusp.
- Can also apply previous data used for native artery stenosis

c. SYNTHETIC GRAFTS:

- Can loosely apply the previous data to determine whether a > 50% diameter reduction exists.
- Anastomotic sites should be evaluated well for aneurysm and/or stenosis.

d. Observe for graft occlusion.

9. INTRAOPERATIVE MONITORING

- Used for checking patency of the anastomotic sites.
- Evaluate any suspicious stenotic or turbulent areas that can occur in vein bypass grafts (e.g., valve cusp sites or suspected branch sites.)
- Further description of this protocol, see Carotid Duplex section.

Chapter 13 Duplex / Color Flow Imaging (Abdomen)

1. **CAPABILITIES**

   a. Aorto-iliac vessels: evaluate vessels for stenosis, status of bypass grafts, aneurysmal disease.

   b. Renal artery: to document >60% diameter reduction.

   c. Kidney: to help in evaluation of nephrosclerotic disease, and transplants.

   d. Mesenteric arteries: to document significant stenosis, or evaluation of mesenteric bowel ischemia.

   e. Liver: suspected portal hypertension; pre/post liver transplants; other.

2. **LIMITATIONS**

   a. Size of patient

   b. Bowel gas

   c. Previous abdominal surgery (scar tissue)

   d. Shortness of breath and rapid respirations

   e. Patient in non-fasting state

3. **PATIENT POSITIONING**

   a. Supine, with minimal head elevation.

   b. Left lateral decubitus (LLD) for access to the right flank and the right lateral decubitus (RLD) for access to the left flank.

   c. Requires creative patient positioning and scanning approaches in order to optimize the study.

4. **PHYSICAL PRINCIPLES**

   Previously described in "Duplex scanning / color flow imaging of the upper extremities."

5. TECHNIQUE: General Remarks

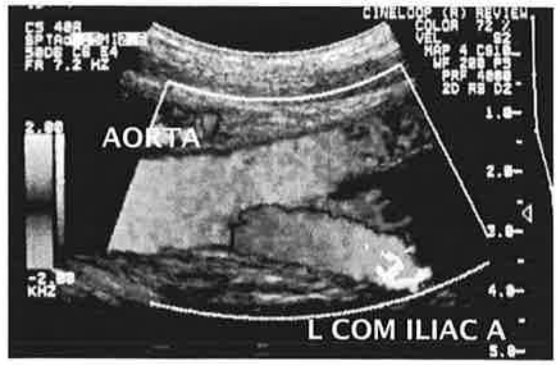
a. Transducer selection related to size of patient; most adult patients require 5, 3, or 2.25 MHz transducer.

b. Combine longitudinal and transverse approaches; evaluate gray scale and color flow patterns, observing for aneurysm, plaque etc.

c. Use longitudinal approach to assess Doppler signal PSV and EDV.

6. AORTO-ILIAC ARTERIES: Technique/Interpretation

<u>TECHNIQUE:</u> (See Technique - general remarks)



a. Observe celiac and SMA
b. Observe renal arteries
c. Aorta: Proximal, mid, distal
d. Common iliac A bilaterally
e. Bil external iliac A Prox - distal
f. Internal iliac artery bilaterally
g. See technique section in Chapter 12

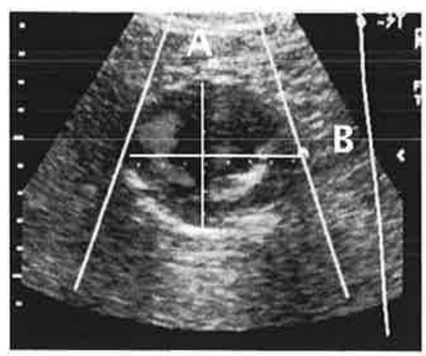
<u>INTERPRETATION:</u>

a. **Aorto-iliac arteries** - stenosis

Please refer to the "Duplex scanning/ color flow imaging of the LE arteries.

b. **Aorto-iliac arteries** - aneurysm

- A dilation > 3cm OR an increase in diameter of 50% greater than original artery.



A = 5.6 cm
B = 6.1 cm

- Majority of abdominal aortic aneurysms (AAA) are atherosclerotic and infrarenal (below the renal artery branches.)

- Type of aneurysm: true, (fusiform, saccular); false (thrombus present?)

Most frequent complication: Rupture of the aortic aneurysm; embolization of the peripheral aneurysms.

7. RENAL ARTERIES: Technique/Interpretation

Patient history: Many of these patients will present with hypertension (controlled or not). Many patients who have hypertension have **renovascular hypertension**, usually caused from renal artery stenosis (which can be secondary to atherosclerosis, fibromuscular dysplasia) or occlusion.

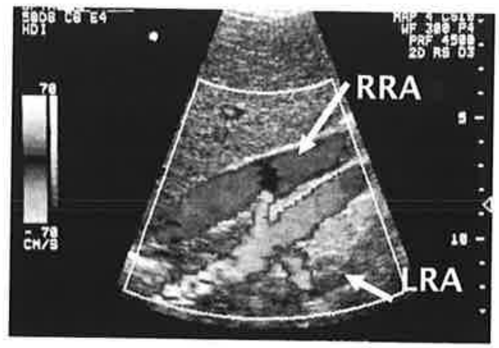
<u>TECHNIQUE:</u> (See Technique - general remarks):

a. Celiac artery and SMA velocity data.

b. Obtain aorta PSV near SMA level

c. In transverse, locate renal arteries.

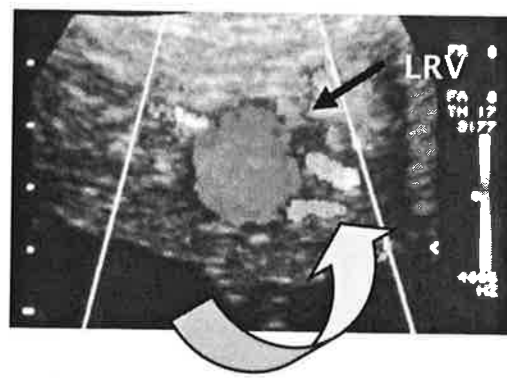
Left renal vein (LRV) is a landmark for identifying left renal artery.



d. Obtain kidney size (length) observe morphology bilaterally.

e. Obtain PSV and EDV bilaterally of:

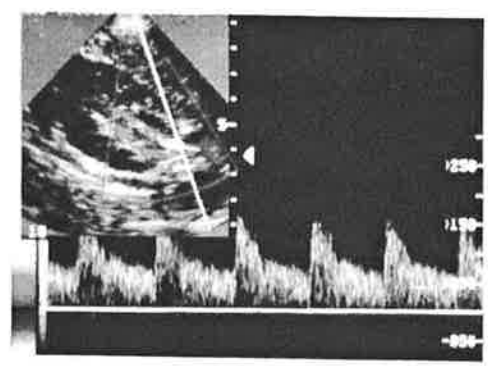
- Renal artery - proximal, mid and distal bilaterally.
- Upper / lower pole of the kidney in segmental arteries.



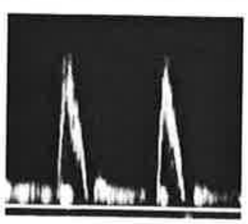
f. Observe for secondary or accessory renal arteries.

<u>INTERPRETATION:</u>

a. Renal arteries and kidney arteries are normally <u>low resistant</u> in quality as are: celiac, hepatic, and splenic arteries.



b. The aorta is usually <u>higher resistant</u> in quality, as are a fasting SMA and IMA.



c.

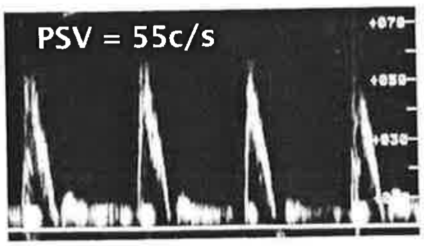
Renal to Aortic Ratio (RAR):

$$\frac{\text{Highest Renal Artery PSV}}{\text{Aorta PSV}}$$

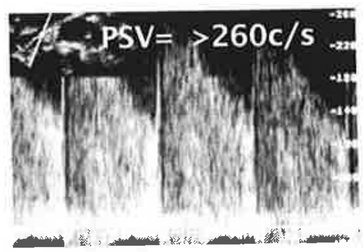
NORMAL = < 3.5
ABNORMAL = ≥ 3.5 Suggests ≥ 60% diameter reduction

Cannot use the RAR:
1) If AAA detected 2) If aortic PSV > 90cm/s or < 40cm/s

With these conditions, look for a renal artery PSV of 180-200cm/sec, that is also followed by post-stenotic turbulence.



AORTA



ABNORMAL RENAL A.

RAR:

$$\frac{\geq 260}{55} = >4.7$$

INTERPRETATION - Kidney (Can also apply to renal arteries)

a. Kidney arteries are normally low resistant in quality.

b. Observe the kidney for morphologic abnormalities (cyst, cortex thinning, other defects.)

c. Normal pole to pole length varies from 10 - 12cm (depends on patient size)

d. Ratios such as the following are utilized:

END DIASTOLIC RATIO (EDR)

Parenchymal Resistance Ratio (PR)

$$\frac{\text{End Diastolic V}}{\text{PSV}}$$

Normal = > 0.2
Abnormal = < 0.2

RESISTIVITY INDEX (RI)

$$\frac{\text{PSV - EDV}}{\text{PSV}}$$

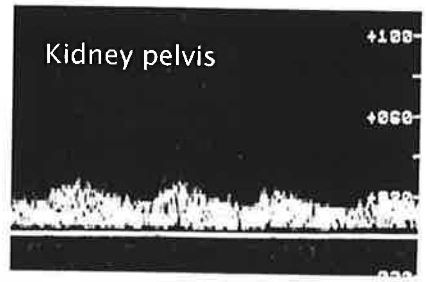
Normal = < 0.75
Abnormal = ≥ 0.75

Abnormal calculations indicate an increase in distal resistance (e.g. nephrosclerotic disease).

Example data: PSV of 45cm/sec; EDV of 5 cm/sec.

- The EDR = 0.1
- The RI = 0.88

e. Proximal high grade stenosis/ occlusion of renal artery may result in dampened, weak Doppler signals distally, but still of low resistant quality. Another term for this is tardus-parvus.



8. ## MESENTERIC ARTERIES: Technique / Interpretation

Patient history: Patients presenting with a history of dull, achy or crampy abdominal pain 15 - 30 minutes after meals are suspect for **mesenteric ischemia.** The pain, known as mesenteric angina, may be due to a stenosis or occlusion of the SMA, celiac, or IMA (inferior mesenteric artery)

Chronic and acute mesenteric ischemia: difficult to diagnose; require arteriogram.

TECHNIQUE:

a. See Abdominal Vessels: Technique - general remarks.

b. Perform exam on FASTING PATIENT.

c. Obtain PSV and EDV of the following: (longitudinal approach)

- Celiac artery (may use tx approach)
- Proximal, mid, distal SMA
- IMA
- Aorta (check for abnormality)

d. Although not routinely done, the food challenge test consists of patient ingesting high caloric liquid (e.g., Ensure). Repeat exam: 20-30 minutes (sooner if symptomatic); hyperemic response begins after about 10 minutes, with maximal response in approximately 30 minutes.

e. Post-prandially, the PSV and EDV of the SMA are obtained.

f. Document:

- Amount of high caloric liquid ingested (usually 1-2 cans),
- Onset, type and duration of symptoms,
- Time began the post-prandial study.

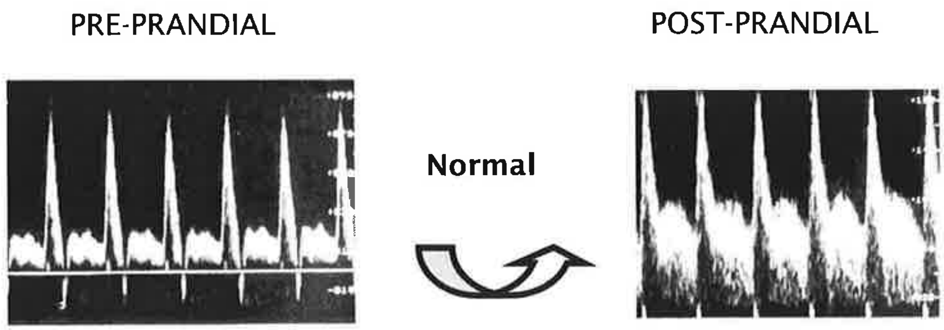
INTERPRETATION:

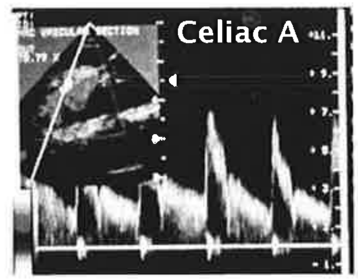
a.

Description	SMA	CELIAC
Pre-prandial (fasting)		
PSV	High	High
EDV	Low	High
Flow reversal	Yes	No
Post-prandial (food challenge)		
PSV	Marked increase	No change
EDV	Marked increase	No change
Loss of flow reversal	Yes	N/A
Velocity criteria		
Normal PSV*	110-177 cm/sec	50-160 cm/sec
Stenosis Criteria*	PSV $\geq$ 275 cm/sec predicts $\geq$ 70% diameter reduction	PSV $\geq$ 200 cm/sec predicts $\geq$ 70% diameter reduction

*Criteria /other info from Gregory Moneta, MD, various papers; Oregon Health & Science University Hospital, Portland, Oregon

b. SMA Doppler spectral waveform examples:





c. Celiac Artery branches into the hepatic and splenic arteries. The liver and spleen have fixed metabolic requirements—not likely influenced by the post-prandial state.*

d. Due to small caliber, difficult to locate IMA off distal aorta. If easily observed, often suggests SMA occlusion.

e. Observe for proximal stenosis of the SMA and/or celiac A.

f. In most cases, 2 of 3 mesenteric vessels have to be abnormal to be consistent with chronic mesenteric ischemia.

g. Extrinsic compression of celiac artery origin by the <u>median arcuate ligament of the diaphragm</u> is frequently seen.

- A reversible celiac artery stenosis occurs during expiration (ligament presses down over artery); rarely the cause of clinical symptoms.
- The high velocity signals, indicative of stenosis, are substantially improved with deep inspiration.

9. ORGAN TRANSPLANTS (Allografts)

(B-mode and Doppler analysis, utilizing a 5 MHz (or 3 MHz) transducer, combine to provide useful information.)

Liver Transplant:

a. Treats patients with end-stage liver disease.

b. Pre- and post-operative evaluation includes documenting patency of: portal vein, hepatic veins, IVC and hepatic artery. Observe for tissue abnormalities.

c. Post-op complications can include thrombosis of vessels such as: portal vein, IVC, and/or hepatic artery leading to hepatic infarction.

d. Acute rejection will cause liver dysfunction. Many suggest rejection is a cellular process.

<u>Renal Transplant:</u>

a. 5 MHz transducer for evaluation.

b. Both living-related or cadaveric donor kidneys used.

c. Transplanted RA anastomosed to EIA or IIA; transplanted RV anastomosed to EIV.

d. Post-op evaluation includes:

- Doppler evaluation as previously described in this chapter
- Careful B-mode observation for conditions such as hematoma, and parenchymal echogenicity.

e. B-mode signs of rejection include: increased renal transplant size, increased cortical echogenicity.

f. Acute rejection often increases arterial resistance; although other conditions can as well.

g. Many consider biopsy to be most reliable method for rejection diagnosis; others use the indicator of increased arterial resistance.

Chapter 14 Miscellaneous Conditions / Tests

A. ARTERIOVENOUS FISTULAE (AVF)

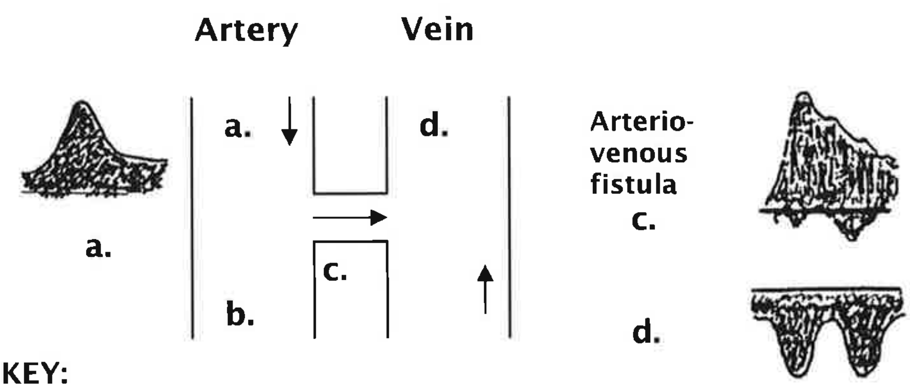
1. Abnormal connection between high-pressure arterial system and low-pressure venous system; marked anatomic and hemodynamic changes.

2. Congenital or traumatic.

3. Fistula close to heart, potential for cardiac failure increases

4. Fistula located peripherally, more likely to cause ischemia.

5. AVF may involve proximal and distal arteries/veins as well as collateral arteries/veins. Diameter and length predicts resistance it offers.



KEY:

a. Proximal arterial flow (proximal to the AVF) has greatly increased diastolic flow because the fistula reduces resistance.

b. Depending on size of fistula, distal arterial flow may resume its normal triphasic pattern or may be somewhat reduced.

c. Flow through the AVF has higher velocities; lower resistant flow

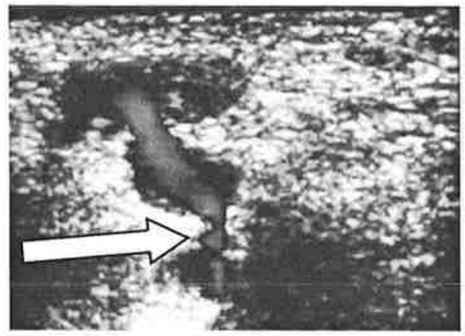
d. Venous outflow takes on the flow quality of the fistula's low resistant, more pulsatile flow.

B. COMPRESSION OF PSEUDOANEURYSM

(The following is a very general, simplified summary of this particular procedure)

1. Under guidance of B-mode ultrasound and Doppler color flow image, transducer compression of certain femoral pseudoaneurysms performed.

2. Size and location of pseudoaneurysm, as well as location of "neck" very important to determine.



> <u>Key to Success</u>: Whether aneurysm "neck" (communicating channel) between native artery and pseudoaneurysm can be <u>uniformly</u> and <u>completely</u> compressed.

3. <u>Technique:</u>

- Patient signs informed consent.
- Ideally, anticoagulant therapy discontinued.
- Dr. present during procedure.
- Patient somewhat sedated due to discomfort.
- No standardized protocol at this time, however alternate compression/ rest periods occur, with compression lasting about 10-15 min. each
- Image used to determine if compression maneuver is working, (i.e. thrombosis of aneurysm)
- Distal monitoring of the great toe arterial pulsations done to make certain there is continued inflow during compression maneuver.
- Procedure lasts up to 30-60 minutes. Varies widely.

4. Percentage of successful compressions varies. Protocol standardization is important. Increased experience determines overall effectiveness. If procedure unsuccessful, patient usually undergoes surgical correction.

C. POPLITEAL ARTERY ENTRAPMENT SYNDROME

1. Popliteal artery possibly compressed by medial head of gastrocnemius muscle (anomalous origin), or fibrous bands.

2. Often found in young men; Repeated trauma to artery may result in development of aneurysm, thrombosis, emboli.

3. Patients present with symptomatic arterial occlusion or intermittent claudication.

4. Flow to the great toe is monitored with an "end point detector" such as PPG. Diminished pulsations considered abnormal.

5. With knee extended and active plantar flexion of the foot (or with passive dorsiflexion of the foot) against resistance, PPG pulsations may diminish or obliterate, which could be suggestive of popliteal artery entrapment.

D. PREOPERATIVE ARTERIAL MAPPING

Epigastric Artery (EA):

1. Deep superior EA is terminal branch of internal mammary artery.

2. Both arteries and perforators, take blood to the rectus abdominis muscle, (long strap muscle) vertically oriented, each side of midline.

3. Rectus abdominis muscle, sub-q fat, arteries, perforators, overlying skin: called the transverse rectus abdominis myocutaneous (TRAM) flap.

4. Reason for mapping: Surgeon wants to use the best arterially supplied muscle section for TRAM flap for autogenous breast reconstruction.

Internal Mammary Artery (Internal Thoracic Artery):

1. Arises off arch of subclavian artery, descending on posterior side of cartilage of upper 6 ribs, about 1cm from sternum.

2. Reasons for mapping:

   Utilized as a recipient site for free flaps in breast reconstruction.

   Second important use for this artery is as graft to the left anterior descending (LAD) coronary artery.

Radial Artery Mapping:

1. Determine suitability for use as graft for coronary artery bypass.

2. Technique:

   - Assess patency of palmar arch utilizing modified Allen Test (Chapter 6).
   - If positive, no need to proceed. Removal of radial artery will compromise the hand. If negative, may proceed with exam.
   - Using duplex, (7.5 or 5 MHz) assess brachial and radial arteries
   - Observe for abnormalities such as: focal elevated PSV, abnormal Doppler quality, intimal thickening, aneurysm, calcification.
   - Measure diameter of the radial artery (proximal, mid, distal).

3. Limitations may include: wall calcification

E. PREOPERATIVE VEIN MAPPING

1. Indications:

   - Determine suitability for use as extremity or coronary bypass
   - Determine suitability of veins for use in dialysis access/graft

2. <u>Technique - LE:</u>

   - Transducer: 7.5 MHz or higher
   - GSV (sometimes LSV/SSV) mapped (proximal to distal)
   - Measurements obtained incrementally along extremity: high, mid, distal thigh; knee, BK, proximal, mid and distal calf, ankle

   <u>Technique - UE:</u>

   - Cephalic and basilic veins are mapped
   - Measurements obtained incrementally along extremity: proximal, biceps, above elbow; elbow, forearm: proximal, mid, distal levels, wrist

3. **<u>VEIN MAPPING</u>: <u>GENERAL TECHNIQUES</u>**

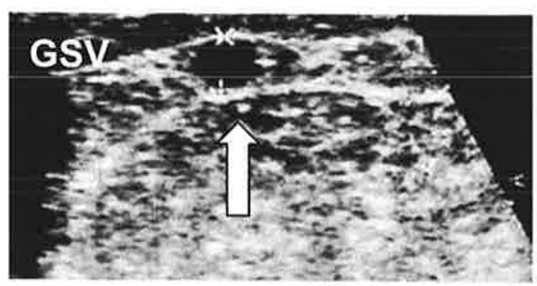
   - Proximal tourniquet assists with expansion of the vein size
   - Vein wall coaptation (compressibility) is evaluated
   - "Outer edge to outer edge" diameter measurements (mm), incrementally along extremity. Vein length (cm) obtained; continuity observed.

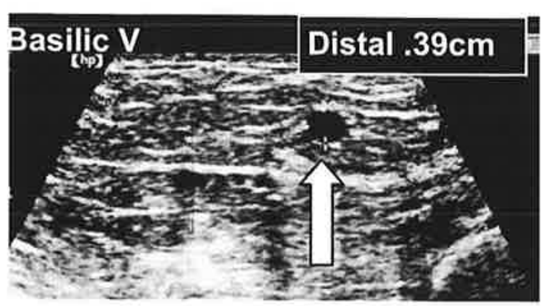
4. Normal findings include:

   Compressibility of veins without evidence of thrombosis.

   Acceptable diameter measurement criteria varies; However, vein dimensions should be at least 2-3mm; basilic vein is often larger than cephalic vein.

5. Comment made re: presence of vein wall thickening or mural calcifications; findings which could prevent use of vein as bypass graft.





F. **THORACIC OUTLET SYNDROME (Arterial component)**

1. Occurs with neurovascular bundle compression by shoulder structures (cervical rib, clavicle, scalene muscle); occurs in certain arm positions.

2. Etiology not well understood; most symptoms due to neurogenic compression of brachial plexus (97%); small percentage due to subclavian vein or artery compression.

3. Symptoms include:

   - Numbness/tingling of arm
   - Pain/aching of shoulder/forearm
   - Exercise/upward positions increase discomfort/symptoms.
   - Note: 25-30% of population have asymptomatic compression.

4. PPG and/or Doppler waveforms utilized to monitor any changes.

   a. PPG attached to index finger; or CW Doppler on radial artery
   b. Resting waveforms obtained; patient's arm placed in various positions while arterial pulsations monitored.
   c. Normal: resting waveforms maintained

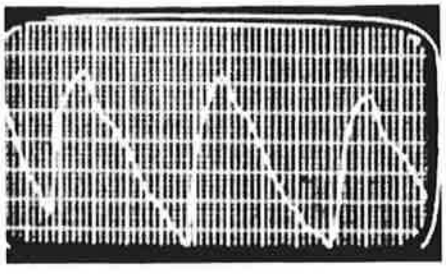
      Abnormal: attenuation or flattening of the waveforms in one or more of the positions.

5. Arm positions (patient sits erect off side of the exam table).

   a. Resting position - hand in lap
   b. Arm at 90° angle
   c. Arm at 180° angle
   d. Exaggerated military stance
   e. Adson Maneuver positioning:
      - Same position as d., head turned sharply to right.
      - Same position as d., head turned sharply to left.
   f. Causative position as described by patient.

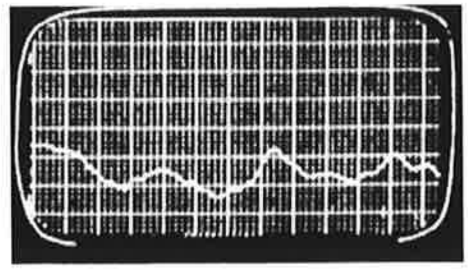
6. Example plethysmographic tracings:

NORMAL



Arm position is at 90° with face away

ABNORMAL



7. Treatment conservative (shoulder exercises) however, surgical treatment done by rib resection with/without scalene splitting.

G. TRAUMA

1. Arterial injuries can result from blunt trauma (e.g., long bone fractures which injure vessels) or penetrating trauma (e.g., stab wounds).

2. Clinical presentation may include open wound, hemorrhage and/or hematoma. There may/may not be distal pulses.

3. Intimal tear can also occur with media and adventitia remaining intact. Pulses may remain palpable initially, but events such as thrombus formation, intimal flap or dissection can occur.

Chapter 15 Miscellaneous Diagnostic Tests / Treatment

A. ARTERIOGRAPHY

1. Method

a. Percutaneous puncture of superficial artery: insertion of thin catheter.

b. Most common arteries used: CFA (safest approach), axillary, brachial.

c. After proper positioning, contrast agent injected into catheter, flows with moving blood; gives picture of lumen.

d. Rapid film changer technique used to expose the films sequentially as contrast agent moves through vessel.

e. Using fluoroscopy, digital information obtained / stored for later manipulation / interpretation.

f. Catheter removed; pressure held on puncture site; patient supine for 6-8 hours. Sandbag placed on top of dressing to help avoid bleeding.

2. Interpretation

a. Primarily an anatomic study NOT a functional study.

b. Hemodynamically significant stenosis usually defined as a 50% diameter reduction.

c. Normal anatomy seen on films as contrast media fills vessel.

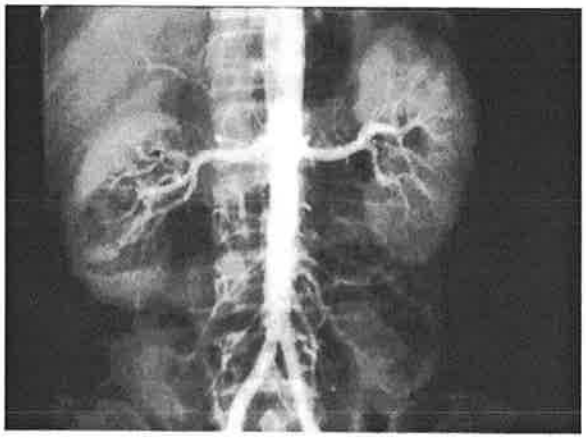
d. Interpretation based on how much (if any) of artery does not fill with blood containing contrast agent. Extent and location of filling defect is determined.

e. Atherosclerotic plaque appears as irregular or smooth.

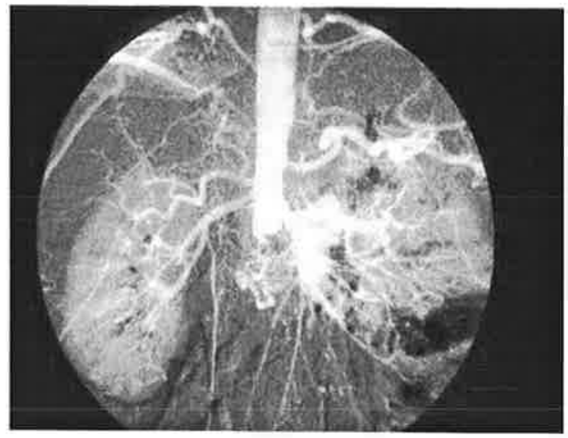
f. Vessel occlusion: no filling seen; collaterals often present with long standing occlusion.

g. Miscellaneous findings:

- Aneurysm appears as dilated artery
- Vasospasm: severe narrowing, usually without occlusion.
- Fibromuscular dysplasia: multiple arterial stenoses caused by medial hyperplasia, appearing as "string of beads."



NORMAL: Aorta, renal, arteries, common iliac arteries



ABNORMAL: Aortic occlusion with collaterals; renal arteries patent

3. **<u>Limitations</u>**

   a. May be contraindicated in patients allergic to contrast agent or in kidney failure.

   b. Inaccurate in its hemodynamic assessment because of inability to provide many images in multiple planes in "real time." Two-dimensional view is standard.

4. <u>Complications</u>

   a. Puncture site hematoma
   b. Pseudoaneurysm
   c. Local arterial occlusion
   d. Neurologic complications

B. MR ANGIOGRAPHY (MRA)

1. **<u>Method</u>**

   a. Employs radio frequency energy and a strong magnetic field to produce images in multi planes.

   b. MRI instruments quantitate blood flow and construct images that look like angiograms (MRA). Flowing blood well distinguished from soft tissue without using contrast agents.

2. **Limitations**

a. Presence of metallic clips, pacemakers, monitoring equipment.

b. Can overestimate stenosis due to slow flow or turbulence.

d. Expensive

e. Claustrophobia may limit some patients.

3. **Interpretation**

a. Useful for AAA, dissections, peripheral artery evaluation.

C. COMPUTERIZED TOMOGRAPHY (CT)

1. **Method**

a. Employs ionizing radiation to obtain cross sectional images of the aorta and other body structures.

b. IV contrast allows more discrete evaluation.

2. **Limitations**

a. Patient motion and presence of metal surgical clips.

b. One plane used

c. Limited application in PAD due to smaller vessels

3. **Interpretation**

a. Identifies size of aorta; extent and size of aneurysm.

b. Helps define relationship of aorta to renal artery origins

D. TREATMENT—MEDICAL

Lifestyle modification:

1. Stop smoking.

2. Consistent exercise to enhance collateral development.

3. Weight control and a low cholesterol diet may enhance normal endothelial cell metabolism.

4. Protection to prevent injury and/or infection to extremity of interest.

Pharmacologic:

1. Aspirin: an antiplatelet drug that decreases platelet aggregation resulting in decreased thrombotic activity.

2. Medications that help decrease blood viscosity.

3. Antihypertensive drugs may serve to decrease shearing forces against vessel walls.

E. TREATMENT—SURGICAL

Decision to operate depends on extent of disease and primarily the patient's symptoms.

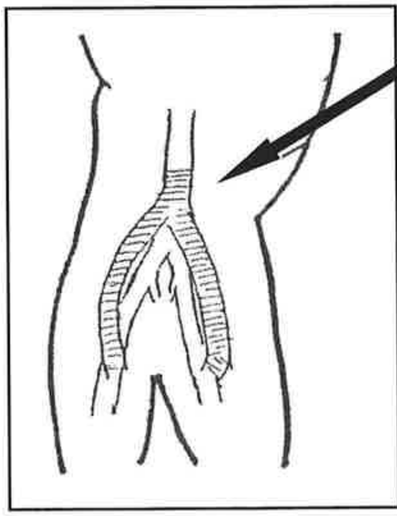
1. Endarterectomy is surgical removal of atherosclerotic material, usually includes portion of intimal lining.

2. In presence of significant stenosis or occlusion, bypass graft provides alternate pathway for blood to travel.

3. Successful bypass grafts include the following components: good inflow, conduit and outflow.

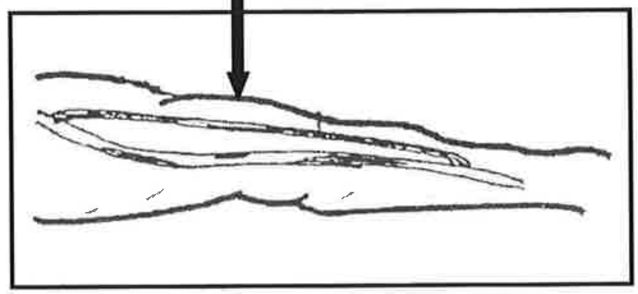
   a. Common bypass grafts:

      - Aorta to both iliac arteries (also used for AAA)
      - Aorta to bi-femoral (also used for AAA)
      - Femoral to popliteal
      - Femoral to PTA; to ATA, to peroneal



Aorta to bi-femoral bypass graft

Reverse saphenous vein graft



b. Other bypass grafts include: renal artery, SMA.

C. TREATMENT—ENDOVASCULAR

1. **Angioplasty** (Percutaneous Transluminal Angioplasty [PTLA] used to dilate focal plaque formation in vessel).

   a. Same technique used for **arteriography,** except a balloon tipped catheter is utilized. Under fluoroscopy:

   - Catheter tip is brought to the region of stenosis.
   - Balloon is slowly inflated, pushing plaque against the walls of vessel, dilating artery lumen.
   - Balloon deflated, catheter removed.

   b. Can not be done on all vessels nor on all types of lesions. Procedure usually performed in vessels with focal stenosis such as: renal, iliac, femoral, or popliteal artery;

   General observations of catheter intervention

   a. Serious complications occur in less than 5% of cases. 2-3% of PTLA patients have complications requiring surgery or alter hospital stay.

   b. Minor discomfort with the procedure.

   c. Morbidity, mortality, cost of angioplasty are low.

2. **<u>Stent / Stent Graft</u>**

a. To maintain intraluminal structure / patency of artery. Acts as a type of "scaffold."

b. Insertion and deployment technique varies with stent type.

c. Similar techniques, as used in **arteriography**, are utilized with insertion of stent introducer sheath.

d. <u>Limitations</u> (depending upon stent site):

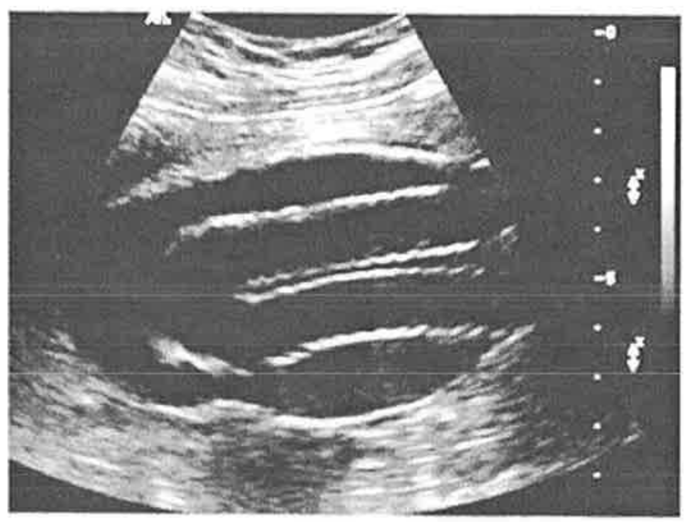
- Abdominal gas
- Inability of patient to lie flat
- Complications similar to those for arteriography.

e. Problems include: restenosis due to intimal hyperplasia, stent migration, graft limb compression, twisting, dislodgment, leaks.

f. Utilized in vessels such as: aorta, renal, iliac, femoral; repair of AAA with stent grafts.

Aorta bifurcated stent graft

Location: aortic aneurysm



Chapter 16: Cerebrovascular Gross Anatomy

A. ANTERIOR CIRCULATION

1. <u>Common carotid artery (CCA)</u>

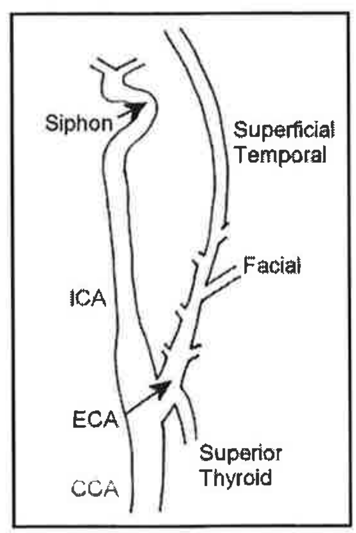
- Right CCA is a branch of the right Innominate, while the left CCA is a branch off the aortic arch.
- The majority of blood flows into the internal carotid

2. <u>Internal carotid artery (ICA)</u>

- Originates from common carotid artery (CCA)
- Travels into base of skull without branching
- Intracranial branches include:
  - ◊ Ophthalmic artery; originates near carotid siphon, a significant curve of ICA
  - ◊ Posterior communicating artery
- Terminates in the middle cerebral artery (MCA) and anterior cerebral artery (ACA)
- Distributes blood to low-resistance vascular beds

3. <u>External carotid artery (ECA)</u>

- Originates from the CCA
- Has eight major branches, the first branch is usually the superior thyroid artery.
- Distributes blood to high-resistance vascular beds



B. POSTERIOR CIRCULATION

1. <u>Vertebral arteries</u>

   a. First branch off the subclavian artery
   b. Right usually smaller than left
   c. Unite after entering skull to form basilar artery

2. <u>Basilar artery</u>

   a. Formed by confluence of vertebral arteries
   b. Divides into posterior cerebral arteries (Fig. 2)

C. CIRCLE OF WILLIS

A hexagonal arrangement of: distal internal carotid (ICA), anterior cerebral arteries (ACA) joined together by the anterior communicating artery (AComm), posterior cerebral arteries (PCA) joined together by the posterior communicating arteries (PComm) (Fig. 3)

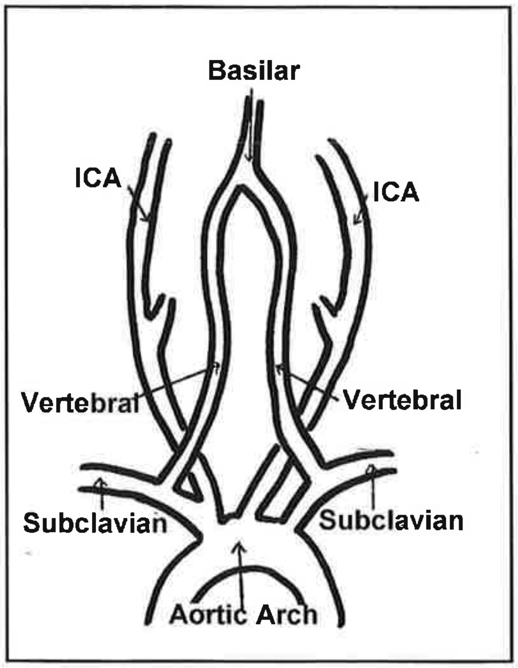


Fig. 2. Vertebrobasilar System (Posterior view)

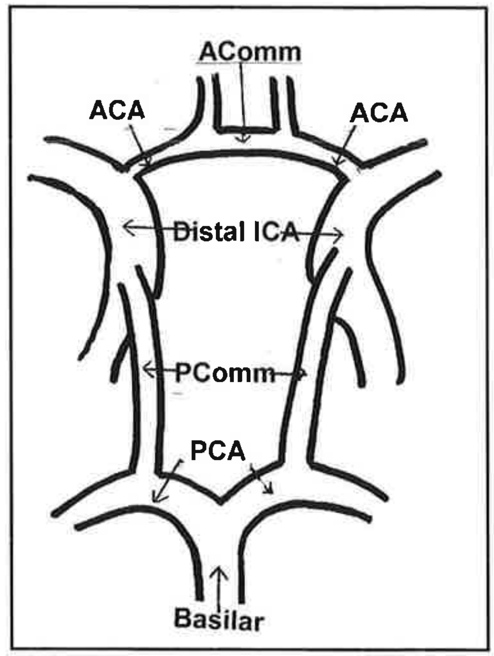
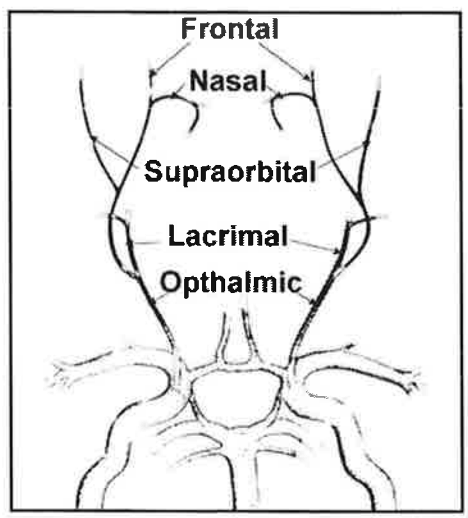


Fig. 3. Circle of Willis (Grossly Magnified)

D. PERIORBITAL CIRCULATION

1. Supraorbital artery: arises from ophthalmic artery; travels anteriorly & superiorly to the globe; joins ECA via some of its branches (e.g., [STA] superficial temporal artery).

2. Frontal artery: arises from ophthalmic artery; exits orbit medially to supply mid-forehead; joins ECA via some of its branches.



Source: Medasonics,
Mountain View, California 1984.

E. Collateral pathways

1. Largest intra-arterial connection: **circle of Willis**

2. Important anastomoses include:

   a. ECA-ICA connections via orbital and ophthalmic arteries
   b. Occipital branch of ECA with atlantic branch of vertebral
   c. Deep cervical and ascending cervical branches of subclavian to branches of lower vertebral artery

F. PHYSIOLOGY and HEMODYNAMICS

Arterial Hemodynamics is covered in Chapter 2. This brief review is important to understand cerebrovascular blood flow.

PRESSURE/FLOW RELATIONSHIPS

1. The Bernoulli principle: total fluid energy along a streamline of fluid flow is constant

Velocity E & Pressure E are inversely proportional:
↑ velocity = ↓ pressure
↓ velocity = ↑ pressure

- Pressure gradients or areas of flow separation are set up; can occur in a vessel because of a geometry change, with or without intra-luminal disease

2. Poiseuille's Law combined with Resistance equation:

- Quantity of flow (Q) is related to the pressure gradient across an arterial segment (P), radius of the vessel (r), viscosity of the fluid (η), and length of the vessel (L):

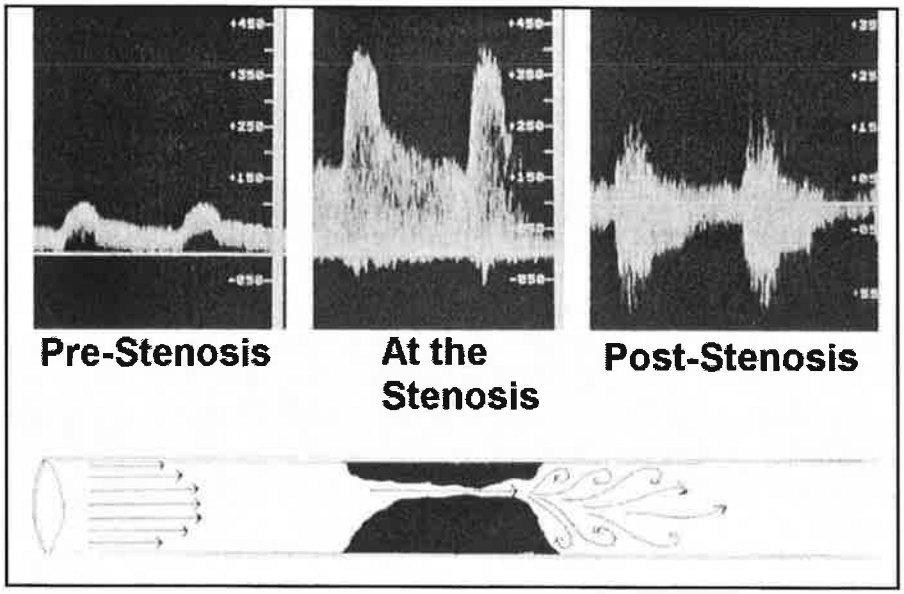
$$Q = (P)\,\pi\, r^4 / 8\,\eta\, L$$

- Flow is directly related to: (P) and r^4
- Flow is inversely related to: η and L

EFFECTS of STENOSIS on FLOW CHARACTERISTICS

1. **Velocity acceleration** results because velocity and area are inversely proportional; acceleration causes increased energy losses.

2. Blood flow must change **direction** as the flowstream narrows entering the stenosis & enlarges as it exits; eddy currents, **turbulence**, & vortices cause energy loss through inertia.



Chapter 17: Cerebrovascular Testing
(Signs, Symptoms, History, Disease Mechanisms, Physical Exam)

Note: You must obtain appropriate indications for testing prior to performing the study.

A. SIGNS and SYMPTOMS

1. Transient symptoms

a. Transient ischemic attack (TIA):

- A fleeting neurological dysfunction
- Symptoms last less than 24 hours
- Usually embolic from heart or carotid artery

b. Vertebral basilar insufficiency (VBI)

2. Permanent symptoms: Cerebrovascular accident (CVA)

a. Symptoms last more than 24 hours

b. Complete recovery does not occur

B. PATIENT HISTORY

Risk factors and contributing diseases applicable to cerebrovascular are similar to those covered in arterial disease. (See Chapter 3 for more information).

C. MECHANISMS of DISEASE

1. Atherosclerosis:

a. **Atheromatous** plaque: a form of arteriosclerosis; localized accumulations of lipid-containing material (atheroma), smooth muscle cells, collagen, fibrin and platelets (Fig. 6)

b. Formed within or beneath the intima (Fig. 7)

c. Causes thickening, hardening, & loss of elasticity of walls

d. Can result in decreased perfusion to brain

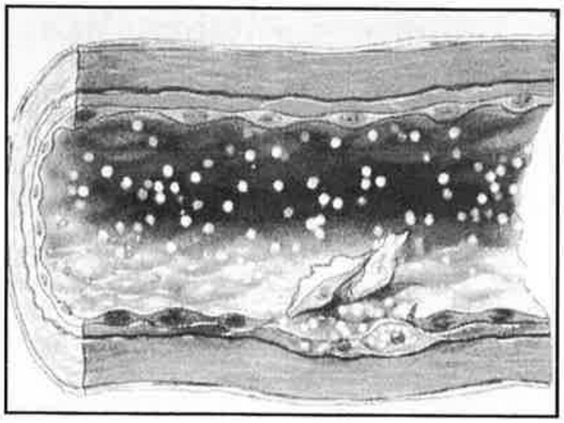


Fig. 6. Damage to endothelial wall results in plaque formation

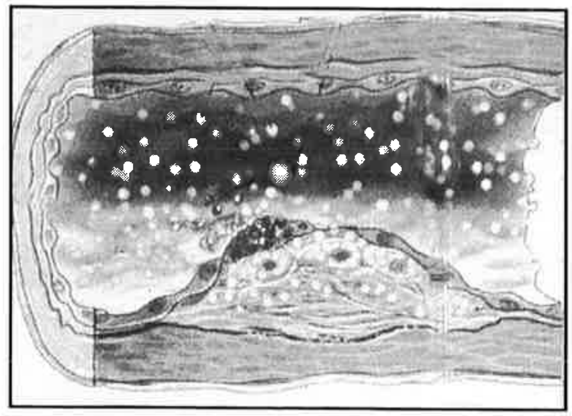


Fig. 7. Formed within or beneath the intima

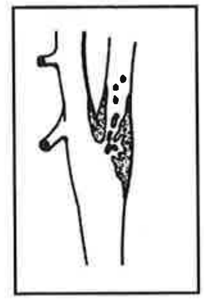
*Drawings' source: Discover Magazine, 1991

TYPES OF **ATHEROMATOUS** PLAQUE

- Fatty streak: thin layer of lipid material on intimal layer.
- Fibrous plaque: accumulation of lipids, collagen and elastic fibers.
- Complicated lesion: a fibrous plaque that includes fibrous tissue, more collagen, calcium and cellular debris.
- Ulcerative lesion: deterioration of the normally smooth surface of the fibrous cap; may result in distal embolization.
- Intra-plaque hemorrhage: evident on B-mode as a sonolucent area within plaque.

2. **Thromboembolic**

Obstruction of a blood vessel by a piece of thrombus



a. Thrombus: Large amounts of red blood cells trapped within a fibrin network; clumps of platelets may also be evident

b. Embolism: Piece of thrombus breaks loose & travels distally until it lodges in a small vessel

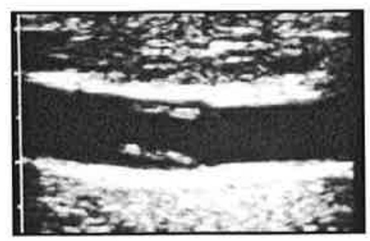
3. **Aneurysm**

a. Localized dilatation of a blood vessel due to congenital defects or weakness of the wall (trauma, infection, atherosclerosis)

b. Rarely seen in cervical carotid artery

c. Pulsatile mass in neck usually a very tortuous CCA

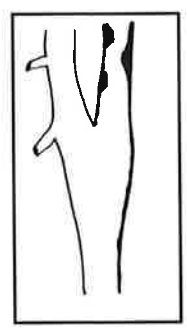
4. **Non-atherosclerotic conditions**



a. **Dissection**

- Results from a sudden tear in the intima
- Creates a false lumen which may gradually extend proximally or distally
- Blood in the false lumen may thrombose

b. **Fibromuscular dysplasia (FMD)**:



- Most commonly caused by dysplasia of media along with overgrowth of collagen
- Characteristic bead-like appearance on angiography
- Often seen in young women

c. **Carotid body tumor (CBT)**

- Carotid body: a small structure located just above the carotid bifurcation (Fig. 11)
  - Helps control respiration by sensing changes in oxygen tension of blood

- CBT is a highly vascular structure that develops between the ICA & ECA, and is usually fed by the ECA.
- Surgical excision may require ligation of ICA and/or ECA

d. **Neointimal hyperplasia**

- Intimal thickening from rapid production of smooth muscle cells (Fig. 12)
- A response to vascular injury/reconstruction, e.g. post carotid endarterectomy
- Denuding of endothelium leads to platelet accumulation, endothelium regeneration, and smooth muscle cell proliferation
- Significant stenosis may occur within 6 to 24 months

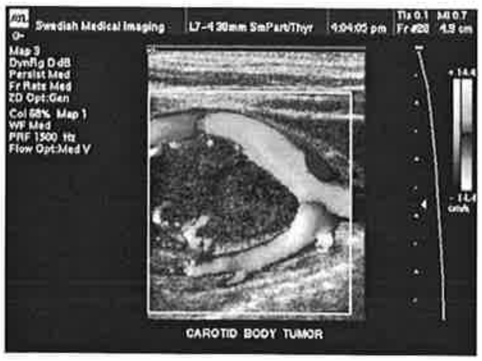


Fig. 11. Carotid body tumor

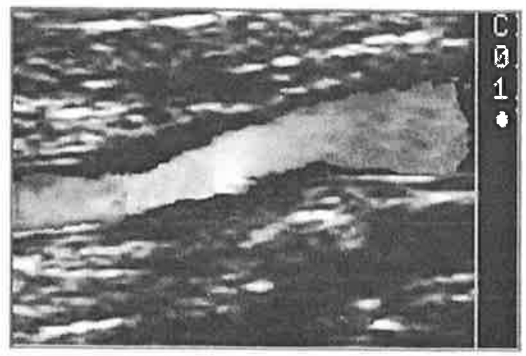


Fig. 12 Neointimal hyperlasia

D. PHYSICAL EXAMINATION

1. Palpation

a. Rhythmic throbbing of artery in time with the heart beat usually felt (palpated) by the 2nd, 3rd (and 4th) fingers.

b. Common sites: common carotid, superficial temporal, subclavian, axillary arteries

2. Auscultation

a. Listening through a stethoscope

b. **Bruit**:

- A noise heard as the result of turbulent flow
- Frequently associated with a hemodynamically significant lesion
- Bruit may not be evident with a very tight stenosis, i.e. >90%
- Common sites for bruit evaluation: carotid, subclavian

3. **Blood Pressure (BP) measurements**: It is recommended to obtain bilateral BP measurements to detect proximal obstruction, i.e., subclavian steal

4. Patient History/Symptoms

Anterior Circulation (Carotid)

Hemispheric (lateralizing) symptoms: Since the left hemisphere of the brain controls the right side of the body, and vice versa, a left hemispheric CVA results in **neurological deficits** on the right side of the body.

> NOTE:
> Specific eye symptoms, e.g., amaurosis fugax, are suggestive of ipsilateral ICA disease.

SYMPTOMS FREQUENTLY SEEN WITH ICA LESION:

- Unilateral paresis: weakness or slight paralysis on one side of body
- Unilateral paresthesia: prickling or tingling of the skin
- Aphasia: inability to speak
- Amaurosis fugax: temporary, partial or total blindness, usually of one eye

(Myopia, commonly referred to as nearsightedness, and homonymous hemianopia, defective vision or blindness in the right or left halves of the visual fields, are not consistent with ICA lesions.)

SYMPTOMS FREQUENTLY SEEN WITH MCA LESION:

- Aphasia or dysphasia.
- More severe facial and arm hemiparesis or hemiplegia (rather than in the leg)
- Behavioral changes

SYMPTOMS FREQUENTLY SEEN WITH ACA LESION:

- More severe leg hemiparesis or hemiplegia
- Incontinence
- Loss of coordination

Posterior Circulation (Vertebrobasilar)

SYMPTOMS FREQUENTLY SEEN WITH VERTEBROBASILAR LESION:

- Vertigo: difficulty in maintaining equilibrium
- Ataxia: muscular uncoordination, i.e. inability to control gait
- Bilateral visual blurring or double vision (Diplopia)
- Bilateral paresthesia or anesthesia
- Drop attack: falling to the ground without a loss of consciousness

SYMPTOMS FREQUENTLY SEEN WITH PCA LESION:

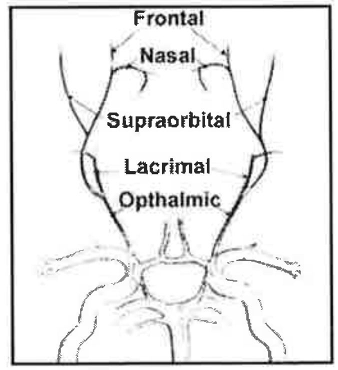
- Dyslexia
- Coma
- Paralysis usually does not occur

Non-localizing symptoms

- Dizziness: sensation of whirling with a tendency to fall
- Syncope: transient loss of consciousness
- Severe headache

Chapter 18 Ocular Pneumoplethysmography (OPG-Gee)

1. CAPABILITIES



Detects hemodynamically significant lesions of ICA by evaluating flow in one of its terminal branches.

2. LIMITATIONS

a. Cannot establish exact location of disease

b. Non-diagnostic when collaterals exist or with a non-hemodynamically significant lesion

c. Cannot differentiate occlusion from tight stenosis

d. Patients with severe hypertension

3. PATIENT POSITIONING

The patient is supine.

4. PHYSICAL PRINICPLES

a. Plethysmography records changes in volume, i.e. changes in blood flow during systole and diastole.

b. Measures ophthalmic artery systolic pressure (OSP) by applying a vacuum to the eye

- Vacuum distorts shape of globe
- Intraocular pressure increases obliterating arterial inflow
- Strip chart recordings made as vacuum is slowly decreased & wave forms appear

5. TECHNIQUE

a. Identify contraindications:

- Allergies to local anesthetics
- Eye surgery within last 6 months
- A past spontaneous retinal detachment
- Acute or unstable glaucoma

(Myopia, an error in refraction commonly called Nearsightedness; or conjunctivitis, an inflammation of the conjunctiva, are not contraindications.)

b. Calibration / standardization manually performed by pressing the STD button (Standardization deflections should have amplitude of 10 mm above and below baseline on the chart.)

c. Eye cups placed on lateral sclera (whites of the eye) after anesthetic drops

d. Patient with a systemic systolic pressure of <140 mm Hg requires vacuum of 300 mm Hg

e. Patient with a systemic systolic pressure of >140 mm Hg requires vacuum of 500 mm Hg

f. Chart recorder started with speed at 5mm/sec; foot switch activates vacuum

g. If ocular pulsations immediately evident when using 300 mm Hg setting, select 500 mm Hg setting, continuing exam procedure

6. INTERPRETATION

a. Identify first eye pulse and determine OSP

b. Diagnostic criteria:

<u>Normal:</u> OSPs should not differ by $\geq$5 mm Hg

<u>Abnormal</u>: OSPs that differ by $\geq$ 5 mm Hg

Chapter 19 Carotid Duplex/ Color Flow Imaging

1. CAPABILITIES

a. Localize lesion in extra-cranial carotid arteries

b. Differentiate occlusion from stenosis

c. Document progression of disease

d. Identify surface characteristics

e. Evaluate pulsatile mass

2. LIMITATIONS

Poor visualization secondary to:

- Presence of dressings, skin staples, or sutures
- Size or contour of neck
- Depth or course of vessel
- Acoustic shadowing from calcification →

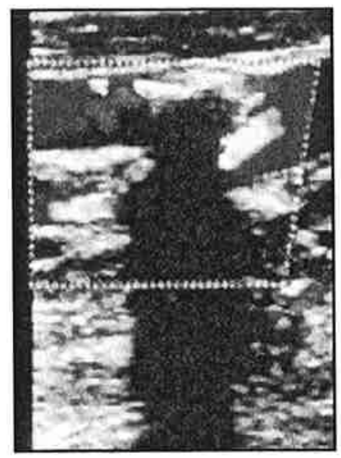


Fig. 14. Calcific shadowing

Over-estimation of the disease process from:

Accelerated flow mistakenly attributed to stenosis instead of:

- Cardiac output
- Tortuous vessel
- Compensatory flow
- Inappropriate Doppler angle

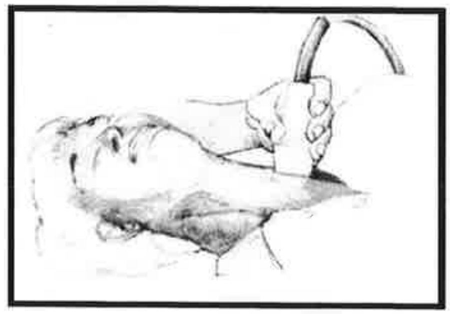
Under-estimation of the disease process:

Accelerated flow not present and/or not detected:

- Jet of accelerated flow missed
- Long, smooth plaque formation
- Stenosis at area of dilatation, i.e. carotid bulb
- Inappropriate Doppler angle

3. **PATIENT POSTIONING**

Patient in comfortable supine position, neck slightly hyperextended, and head turned slightly



4. **PHYSICAL PRINCIPLES**

a. Spectral analysis

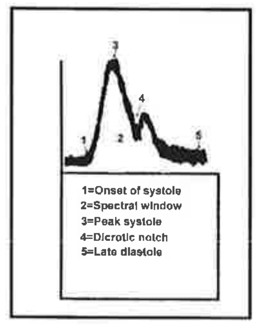
- Information displayed electronically on a monitor
- Utilizes Fast Fourier Transform (FFT) method: Individual frequencies/ velocities displayed with time on horizontal axis and various frequency shifts/velocities on vertical axis
- Displays true frequency/velocity shifts
- Commonly used with imaging modalities

b. Continuous-wave Doppler (CW)

- Two piezo-electric crystals: one constantly sending ultrasound, one constantly receiving reflected waves
- No range resolution
- Fixed sample size

c. Pulsed Doppler

- One crystal sending and receiving ultrasound
- Has range resolution
- Variable sample size
- Well-defined spectrum

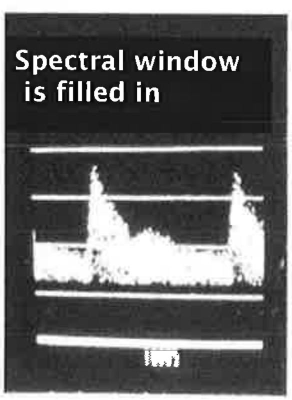


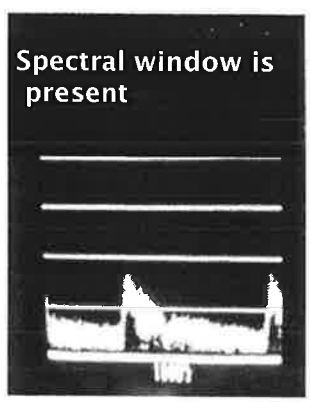
NORMAL SPECTRAL ANALYSIS

CW DOPPLER

PULSED DOPPLER

ICA





d. Color Doppler

- Assigns color to display average frequencies (e.g. hue, brightness of
- color) and direction (phase) of moving blood
- Pulsed Doppler beams evaluate multiple sample sites throughout a specific area
- Scan rates are slower because of multiple transmit/received pulse cycles in each 'color line of site'
- Several processing methods exist to produce color duplex

5. TECHNIQUE

Sagittal / longitudinal View

a. Vessels followed from clavicle to mandible using anterior, oblique, lateral, and/or posterior oblique views (Fig. 15)

b. Evaluate flow patterns of CCA, ICA, and ECA

c. Evaluate vertebral arteries

- Use posterior-lateral approach
- Identify artery by vertical shadows running through it [from transverse processes of vertebrae] (Fig. 16)
- Evaluate direction of flow

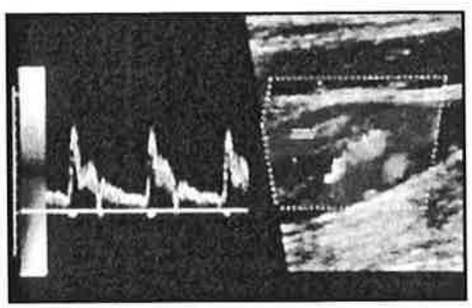


Fig. 15. Evaluation in sagittal view

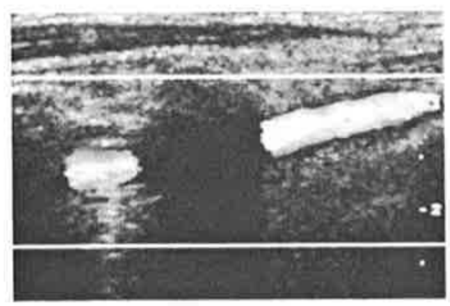


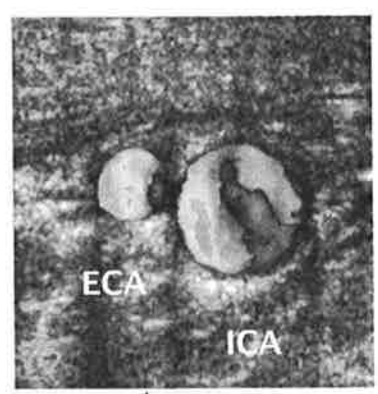
Fig. 16. Landmark for vertebral artery

Transverse / cross- sectional View

a. Vessels followed from clavicle to mandible

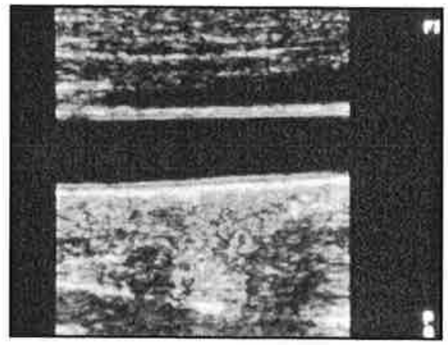
b. Document plaque formation

c. Percent stenosis may be calculated



Left BIF

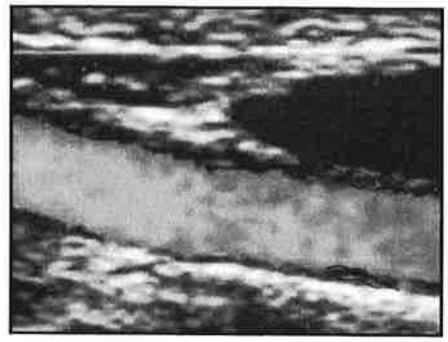
6. INTERPRETATION



B-mode: Normal vessel

Absence of wall irregularities
or soft tissue abnormalities

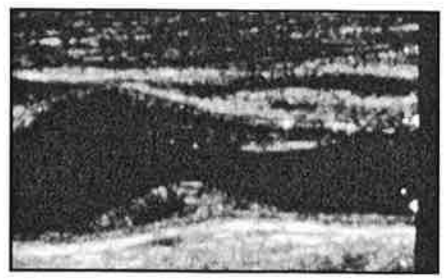
B-mode: Abnormal vessel



- Hypoechoic & homogenous: low level echoes of similar appearance, i.e. fatty streaks, found in persons of all ages

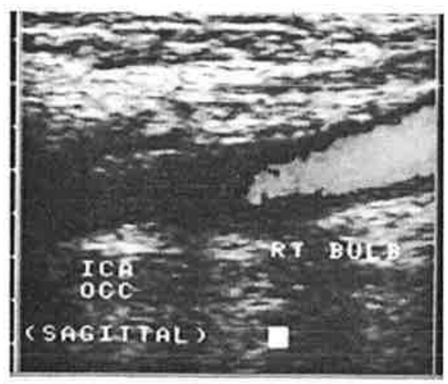
- Homogenous: low-to-medium level echoes of similar appearance, i.e. fibrous (soft) plaque

- Echoic & heterogeneous: all levels of echoes (soft and dense areas), i.e. complex plaque or intraplaque hemorrhage (sonolucent area inside plaque).



- Hyperechoic: very bright/highly reflective echoes; acoustic shadow from calcium deposit may result in erroneous calculation of % stenosis

- Thrombosis: same echogenicity of flowing blood on B-mode



- Surface Characteristics: smooth, slightly irregular, or grossly irregular surface

- Stenosis should be visible from at least 2 projections

- Occluded artery

  ◊ Varying degrees of echogenic material
  ◊ Vessel completely filled with echoes (Fig. 17)
  ◊ Vessel motion: horizontally or piston-like

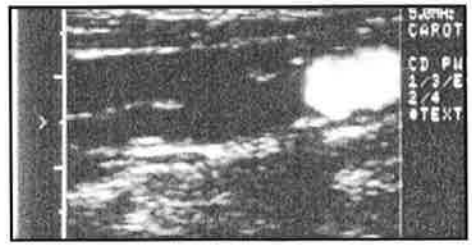
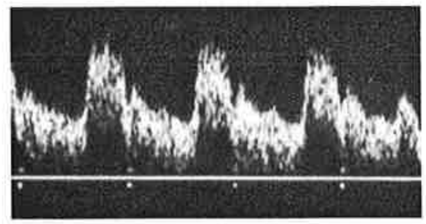


Fig. 17. Vessel filled with echoes

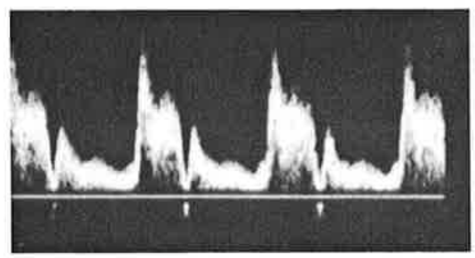
NORMAL DOPPLER SIGNALS

a. ICA



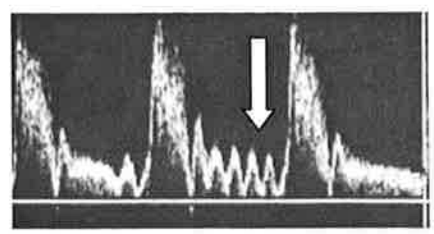
- More high-pitched and continuous than ECA
- Wave form has rapid upstroke and down stroke with a high diastolic component

b. ECA



- Signal more pulsatile; very similar to peripheral vessels
- Rapid upstroke and down stroke with low flow in diastole

- Dicrotic notch clearly seen; oscillations in waveform seen with tapping STA →



c. CCA: Flow characteristics of both ICA and ECA.

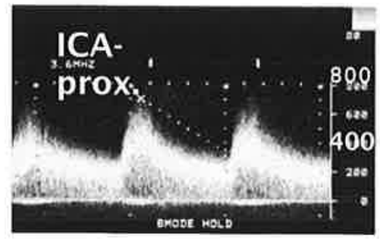
ABNORMAL DOPPLER SIGNALS

a. Stenosis

- Characterized by higher pitched sound and waveform with higher amplitude (from elevated frequencies)

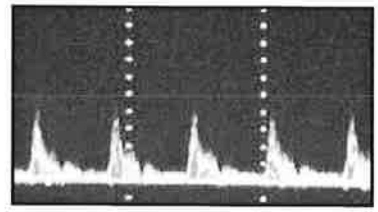
- Spectral broadening evident (representing turbulence)

- Loss of spectral window also represents loss of laminar flow



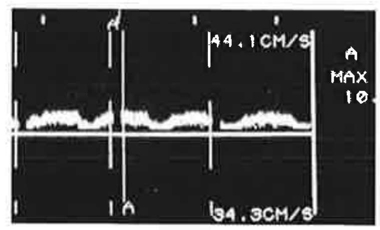
b. Distal to a stenosis: disturbed flow patterns e.g. turbulent, bi-directional; then can become dampened and monophasic/continuous

c. Consider disease at carotid siphon when high resistant flow patterns evident in ICA



d. Diminished CCA velocities bilaterally may indicate poor cardiac output or stroke volume

e. Diminished velocities unilaterally suggests proximal disease, e.g. innominate or common carotid artery



f. Essential to compare flow characteristics bilaterally; and proximal-to-distal segments of ipsilateral vessel

NOTE: One example of categorizing disease follows

% STEN	PSF	EDF	PSV	EDV
Normal #	< 4 KHz	NA	<125 cm/sec	NA
1-15% #	< 4 KHz	NA	<125 cm/sec	NA
16-49% #	< 4 KHz	NA	<125 cm/sec	NA
50-79%	> 4 KHz	< 4 KHz	>125 cm/sec	<140 cm/sec
* 80-99%	> 4 KHz	> 4 KHz	>125 cm/sec	>140 cm/sec
Occluded	Absent	Absent	Absent	Absent

Categories differ in the amount of spectral broadening; based upon criteria developed by D.E. Strandness, MD, University of Washington, Seattle, WA

Additionally, these categories of disease are based on a 60° angle of insonation.

CRITERIA CRITICAL IN DETERMINING AN OCCLUSION

- CCA may have a very low or absent diastolic component (Fig. 18A)
- Evidence of collateralization e.g., ECA may exhibit high flow in end diastole
- Absent ICA Doppler signal or pre-occlusive thump (Fig. 18B)

NOTE: Although an absent signal may indicate occlusion, a tight stenosis (also termed "string sign") cannot be ruled out.

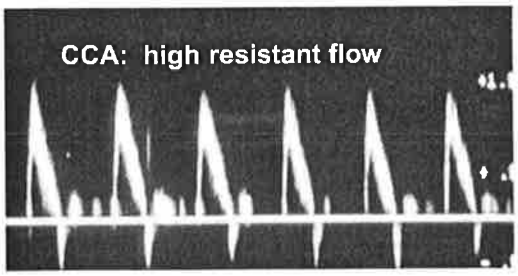


Fig. 18A

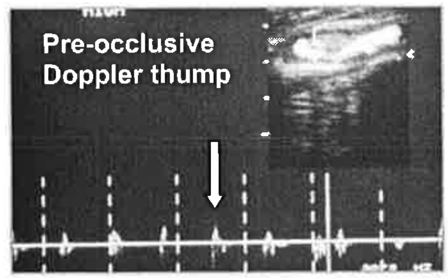
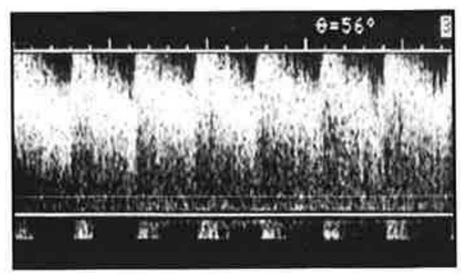


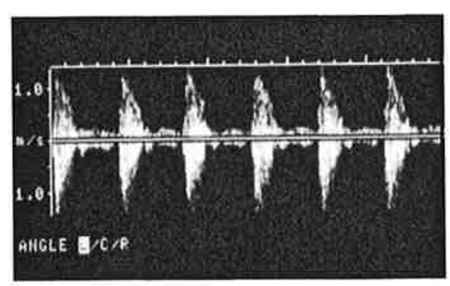
Fig. 18B

ALIASING



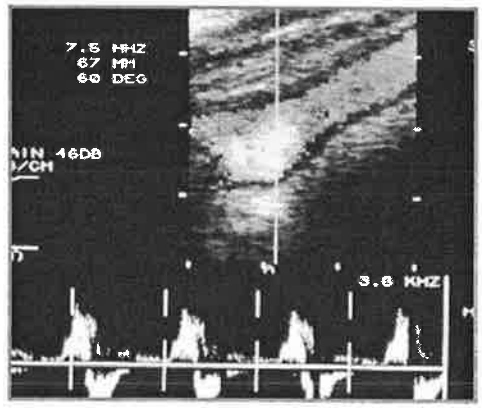
- Misrepresentation of Doppler signal due to limitations of the equipment being used, i.e., a low PRF
- Maximum frequency is ½ PRF; flow information greater than ½ PRF cannot be displayed (Nyquist limit)
- Wave form has flat, crew-cut appearance
- Methods of increasing the PRF/ Nyquist limit include:
  - ◊ Decrease the baseline; increase Doppler scale
  - ◊ Change transducer frequency
  - ◊ Alter angle of insonation
  - ◊ Decrease depth
  - ◊ Use continuous-wave Doppler

MIRROR IMAGING



- Doppler shifts above and below baseline
- Display duplicate spectrum or color-flow Doppler
- Artifact from strong reflectors or too much gain

HELICAL FLOW



- Occurs when flow moves into a wider portion of the vessel (e.g., carotid bulb)
- Doppler shifts above / below baseline
- Spectral waveforms are different
- Doppler angle constantly changing
- Flow is not laminar; spectral broadening present

7. INTRAOPERATIVE MONITORING

a. Capabilities

- Identification of defects secondary to surgery and/or areas of platelet aggregation
- Evaluates hemodynamic significance of wall irregularity

b. Technique

- Sterile sleeve/plastic bag containing acoustic gel
- Wound filled with sterile saline
- Areas of flow disturbance identified with color-flow Doppler
- Gray scale imaging critical in detecting subtle wall defects

Chapter 20 Transcranial Doppler (TCD)

1. CAPABILITIES

a. Detect intracranial stenoses, occlusions, and assess collateral circulation

b. Evaluate onset, severity, and time course of vasoconstriction from subarachnoid hemorrhage

c. Evaluate intracranial arteriovenous malformations

d. Assess patients with suspected brain death

2. LIMITATIONS

a. Recent eye surgery may eliminate transorbital approach

b. Adequate penetration of temporal bone from hyperostosis

c. Inaccurate vessel identification with nonimaging technique

3. PATIENT POSITIONING

Patient supine and avoids speaking during examination

4. PHYSICAL PRINCIPLES (see previous description for principles of Pulsed Doppler)

<u>IMPORTANT POINTS</u>

- 2 MHz pulsed Doppler used with spectrum analysis
- Zero angle of insonation is assumed
- Three acoustic windows used: transtemporal, transorbital, and transforaminal / suboccipital
- Accurate vessel identification requires:
  - —Depth of sample volume
  - —Direction of blood flow
  - —Velocity of the blood flow
  - —Relationship of flow patterns to one another
- Time averaged maximum velocity (TAMV) or mean velocity used

5. TECHNIQUE

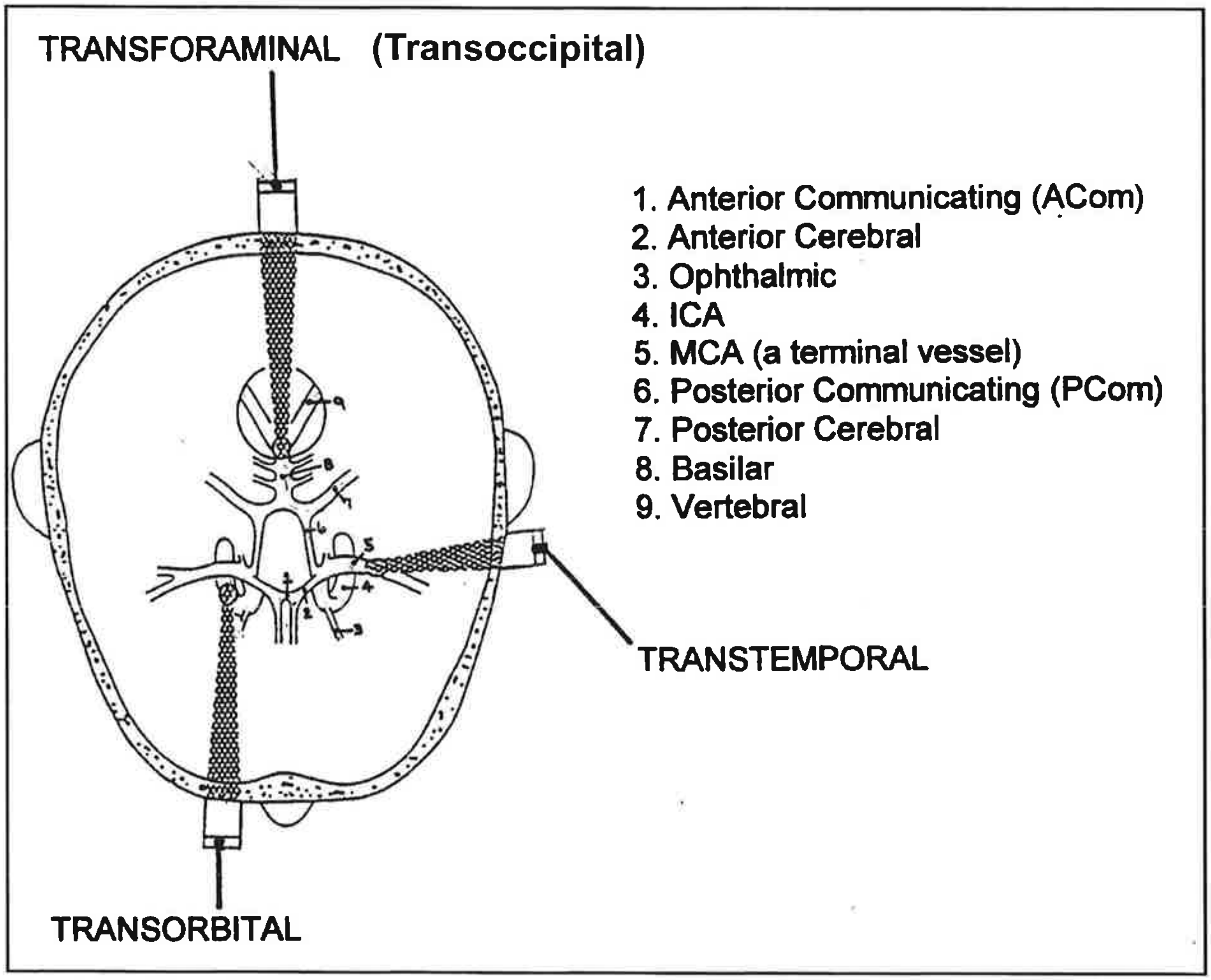
a. Unilateral transtemporal approach for MCA, ACA, PCA, and terminal ICA.

b. Ipsilateral transorbital approach to evaluate ophthalmic artery and carotid siphon

c. Preceding two steps completed contralaterally

d. Transforaminal / suboccipital approach used to evaluate the intracranial vertebral and basilar arteries

e. Inappropriate Doppler angle usually negates ability to insonate posterior communicating (PCom) arteries. However, flow direction is normally from anterior to posterior .



(Used with permission from Keith Fujioka, BS, RVT, Pacific Vascular Inc., Bothell, WA.)

f. Use the following guidelines:

VESSEL	WINDOW	DEPTH	DIRECTION	VELOCITY (TAMV or Mean)	ANGLE
MCA	Transtemp	30 - 60 mm	Antegrade (toward the probe)	55 +/- 12 cms/sec	Anterior & Superior
Term. ICA	Transtemp	55 - 65 mm	Bi-directional	55 +/- 12 cms/sec	Same as above
ACA	Transtemp	60 - 80 mm	Retrograde (away from the probe)	50 +/- 11 cms/sec	Anterior & Superior
PCA	Transtemp	60 - 70 mm	Antegrade	39 +/- 10 cms/sec	Posterior
ICA (at siphon)	Transorbit	60 - 80 mm	Parasellar: antegrade Supraclinoid: retrograde Genu: both	47 +/- 14 cms/sec	Varies
Ophth.	Transorbit	40 - 60 mm	Antegrade	21 +/- 5 cms/sec	Medial
VA	Transforam	60 - 90 mm	Retrograde	38 +/- 10 cms/sec	Right & left of midline
BA	Transforam	80 - 120mm	Retrograde	41 +/- 10 cms/sec	Midline

(Used with permission: Vascular Laboratory Policies and Procedures Manual, C. Rumwell and M. McPharlin, Editors, Davies Publishing, Inc. [Chapter author: Colleen Douville, BA, RVT]

6. INTERPRETATION

a. Based on flow characteristics:

- Direction
- Velocity
- Turbulence
- Pulsatility
- Systolic upstroke

b. Collateral pathways

- Cross-over: Antegrade flow in anterior cerebral artery (ACA) from cross-over collateralization, e.g. flow from contralateral ACA via anterior communicating artery
- External-to-internal: Retrograde flow in ophthalmic artery (OA)
- Posterior-to-anterior: Increased flow in posterior cerebral artery (PCA), reversing direction of flow in the PCom artery

c. Factors that may alter intracranial blood flow, e.g. age, sex, hematocrit, blood gases, metabolism

d. Occlusion:

- Most accurate in ICA or MCA
- Criteria similar to those used in other vessels

e. Vasospasm:

- Most accurate in MCA
- Serial recordings necessary
- Mean velocities >120 cm/sec. Severe vasospasm = >200cm/sec.

f. **Arteriovenous malformation:**

- Increased systolic and diastolic flow velocities
- Very low pulsatility indices
- Reduced flow in adjacent arteries

7. INTRAOPERATIVE MONITORING

a. Capabilities: Identification of flow abnormalities may warrant change in surgical technique, e.g. carotid endarterectomy or coronary artery bypass grafting

b. Technique

- Headset utilized for continuous monitoring
- Not working in sterile field

c. Interpretation

- Significant decrease in MCA flow velocities during cross-clamping of vessel may indicate need for shunting
- Audible signals related to micro-emboli may alter surgical technique

Chapter 21 Miscellaneous
(Conditions, Diagnostic Tests, and Treatment)

A. CONDITION: SUBCLAVIAN STEAL

1. Subclavian occlusion results in retrograde flow in ipsilateral vertebral artery (Fig. 19)
2. Patients usually asymptomatic
3. May have blood pressure difference of $\geq$ 15-20 mmHg
4. May have decreased pulses in affected arm
5. Arm claudication rare
6. Ipsilateral vertebral artery feeding high-resistance vascular bed (Fig. 20)
7. Surgical treatment may include a bypass graft or endarterectomy

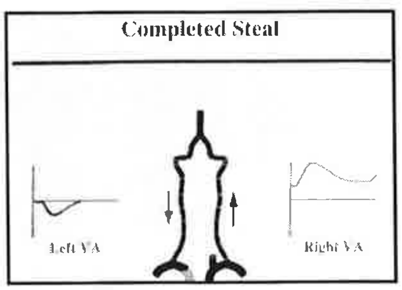


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Fig. 19. Retrograde flow

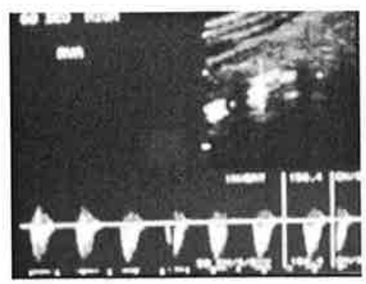


Fig. 20. High-resistance flow

B. CONDITION: TEMPORAL ARTERITIS

1. Inflammation of the arterial wall of the superficial temporal artery or its frontal and/or parietal branches
2. Inflamed arterial segments usually larger in diameter with homogeneous thickening evident on B-mode
3. An anechoic 'halo,' from edema of the intima, may be seen
4. Intimal thickening may result in hemodynamically significant stenosis where PSV's are doubled
5. Protocol still undergoing development

C. MISCELLANEOUS DIAGNOSTIC TESTS

(Additional Information in Arterial Section, Chapter 15)

1. Arteriography

a. Interpretation

—Normal anatomy should be visualized (Fig. 21)
—Abnormalities include filling defect, absent vessel, aberrant anatomy (Fig. 22)
—Percent stenosis calculated (Fig. 23)

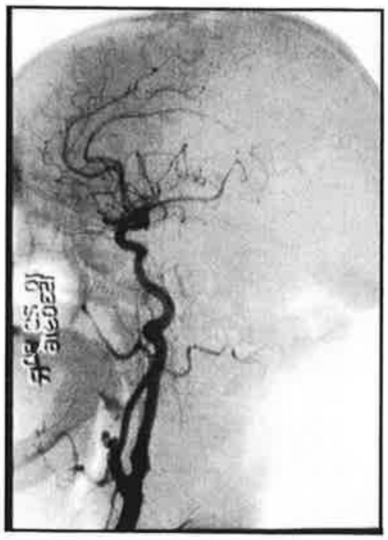


Fig. 21. Normal anatomy

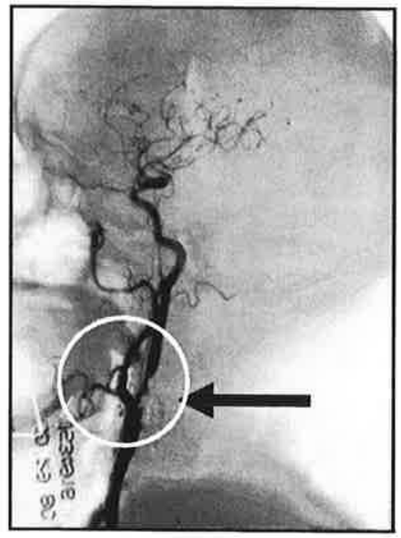
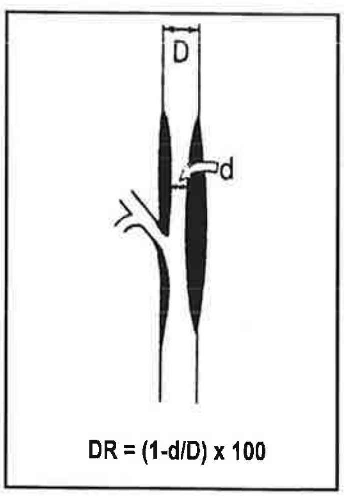


Fig. 22. Filling defect



EXAMPLE OF CALCULATING DIAMETER REDUCTION

Diameter reduction: D= 8mm, d=2mm

[1-(d/D)] x 100 =

[1-(.25)] x 100 =

.75 x 100 = 75%

Fig. 23. Calculate % stenosis based on NASCET study (North American Symptomatic Carotid Endarterectomy Trial)

2. **MR Angiography**

   a. Extremely sensitive to the presence of stenosis but tends to overestimate the disease process

   b. May be used when a carotid duplex study is equivocal or technically limited

3. **CT**

   a. Most frequent application in cerebrovascular disease: evaluate the nature of cerebral infarctions, intracranial aneurysms, hemorrhage, and AVMs

D. TREATMENT— Medical

Lifestyle modifications

1. Stop smoking

2. Weight control and low cholesterol diet

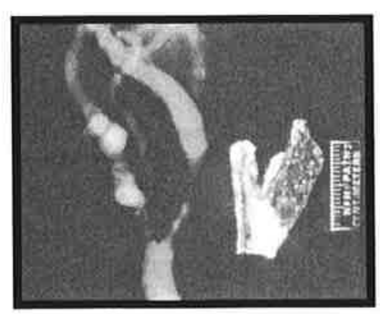
Pharmacological

Aspirin: an antiplatelet drug that decreases platelet aggregation resulting in decreased thrombotic activity

E. TREATMENT—Surgical

1. Stenosis: Endarterectomy, surgical removal of atherosclerotic material

2. Occlusion: Usually receives no surgical intervention

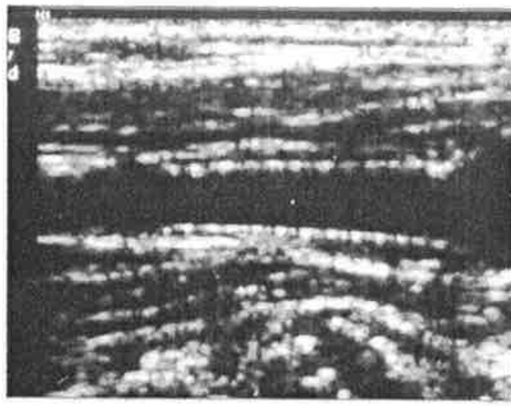


F. TREATMENT—Endovascular (Stent)

Previously described in Arterial Section, Chapter 15

1. Designed to maintain the intraluminal structure and patency of the artery; acts as a type of 'scaffold'

2. Increasing use of stents in carotid artery stenosis; may expect some flow acceleration post-stenting.

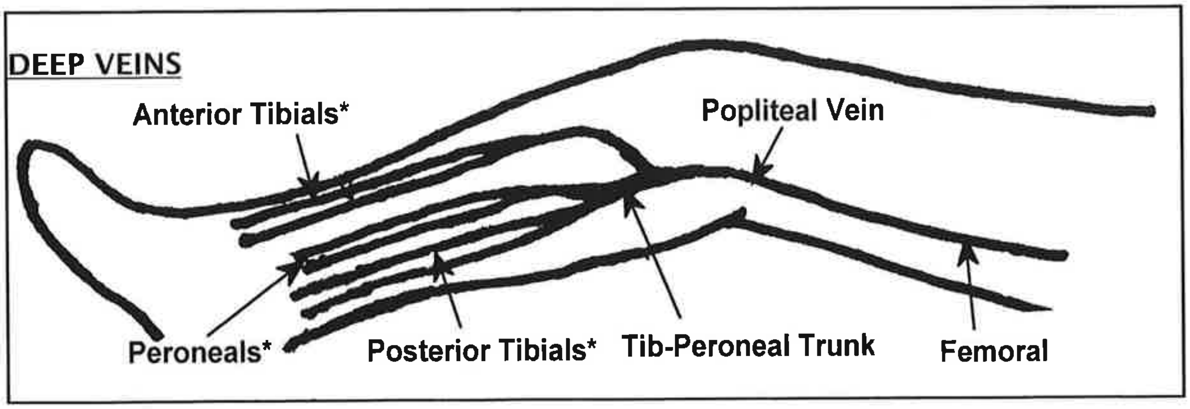


Example:
ICA Stent

Chapter 22: Venous Gross Anatomy

A. LOWER EXTREMITY VEINS

Note: The anatomical relationship of the veins to the heart is the same as for the arteries. Veins located at the ankle are considered distal; while veins located closer to the heart (e.g., femoral) are considered more proximal.



* The paired, deep veins of the calf follow the corresponding arteries; are called venae comitantes [corresponding veins]

1. <u>Paired peroneal veins (PerV):</u>
   a. Formed by confluence of venules
   b. Empties lateral leg
   c. Carries blood cephalad into tibial-peroneal trunk
2. <u>Paired posterior tibial veins (PTV):</u>
   a. Formed by confluence of venules
   b. Empties back of leg
   c. Carries blood into tibial-peroneal trunk
3. <u>Paired anterior tibial veins (ATV):</u>
   a. Formed by confluence of venules
   b. Empties front of leg

4. <u>Popliteal vein (PopV):</u>

   a. Formed by union of ATV & tib-peroneal trunk

   b. Usually just below knee

   c. Becomes femoral vein (previously called superficial femoral vein) when passes through adductor hiatus in lower thigh

5. <u>Femoral vein (FV):</u>

   Popliteal vein becomes FV when vein passes through adductor hiatus

6. <u>Common femoral vein (CFV):</u>

   Formed by joining of FV & deep femoral vein

7. <u>External iliac vein (EIV):</u>

   Common femoral vein becomes EIV when vein passes through inguinal ligament

8. <u>Common Iliac Vein (CIV):</u>

   Formed by confluence of external & internal iliac veins

> Because the left common iliac vein passes under the right common iliac artery, extrinsic compression may be evident. This pressure point may account for left sided DVT; also known as May-Thurner Syndrome.

9. <u>Inferior vena cava (IVC):</u>

   a. Formed by confluence of common iliac veins

   b. Commonly at level of 5th lumbar vertebra

   c. Carries blood into right atrium of heart

SUPERFICIAL VEINS

1. Lesser/small saphenous vein (LSV/SSV):

   Ascends back of calf joining popliteal vein

2. Greater/Great saphenous vein (GSV):

   Longest vein in body originating on dorsum of foot; traveling medially to saphenofemoral junction in the groin (about level of CFA bifurcation)

PERFORATORS

1. Carry blood from superficial veins into deep veins (Fig. 2)

2. Posterior arch vein:

   - Has three ankle perforators
   - Plays major role in development of venous stasis ulcers (Fig. 3)

3. LSV/SSV has an important lateral perforating branch

Fig. 2. Blood flows from the superficial veins (S) to the deep venous system (D)

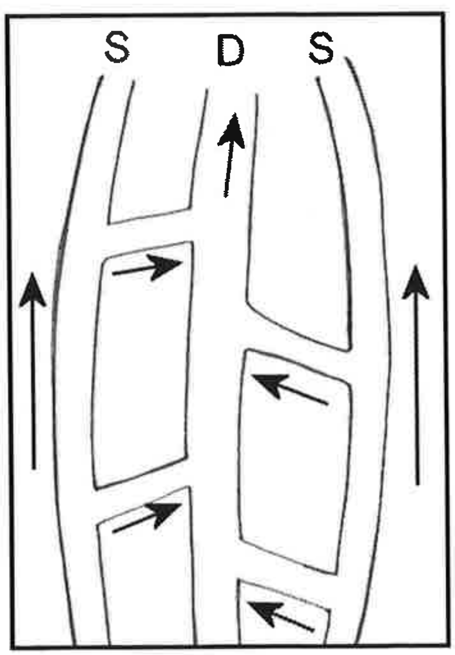
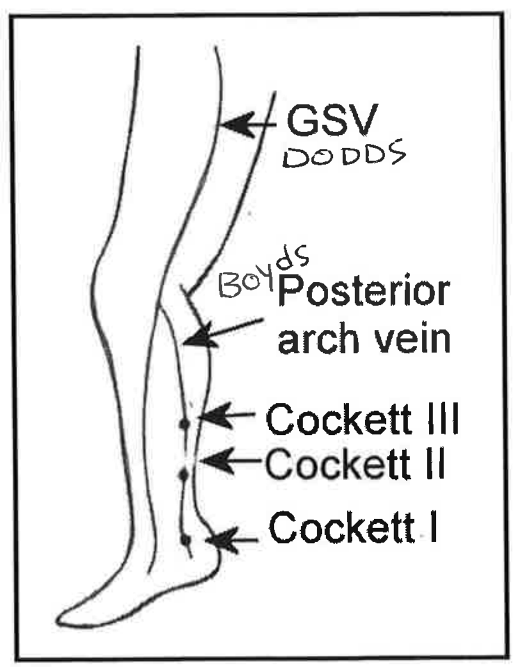


Fig. 3. Important perforators of posterior arch vein



VENOUS SINUSES

1. Intracranial:

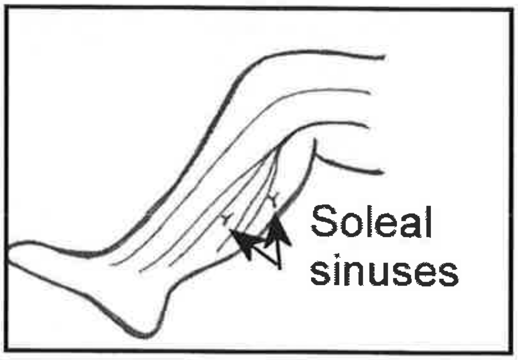
   Spaces between dura mater & periosteum that drain blood into the IJV

2. Lower extremity:

   a. Dilated channels in soleal & gastrocnemius muscles

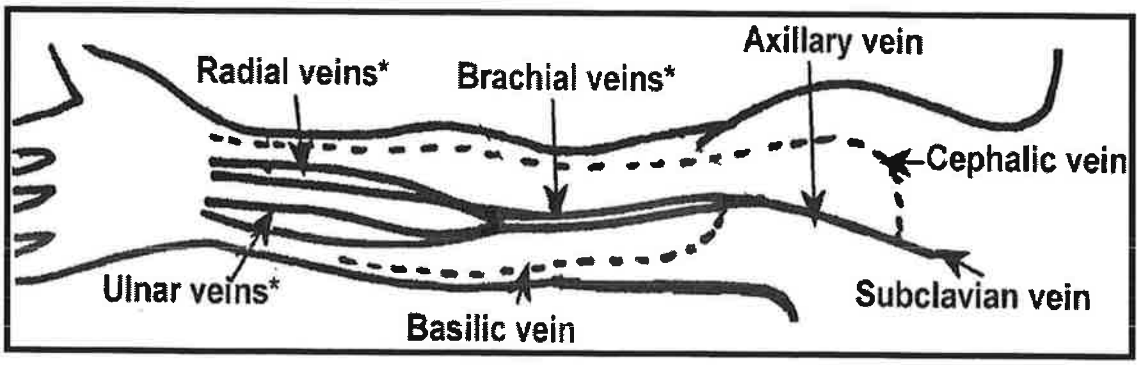
   b. Drains blood into the PTV & PerV

   c. Major part of calf-muscle pump



B. UPPER EXTREMITY VEINS

DEEP VEINS



* The paired, deep veins of the arm follow the corresponding arteries; called venae comitantes [corresponding veins]

1. Paired radial veins (RadV):

   a. Formed by confluence of venules

   b. Empties lateral hand & forearm

2. Paired ulnar veins (UlnV):

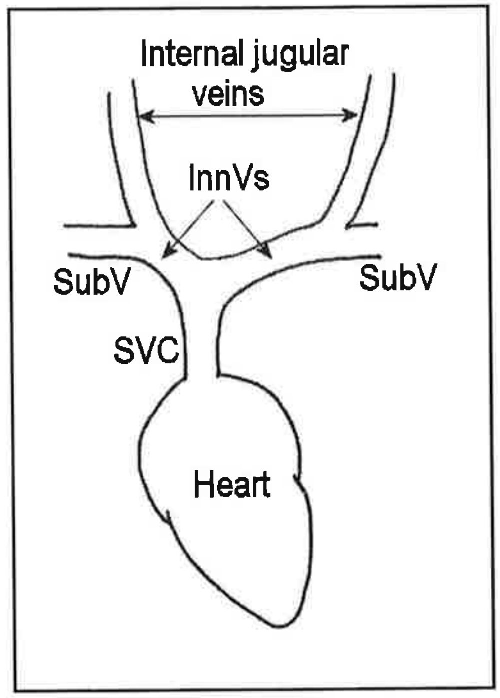
   a. Formed by confluence of venules

3. <u>Paired brachial veins (BraV):</u>

   Formed by confluence of radial and ulnar veins

4. <u>Axillary vein (AxV):</u>

   Formed by confluence of brachial vein & basilic vein (basilic vein in superficial system)



5. <u>Subclavian vein (SubV):</u>

   Formed by confluence of axillary

   vein & cephalic vein (cephalic vein in superficial system)

6. <u>Innominate vein (InnV):</u>

   Formed by confluence of subclavian vein & internal jugular (Also called <u>brachiocephalic)</u>

7. <u>Superior vena cava (SVC):</u>

   a. Formed by confluence of right & left innominate veins

   b. Carries blood to right atrium

SUPERFICIAL VEINS

1. <u>Basilic vein:</u>

   a. Formed by the digital veins

   b. Empties medial aspect of arm

   c. Joins brachial vein to form axillary vein

   d. Can be harvested for arterial bypass conduit

   e. See illustration on preceding page

2. Cephalic vein:

   a. Formed by digital veins

   b. Empties lateral aspect of arm

   c. Joins axillary vein to form subclavian vein

   d. Can be harvested for arterial bypass conduit

   e. See illustration on preceding page

C. CENTRAL VEINS

1. Superior Vena Cava (SVC):

   a. Formed by confluence of innominate veins

   b. Drains head and upper extremity veins

   c. Terminates in right atrium

2. Inferior Vena Cava (IVC):

   a. Formed by confluence of common iliac veins

   b. Drains lower half of body

   c. Terminates in right atrium

3. Portal System:

   a. Portal vein

      - Formed by superior mesenteric and splenic veins
      - Drains abdominal part of digestive tract, pancreas, spleen, & gall bladder
      - Carries blood into sinusoids of the liver (hepato-petal flow)
      - Carries approximately 80% of blood flow to the liver.

   b. Hepatic veins

      - Carries blood from the liver into IVC (flow away from the liver called hepato-fugal)

4. Renal veins - empty into the IVC

D. **STRUCTURAL ANATOMY OF VEINS**

1. Function

   a. Thin walled, collapsible tubes that transport blood from capillaries toward heart

   b. Carry away waste products of cellular activity

   c. Not completely passive structures; have some element of reactivity, which may be referred to as veno-motor tone; contraction of smooth muscle cells can occur in response to stimulation of sympathetic nervous system, i.e. temperature, exercise, stress, trauma,

2. Anatomy

   a. Same three layers as arteries, e.g. intima, media, adventitia (medial layer very thin)

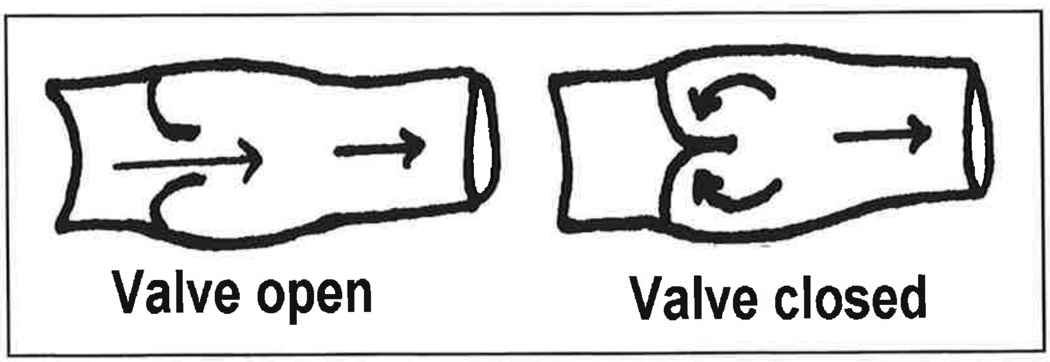
   b. Venous system starts at capillary level with progressive increase in size (venules smallest, vena cava largest)

3. <u>Venous valves</u>

   a. Extensions of intimal layer

   b. Bicuspid structures providing unidirectional flow

   c. Valves of lower extremities more susceptible to disease secondary to the effects of **venous thrombosis**, increased ambulatory venous pressure from gravity, increased intrabdominal pressure, and/or venous obstruction



VEINS WITHOUT VALVES:

- Soleal sinuses
- External iliac vein; contains valves approximately 25% of the time
- Common iliac & internal iliac
- Innominate
- Superior & Inferior vena cava

VEINS WITH VALVES:

- Great saphenous vein has approximately 12 valves; most below knee
- Small saphenous: 6 - 12 valves
- Perforators: each contains a valve
- Infrapopliteal (deep veins): 7 - 12 valves each
- Popliteal and femoral: 1 - 3 valves each
- External iliac vein: contains valves approximately 25% of the time
- Common femoral: 1
- Jugular vein: 1
- Basilic and cephalic
- Variable number in UE deep veins

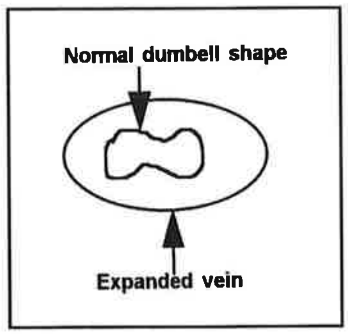
Chapter 23 Physiology and Hemodynamics

A. VENOUS RESISTANCE

1. Peripheral venous and arterial resistances are similar

   - Both arteries and veins carry same amount of blood
   - This paradox is explained by the "collapsible" nature of the venous wall

2. Flattened shape offers more flow resistance than circular shape



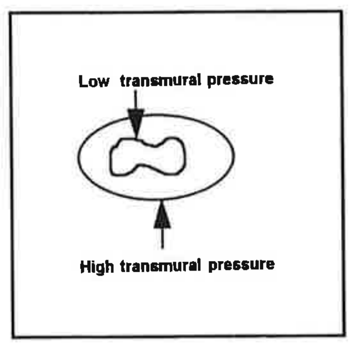
B. PRESSURE/VOLUME RELATIONSHIPS

1. When distended, cross-sectional area of the vein is about 3-4 times that of the corresponding artery. The extra-pulmonary veins carry about 2/3's of the blood in the body

2. Shape of veins determined by transmural pressure (distention pressure), e.g. the pressure within the vein versus pressure outside the vein

   - Low transmural pressure: low volume of blood results in dumbbell shape

   - High transmural pressure: high volume results in circular shape



3. Small pressure changes required to expand or distend vein from normal dumbbell shape to a circular one

4. Once completely distended, greater pressure ranges required to accommodate further increases in volume

C. HYDROSTATIC PRESSURE (HP)

1. Equivalent to the weight of a column of blood pressing against the

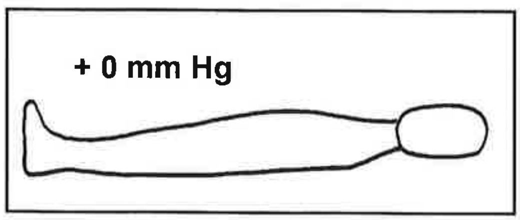
vessels of the body; uses the heart as a reference point (HP is zero at the heart level):

HP = pgh

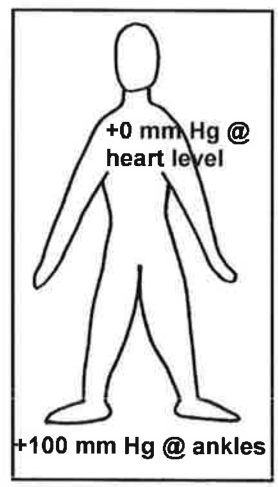
p = specific gravity of blood,
g = acceleration due to gravity,
h = distance from the heart

2. HP is added to the existing circulatory pressure and is related to position:

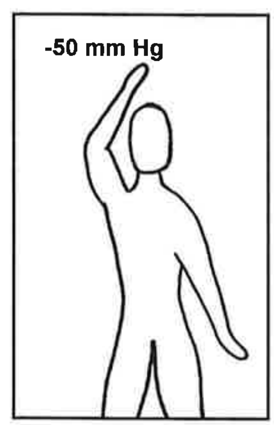
a. Supine: HP exerted on veins and arteries negligible, assumed to be zero (Pressure [P] measured at all levels = actual circulatory P)



b. Standing: HP gradually increases from level to level down the body reaching approximately 100 mmHg at the ankle. Pressure measured is too high (Ankle P = circulatory P + 100 mmHg)



c. Body part above heart: Negative HP. Measured pressure less than circulatory P. However, minimum measured pressure is 0 mmHg because vessels will collapse at 0 mmHg (especially important in venous system). (Arm P = circulatory P - 50 mmHg)

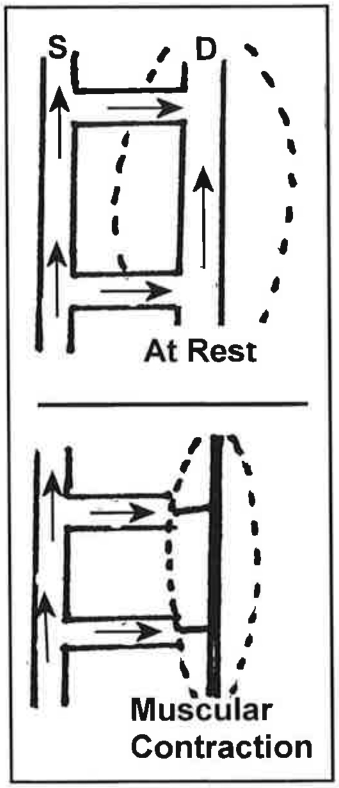


D. FACTORS AFFECTING VENOUS FLOW

1. Venous/skeletal muscle pump/ 'venous heart'

   a. Veins are reservoirs for blood collection

   b. Muscle contraction squeezes vein propelling blood toward heart

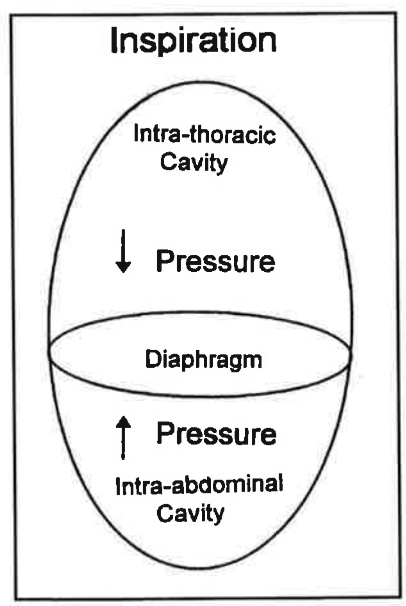


   c. Effective calf muscle pump

   - Blood moves from superficial system (S) to deep system (D)
   - Competent valves prevent reflux
   - Venous volume and pressure decreases
   - Venous return to heart increases

   d. Ineffective calf muscle pump

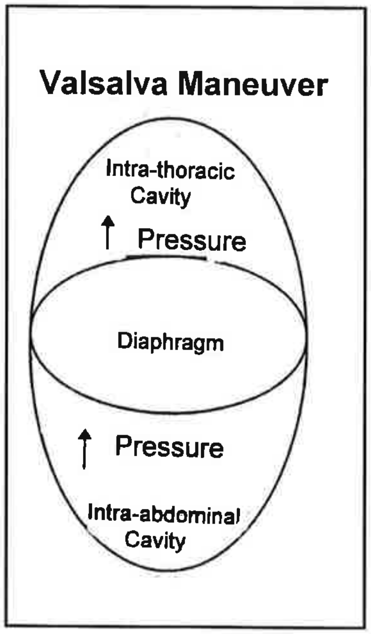
   - Incompetent valves cause reflux
   - Venous volume and pressure increases
   - Results in venous pooling & ambulatory venous hypertension



2. Respiration

   a. Inspiration

   - Decrease in intra-thoracic pressure
   - Increases blood flow from upper extremities
   - Increase in intra-abdominal pressure
   - Decreases blood flow from lower extremities



b. Exhalation

- Increase in intra-thoracic pressure
- Decreases blood flow from upper extremities
- Decrease in intra-abdominal pressure
- Increases blood flow from lower extremities

c. Valsalva maneuver

- Patient takes in deep breath & holds it , then bears down (as if having a bowel movement)
- Intrathoracic & intraabdominal pressure increase significantly
- All venous return halted
- This maneuver equates with proximal compression while performing Doppler assessment of the lower extremities

Chapter 24 Venous Testing
(Signs, Symptoms, Disease Mechanisms)

Note: It is very important to obtain a history and appropriate indications for testing prior to the performance of the study.

A. SIGNS AND SYMPTOMS

1. **Acute deep venous thrombosis (DVT):** Clinical diagnosis based on medical history and physical exam

MOST COMMON FINDINGS of DVT:

- Swelling
- Pain
- Redness
- Warmth

DIFFERENTIAL DIAGNOSES INCLUDE:

- Muscle strain
- Direct injury to the leg
- Muscle tear
- Baker's cyst
- Cellulitis
- Lymphangitis
- Heart failure
- Extrinsic compression

2. **Chronic venous obstruction**

MOST COMMON FINDINGS: CHRONIC VENOUS DISEASE

- Swelling
- Heaviness
- Discoloration
- Ulcers
- Varicosities

3. <u>Skin changes</u>

- Induration of tissue (edema): fluid accumulation
- Redness (erythema): inflammatory process, e.g. cellulitis
- Brownish discoloration (brawny): venous stasis usually in lower leg-to-ankle area (gaiter zone)
- Whiteness (pallor): arterial spasms secondary to extensive, acute iliofemoral thrombosis; limb threatening; called phlegmasia alba dolens
- Bluish discoloration (cyanosis): severely reduced venous outflow from iliofemoral thrombosis markedly reduces arterial inflow; limb threatening; called phlegmasia cerulea dolens

4. **<u>Venous ulcers</u>** <u>VS Arterial ulceration</u>

The following table includes some general guidelines.

Characteristics	Venous Ulcers	Arterial Ulcers
Location	Often near medial malleolus	Tibial area, toes, bony prominences
Appearance	Shallow, irregular shape	Deep, regular shape "punched out" appearance
Skin changes	Stasis changes: inflammation, infection, brawny discoloration, presence of varicosities	Trophic changes: dryness, scaly, atrophy, shiny skin, loss of hair, thickened toenails
Pain	Mild	Severe
Bleeding	Venous ooze	Little

5. **<u>Edema</u>**

a. Edema: body tissue contains an excessive amount of fluid

b. 'Pitting' edema:

- Fluid in subcutaneous tissue
- Depression of skin surface with manual pressure
- Can be related to many conditions (e.g., fluid retention, congestive heart failure, or elevated venous pressure)

6. **Lymphedema:**

a. Normally, lymphatic system drains excess fluid from tissue. Fluid accumulates when lymph nodes and/or lymph vessels are removed or damaged

b. Frequently seen after many types of cancer surgery

c. Non-pitting edema:

B. MECHANISMS OF DISEASE

1. **Risk Factors:** VIRCHOW'S TRIAD

a. Trauma to the vessel/ endothelial damage Example: →

> Paget-Schroetter Syndrome
>
> - Stress / effort thrombosis
> - Involves axillary or subclavian vein
> - Venous component of TOS

b. Venous stasis, e.g. immobility, chronic obstructive pulmonary disease, obesity, pregnancy, previous DVT, extrinsic compression. Example: →

> Superior Vena Cava (SVC) Syndrome
>
> - Obstruction by neoplasm (benign or malignant)
> - Edema & engorgement of vessels evident
> - Pt may have cough and/or difficulty breathing
> - Flow to the UE remains the same during inspiration, i.e. continuous flow

Why surgery? Surgery is a risk factor for DVT. May be due to alteration in endothelial cell function.

c. Hypercoagulability, e.g. certain protein deficiencies, pregnancy, cancer, hormones, i.e. estrogen intake

2. **Acute Thrombosis**

a. Intraluminal thrombi frequently begin at valve cusps or in soleal sinuses secondary to stagnation

b. Thrombi resulting from trauma occur at any site

c. Sequelae:

Chronic venous insufficiency

- Stretching of walls results in damage to valves
- Increased venous pressure causes flow changes

Post-phlebitic syndrome

- Chronic flow changes result in persistent edema, stasis changes and pain
- May also lead to ulceration

Pulmonary embolism

- Thrombus breaks loose
- Travels into pulmonary circulation
- CTA chest and pulmonary angiography are currently the definitive diagnostic tools
- Can be life-threatening

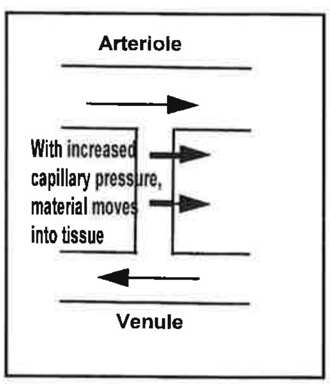
3. Valvular incompetence

a. Valves no longer maintain unidirectional flow

b. Calf muscle pump no longer forces blood cephalad towards heart or from superficial system into deep

c. RESULTS OF INCOMPETENCE

- Increased pressure/venous hypertension
- Ambulatory venous hypertension: increasing pressure when patient stands or walks
- Precipitates edema, varicosities, and ulcers
- Fluid, red blood cells, and fibrinogen may leak into surrounding tissue; hemosiderin, from breakdown of stagnant red blood cells, cause brawny discoloration
- Breakdown of other substances prevents proper tissue nutrition and oxygenation



4. **Varicose veins**

   a. **Primary**: dilated veins secondary to valvular incompetence of superficial system; deep system intact

   b. **Secondary**: dilated veins caused by incompetence of the superficial system resulting from a deep venous obstruction; deep system not intact

5. **Congenital venous disease**

   a. Avalvular (valveless) vein(s)

   b. Arteriovenous malformations (AVM)

   c. Syndromes: (for example)

      - Klippel-Trenaunay: Can include multiple varicosities of the superficial system and hypoplastic or absent deep veins

6. **Portal hypertension**

   a. Elevated venous pressure results from obstruction of blood flow; may result in reverse (hepato-fugal) flow in the portal vein and increased portal venous pressure that impedes blood flow into the liver.

   b. Usually related to some form of advanced chronic liver disease, e.g. hepatitis C; cirrhosis

Chapter 25 Photo-Plethysmography (PPG)

1. CAPABILITIES

To document venous insufficiency / quantitate venous reflux in patients with chronic swelling, venous ulcers and/or varicose veins

2. LIMITATIONS

a. Contraindicated in acute deep vein thrombosis

b. Improper placement of PPG sensor results in inaccurate information, e.g. over a varicose vein

c. Thickening of skin may prevent adequate penetration of infrared light

d. Intact skin required for sensor placement

3. PATIENT POSITIONING

Seated with legs dangling, i.e. non-weight bearing

4. PHYSICAL PRINCIPLES

a. Plethysmography measures volume changes

b. PPG not true plethysmography, but can determine changes in blood content of skin (microcirculation), which reflects intravenous volume

c. Photocell consists of light emitting diode and photo-sensor

d. Diode transmits infrared light into subcutaneous tissues, which is <u>reflected</u> back to photo-sensor; light is not absorbed

e. The cutaneous blood flow determines the reflection.

e. Blood attenuates light in proportion to its content in tissue. Increased blood flow results in decreased reflection. However, that's displayed as an increase/positive deflection on the waveform.

f. Electrical coupling

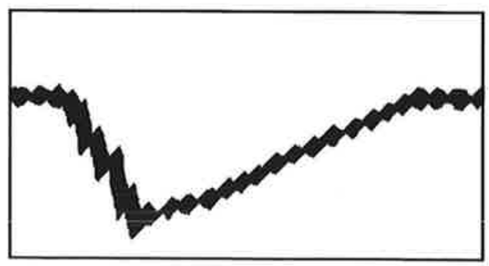
- Method to increase gain and display signal
- Only two types available: DC and AC
- Allows one type of current to pass and blocks the other type

DC Coupling	AC Coupling
• Direct current • Electric voltage that is either positive or negative • Current flows in only one direction • Batteries are DC • Detects slower changes in blood content • Used for venous studies	• Alternating current • Electric voltage that reverses polarity (positive or negative) 60 times a second • Current flows in both directions • Wall plugs deliver 120 volts of AC • Detects fast changes in blood content • Used for arterial studies

g. Calibration

- Cannot be calibrated volumetrically as with venous air plethysmography (APG)
- Important to maintain the same 'size' or 'gain' setting throughout study
- Significant difference in tracing should mean a significant difference in blood volume

h. Display: tiny arterial pulsations usually evident; superimposed on tracing of venous flow



5. TECHNIQUE

a. Sensor applied to lower leg, approximately 5 - 10 cms above medial malleolus (must not be over a varicosity)

b. Strip chart recorder:

- Running at slow speed (usually 5 mm/sec)
- Stylus records on heat sensitive paper

c. Patient instructed to complete series of exaggerated dorsiflexions to empty calf veins

d. Manual compressions of calf can be performed; must be done bilaterally to ensure consistency

e. Tracing continues to be obtained after dorsiflexions or manual compressions to record venous refill time/venous reactive time (VRT)

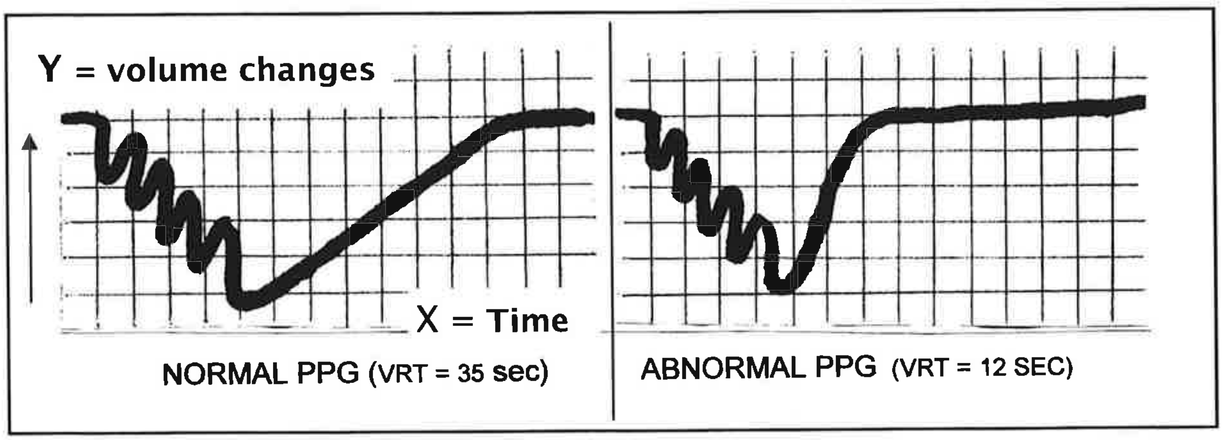
f. If the VRT is <20 seconds, study repeated after **tourniquet** applied to eliminate influence of superficial system

6. INTERPRETATION - Quantitative

a. Normal: VRT $\geq$ 20 seconds without tourniquet

b. Superficial system incompetence: VRT of <20 seconds without tourniquet but normalizes, i.e. >20 seconds, with tourniquet

c. Deep system incompetence: VRT of <20 seconds with and without tourniquet application



(Graphs not drawn to scale)

d. Trouble-shooting

- *Artifact from patient movement?* Study technically impossible
- *Absent deflections or grossly irregular tracings?* Ensure equipment on DC mode (not AC) and/or on PPG setting (not Doppler, VPR, etc. [depending on equipment])
- *Deflections off-the-scale or barely discernible?* Adjust gain/sensitivity/size setting

Chapter 26 Air-Plethysmography (APG)

1. CAPABILITIES

To evaluate venous function, e.g. document venous insufficiency / quantitate venous reflux

2. LIMITATIONS

a. Inability of patient to maintain positions or perform exercise

b. Casts, traction, or heavy non-removable bandages

c. Will not diagnose incompetent perforators or isolated incompetent distal veins

3. PATIENT POSITIONING

Study started with patient supine; patient then assumes a variety of standing positions

4. PHYSICAL PRINCIPLES

a. Plethysmography measures volume changes

b. Pneumatic cuff connected to pressure transducer; volume changes amplified and converted to analog display

c. Documents volume changes secondary to position changes /exercise

5. TECHNIQUE

a. Patient supine; cuff applied to lower leg; air injected into cuff to a specific pressure level; manual calibration completed

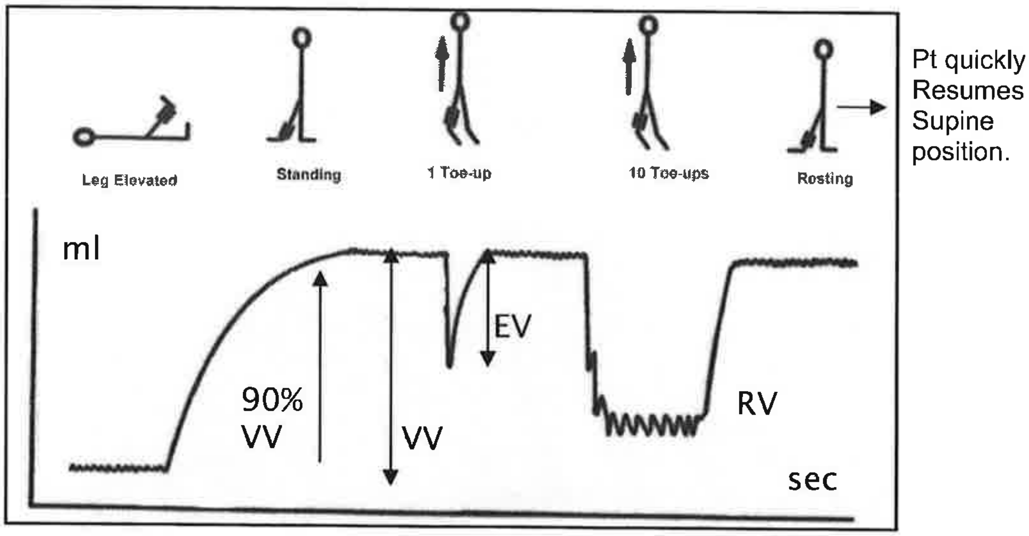
b. Leg passively elevated to empty venous system (zero venous volume)

c. Patient quickly stands bearing weight on contralateral leg (increase in venous volume (VV) documented on strip chart recorder)

d. Patient stands with weight distributed equally over both feet; performs 1 tip-toe exercise to document a decrease in calf VV (calculated as the ejection volume [EV]) and the venous filling time (VFT); 10 tip-toe maneuvers then completed

e. Patient quickly resumes supine position; test leg elevated to empty veins

f. If finding abnormal, study repeated after tourniquet applied to eliminate influence of superficial system



6. INTERPRETATION - Quantitative

a. Venous filling index (VFI) shows rate of venous refilling; calculated using venous volume (VV) and venous filling time (VFT):

- Normal: Want this to be a low number

b. Ejection fraction (EF) measures calf muscle pump function; calculated using the ejection volume (EV) and functional venous volume (VV):

- Normal: Want this to be a high %

c. Residual volume fraction (RVF) is equivalent to ambulatory venous pressure in mmHg; calculated as the percentage of VV remaining after 10 tip-toes movements:

- Normal: Want this to be a low %

Chapter 27 Continuous-Wave Doppler

1. **CAPABILITIES**

   a. Evaluation of deep venous obstruction

   b. Evaluate venous incompetence

2. **LIMITATIONS**

   a. Unable to differentiate abnormal flow patterns from DVT versus extrinsic compression, i.e. tumor, ascites, pregnancy

   b. Normal flow patterns may be evident with partial or well-collateralized thrombosis

   c. Paired deep veins in calf limit diagnosis of an isolated calf clot

   d. Sources of <u>false positive</u> studies include:

      - Extrinsic compression: tight clothing, tumors, ascites, pregnancy, obesity, improper patient positioning, or pain causing muscle contraction
      - Peripheral arterial disease (PAD): decreased venous filling
      - Chronic obstructive pulmonary disease: elevated central venous pressure
      - Improper Doppler angle or probe pressure

   e. Sources of <u>false negative</u> studies include:

      - Collateral development
      - Presence of bifed system (multiple deep veins)

   f. Requires a very experienced technologist

3. **PATIENT POSITIONING**

   a. Supine with body shifted to side being examined

   b. Leg externally rotated with hip and knee flexed

   c. An extreme side-lying position may diminish extrinsic compression on IVC from pregnancy, ascites, or tumor formation

d. Position should facilitate venous filling (extremities lower than heart), i.e. reversed Trendelenburg (approximately 30°)

4. **PHYSICAL PRINCIPLES**: Doppler effect has been previously described

Reminders

- Use 5 MHz probe held at 45-60° to skin surface
- Deep veins found adjacent to corresponding artery
- Correct vessel identification requires hearing accompanying arterial signal
- Examination based on audible venous signals

5. **TECHNIQUE**

a. Begin with asymptomatic side

b. Placing probe at inguinal ligament, identify common femoral artery; angle probe medially to insonate common femoral vein (CFV) and evaluate venous Doppler signals

VENOUS FLOW PATTERNS:

- Spontaneous
- Phasic
- Augment with distal compression
- Augment with proximal release

c. CFV on symptomatic side evaluated for same flow patterns

d. The femoral [superficial femoral], popliteal, and posterior tibial veins are evaluated as previously described going from asymptomatic side to symptomatic side

e. Abnormal venous signals require repositioning and re-evaluation before a conclusion can be reached

Chapter 28 Duplex/Color Flow Imaging

1. CAPABILITIES

PERIPHERAL VEINS:

a. Identify venous thrombosis (help dfferentiate acute from chronic)

b. Evaluate non-occluding/partial thrombus

c. Detect calf lesions

d. Distinguish between extrinsic compression and intrinsic obstruction

e. Evaluate soft tissue masses

f. Detect venous incompetence

g. Document recanalized channels or collateralization

ABDOMINAL AND PELVIC VEINS

a. Assist with documenting elevated systemic venous pressure

b. Identify venous thrombosis

c. Evaluate patency of inferior vena cava interruption devices

d. Assess portocaval shunts

e. Evaluation of some liver diseases

2. LIMITATIONS

Visualization may be sub-optimal secondary to edema, scarring, recent surgery, obesity (lower frequency transducer may help improve visualization)

a. Sources of false positive studies include:

- Extrinsic compression: e.g., tumors, ascites, pregnancy
- Peripheral arterial disease (PAD): decreased venous filling
- Chronic obstructive pulmonary disease: elevated central venous P
- Improper Doppler angle or probe pressure
- Superior vena cava syndrome

b. Sources of false negative studies include:

- Proximal obstruction
- Technically limited studies

c. Peripheral veins: Lower Extremity

May be difficult to thoroughly evaluate infra-popliteal veins secondary to vessel size, depth, and course

d. Peripheral veins: Upper Extremity

Difficult to thoroughly evaluate subclavian and brachiocephlic/ innominate veins secondary to bony structures

e. Abdominal and pelvic veins:

May be difficult to thoroughly evaluate all veins secondary to vessel depth and presence of bowel gas

3. PATIENT POSITIONING

Peripheral veins: Lower Extremity

- Facilitate venous filling, i.e. reversed Trendelenburg
- Diminish extrinsic compression, (e.g.,extreme lateral decubitus position
- For reflux testing: When the patient is standing, he/ she bears weight on the contralateral leg.

Peripheral veins: Upper Extremity

- Supine or low fowlers position
- Arm in 'pledge position'

Abdominal and pelvic veins

- Supine with head slightly elevated.
- Left lateral decubitus with head-of-bed elevated slightly
- Reversed Trendelenburg
- Whatever works!

4. PHYSICAL PRINCIPLES

The physical principles of Doppler and B-mode imaging have been previously described in Chapters 11 and 12. Other areas of particular concern with venous imaging include:

a. Velocity signals obtained in sagittal view

b. Combine transverse and sagittal views

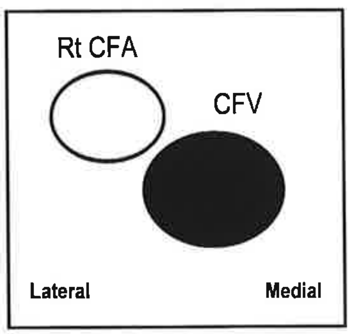
c. Maximize color filling & flow patterns

- Adjust color scale to detect slower velocities
- Change wall filters
- Increase color gains

5. TECHNIQUE: PERIPHERAL VEINS—LE

<u>ACUTE DEEP VENOUS THROMBOSIS (DVT)</u>

a. Common femoral vein (CFV):



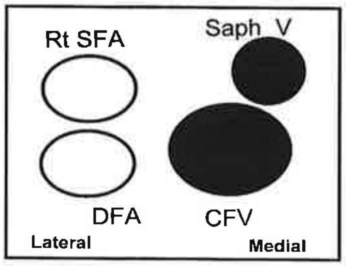
- Formed by confluence of femoral & deep femoral veins
- Begin scanning at inguinal ligament and identify common femoral vein (CFV) and artery
- Should be free of visible thrombus
- In transverse view assess for complete compressibility of vein walls

Coaptation, from a Latin word which means "to fit together" is another word for compressibiity

VENOUS FLOW PATTERNS

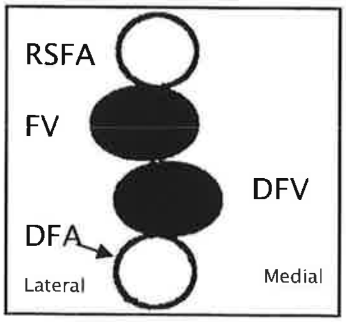
Evaluate venous Doppler signals in sagittal view:

- Spontaneous
- Phasic
- Augment with distal compression
- Augment with proximal release



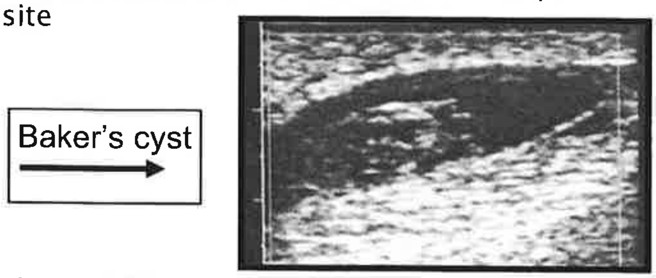
b. Saphenofemoral junction:

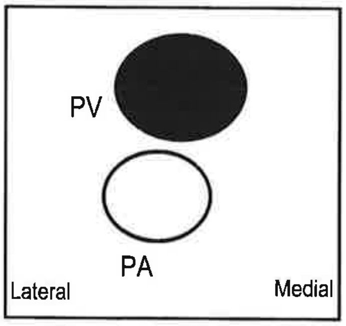
- Where greater/great saphenous joins CFV
- Thrombosis of superficial system at or near deep system may require more aggressive treatment than an isolated superficial system thrombosis



c. Femoral vein (FV):

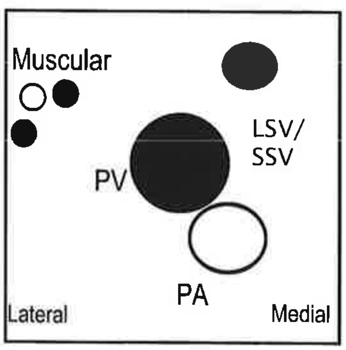
- Confluence of FV and deep femoral vein/profunda femoris vein forms CFV
- Distal third of thigh, FV dives deep passing through adductor canal becoming Popliteal vein; vessel may normally be very difficult to compress at this site

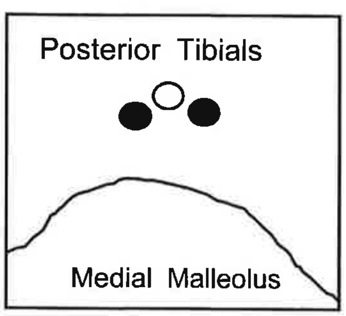




d. Popliteal vein (PV)

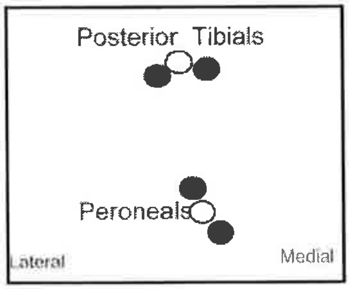
- Formed by confluence of anterior tibial vein and tibio-peroneal trunk
- Scan through out popliteal fossa observing for any cystic structures or masses, i.e. Baker's cyst
- Also evaluate muscular/gastrocnemius veins, lesser /small saphenous vein, and veins of trifurcation
- May need antero-lateral approach to evaluate anterior tibial veins
- The lesser/small saphenous (LSV/SSV)-popliteal junction must be carefully evaluated as a thrombosis of the superficial system at or near the deep system may require more aggressive treatment than an isolated superficial system thrombosis.





e. Posterior tibial veins (PTV):

- Probe between medial malleolus and Achilles tendon to locate the posterior tibial veins (2 veins with one artery)
- Evaluate vessels along medial surface of calf
- Terminates in tibio-peroneal trunk near popliteal fossa



f. Peroneal veins (PerV):

- Evident a few centimeters proximal to the malleolus (2 veins with one artery); deeper than PTVs
- Evaluate vessels along medial calf (with PTVs)
- Terminates in tibio-peroneal trunk

6. TECHNIQUE: PERIPHERAL VEINS—LE

CHRONIC VENOUS INSUFFICIENCY

a. Flow reversal, usually in response to a Valsalva maneuver or during proximal manual compressions, indicates venous reflux

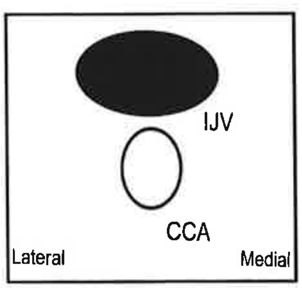
b. Methods to identify reflux include:

- Spectral analysis: reversed venous flow lasting more than 1second noted during proximal compression maneuver
- Color-flow imaging: Color changes noted during proximal compression maneuver

c. Some protocols include evaluation of major venous sites with patient standing and/or blood pressure cuffs applied distal to the transducer

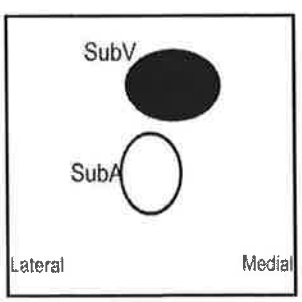
7. TECHNIQUE: PERIPHERAL VEINS— UE

ACUTE DEEP VENOUS THROMBOSIS (DVT)



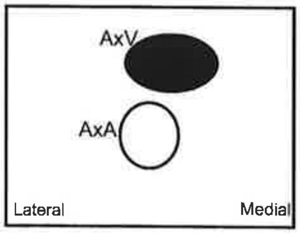
a. Internal jugular vein (IJV):

- Near common carotid artery (CCA) in neck
- Evaluate from mandible to where it joins subclavian vein (SubV)
- Transverse view to evaluate compressibility; with sagittal orientation for Doppler signals



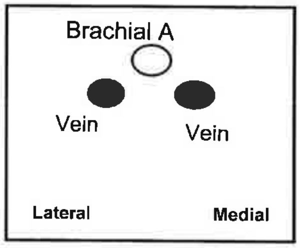
b. Subclavian vein (SubV):

- Formed by confluence of cephalic/ axillary veins
- Joins with IJV to form Innominate vein (bilaterally)
- Evaluated from supraclavicular approach
- Can usually be followed only to outer border of first rib (clavicle prevents further visualization)
- Infraclavicular approach may be used to evaluate distal segment



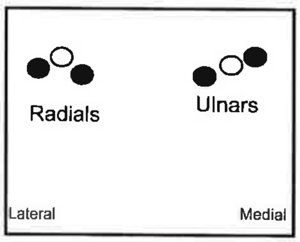
c. Axillary vein (AxV):

- Formed by confluence of the basilic and brachial veins
- Evaluated with arm raised, i.e. pledge position, and probe in axilla (arm pit)



d. Brachial veins (BraV):

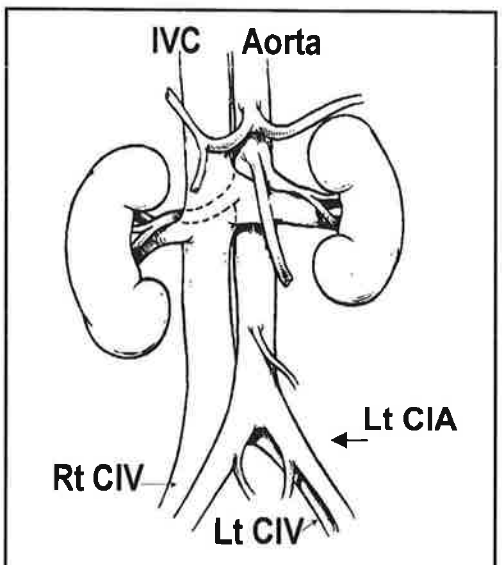
- Formed by radial and ulnar veins
- Location varies considerably from antecubital fossa to axilla



e. Radial & ulnar veins:

- Formed by confluence of palmar arch veins

8. TECHNIQUE: <u>ABDOMINAL AND PELVIC VEINS</u>



Note: Aorta located to left of midline.

a. Inferior vena cava and pelvic veins evaluation begins in transverse view at level of umbilicus

b. Evaluation of other abdominal vessels begins in transverse view at xiphoid process.

c. Accurate identification of vessels depends on careful attention to anatomical landmarks, e.g. pulsating aorta, aorto-iliac bifurcation, left renal vein

IVC = Inferior Vena Cava,

CIV = Common Iliac Vein,

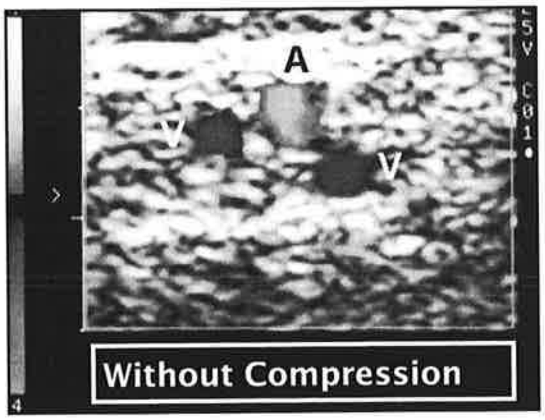
CIA = Common Iliac Artery

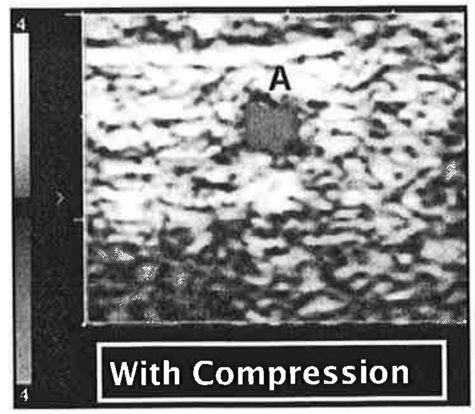
9. INTERPRETATION - Qualitative

NORMAL

a. Lower Extremity: peripheral veins

- Completely compressible





- Normal Doppler venous signals

  ◊ Spontaneity: Signal immediately heard at all sites, except posterior tibial veins

  ◊ Phasicity: Sound varies with respirations (Fig. 33)

  — Lower extremities: increases with expiration and decreases with inspiration

  — Upper extremities: decreases with expiration and increases with inspiration

  ◊ Augmentation with distal compression and proximal release (Fig. 34)

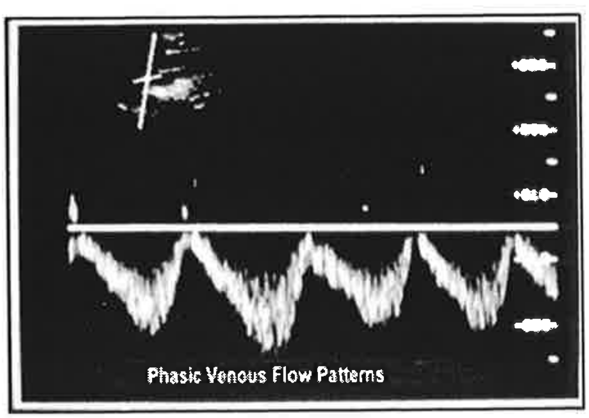


Fig. 33. Normal venous phasic flow patterns

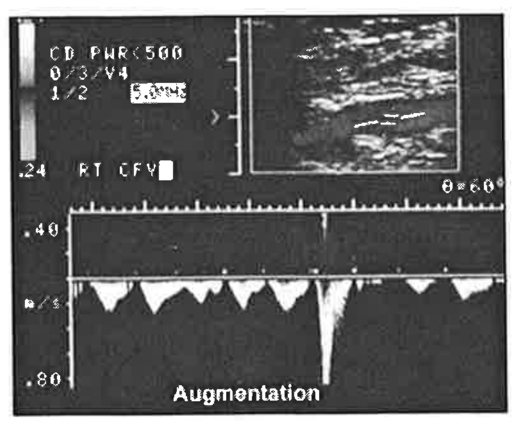


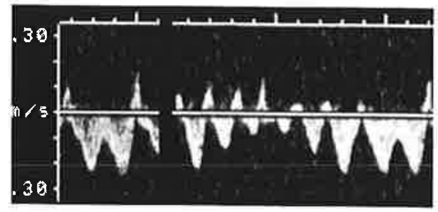
Fig. 34. Augmentation with distal compression

- Miscellaneous Findings:

  ◊ Valsalva maneuver: Flow augments following this maneuver; augmentation during it signifies reflux / retrograde flow

  ◊ Extrinsic compression: Pressure on vessels from surrounding tissues and/or structures can cause abnormal flow patterns

  ◊ Pulsatile venous flow pattern evident with fluid overload or congestive heart failure



b. Upper Extremity: peripheral veins

- May not completely compress secondary to bony structures
- A quick breath through pursed lips should collapse vein
- Pulsatile Doppler signals in subclavian and innominate veins
- Augmentation with distal compression may not be evident

c. Abdominal and pelvic veins

- Dilatation with deep inspiration
- Spontaneous Doppler signals
- Phasic , bi-directional/pulsatile Doppler signals in **IVC**, **renal,** and **hepatic** veins (Fig. 35)
- Minimally phasic, continuous Doppler signals in **portal**, **splenic** and **mesenteric** veins (Fig. 36)
- Note: During inspiration, there is minimal flow fluctuation in the portal vein; flow is variable in the hepatic vein.

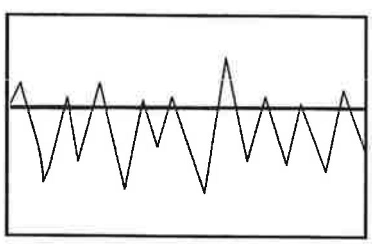


Fig. 35. Phasic, bi-directional flow

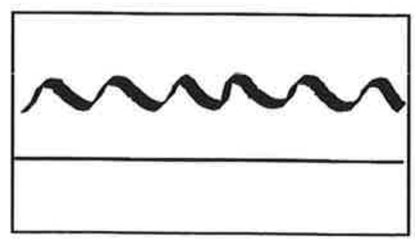


Fig. 36. Minimally phasic, continuous flow

ABNORMAL: <u>ACUTE THROMBOSIS</u>

a. Peripheral veins not completely compressible being filled with very low level echoes (IVC and iliac veins not compressible secondary to depth of vessels)

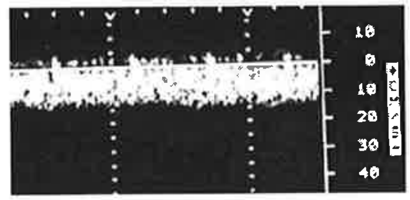
b. Visible thrombus may be evident

c. Vessel is dilated

d. Abnormal Doppler signals

ABNORMAL VENOUS FLOW PATTERNS

- If flow is not spontaneous at the CFV, FV and/or Pop V, an obstruction (DVT or extrinsic compression) distal to or at that site is suggested.
- If flow is not phasic, but rather continuous, a proximal obstruction should be considered.
- If no augmentation with distal compression is seen, think about an obstruction between where you are compressing and where you are listening, or slightly more proximally.
- If there is no augmentation with proximal release, consider a more proximal obstruction.
- If flow increases during proximal compression, that signifies venous reflux.

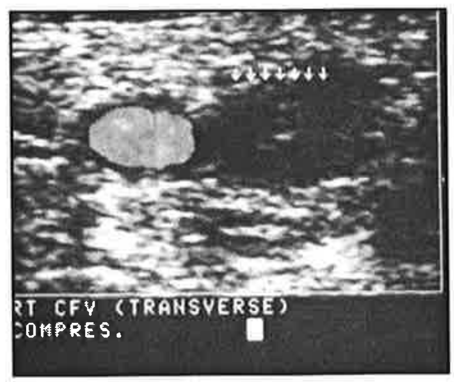


e. Color flow Doppler shows no filling or partial filling of vessel lumen

f. Thrombus formation may be poorly attached

g. Thrombus formation may have spongy texture

h. ROULEAU FORMATION:

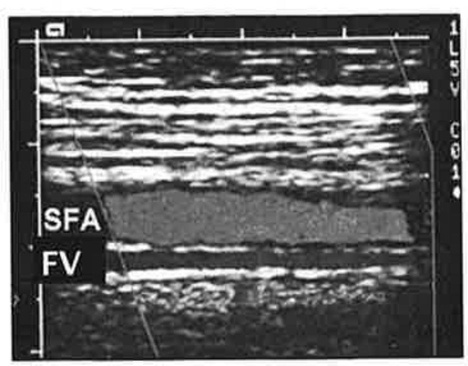


- Very sluggish flow seen as heterogeneous material moving with respirations and augmentation maneuvers
- Red blood cells arranged like rolls of coins or rouleau (from a French word meaning 'roll')
- A compressible vessel with evidence of rouleau formation on B-Mode, could be normal or suggest proximal obstruction (e.g., DVT, increased Central venous pressure, SVC syndrome)

ABNORMAL: <u>CHRONIC CHANGES</u>

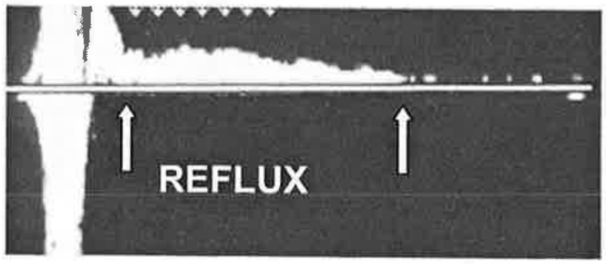
a. B-Mode:

- Highly echogenic
- Visible collateralization or recanalization may be evident
- Vessel not dilated; may retract over time



b. Flow characteristics:

- Abnormal Doppler venous signals may be evident (e.g., continuous, decreased phasicity, or no augmentation)
- Venous reflux lasting > 0.5 or >1 sec. (depends on literature source)



c. Color flow Doppler: venous reflux appears as a shift in color from flow away from probe to flow towards probe during the Valsalva maneuver and/or following compression distal to the transducer

MISCELLANEOUS ABDOMINAL FINDINGS

a. IVC interruption device usually placed below renal veins and may appear as bright echogenic lines

b. Systemic venous hypertension - persistent dilated vessels evident

c. Portal hypertension - Increased portal venous pressure can result in variety of flow alterations:

- Reversed flow in portal vein, i.e. hepato-fugal
- Collateral development

d. Budd-Chiari Syndrome

- Results from hepatic vein occlusion
- Primary site of obstruction may be hepatic vein, sinusoids, or IVC
- Clinical findings may include: hepatomegaly, abdominal pain, sudden onset of ascites

Chapter 29 Miscellaneous (Diagnostic Tests, Treatment)

A. TEST: CONTRAST VENOGRAPHY

1. Capabilities

Although contrast venography is still considered to be "gold standard," the numbers of venograms has markedly decreased due to the accuracy of duplex scanning.

a. Ascending venography: utilized for evaluation of acute DVT, congenital venous disease and/or anomalies, evaluation of chronic venous thrombosis

b. Descending venography used primarily to detect and quantify reversed flow from incompetent venous valves

2. Limitations

a. Highly technical in technique and interpretation

b. Relatively expensive

c. Uncomfortable

d. Adverse effects: allergic reactions or extravasation of contrast media

e. May be contraindicated in patients with severe PAD secondary to risk of extravasation, and patients with allergies to iodine

3. Technique

a. Ascending venography

- Radio-opaque contrast media injected into vein on dorsum of the foot to visualize anatomy
- Serial x-rays taken as media passes through veins identifying filling defects, anatomic variations, development of collateral channels

b. Descending venography

- Radio-opaque contrast media injected into common femoral vein
- Serial x-rays taken as material passes through veins to detect and quantify reversed flow and location of incompetent valves

B. TREATMENT— Medical

Lifestyle modifications

Control risk factors (related to Virchow's Triad)

1. Decrease venous stasis

   a. Limit long periods of inactivity or bed rest

   b. Promote venous drainage

   - Wear **support hose or elastic stockings**
   - Elevate legs
   - Intermittent pneumatic calf compression during and after surgery

2. Prevent injury or infection

3. Aware of hypercoagulability states/factors

Pharmacologic

1. **Anticoagulant therapy**

   Prophylaxis

   Low-dose heparin:

   - Interferes with formation of blood clot
   - Does not lyse an existing thrombus

   Acute DVT and/or PE

   a. Heparin: Loading dose followed by continuous intravenous infusion for 5-10 days

   b. Coumadin

   - Nearing end of heparinization oral anticoagulation started
   - Prescribed for three to six months

c. **Lytic therapy**

- Breakdown of thrombus, i.e. hemolysis
- Urokinase or streptokinase may be recommended in acute iliofemoral DVT

2. Treat venous ulcers with medicated compression dressing, e.g. unna boot; Profore (4 layer) dressing.

C. TREATMENT—Surgical and Endovascular

ACUTE DVT OR PE

1. **Vena caval interruption devices**

   a. Performed to prevent PE in patients who cannot be anticoagulated

   b. Using fluoroscopy, a filter placed in IVC via jugular or femoral vein.

   c. An external caval clip may also be placed around IVC during abdominal surgery to decrease risk of PE

2. Iliofemoral venous thrombectomy

   Performed for impending limb loss, i.e. phlegmasia cerulea dolens, if **thrombolytic** therapy, e.g. streptokinase or urokinase, does not dissolve clot

CHRONIC VENOUS INSUFFICIENCY

1. Vein ligation of incompetent perforators performed

2. Valvular reconstruction or valve transplantation procedures infrequent

VARICOSE VEINS

1. Surgical procedures include: saphenous vein removal or local excision (phlebectomy) of varicosities

Laiser

2. Endovenous procedures treat the veins from the inside, using heat energy that causes the vein to shrink, then slowly disappear

   - Major ablation types are radio frequency ablation and laser ablation
   - Duplex scanning utilized to confirm patency of veins; size and depth of vein to be treated; during procedure to follow catheter and post procedure to confirm vein ablation and absence of thrombosis.

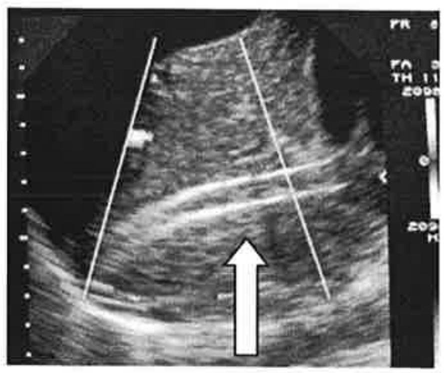
3. Sclerotherapy used to treat small varicosities by sealing them off: Sclerosing agent, e.g. sodium tetradecyl sulfate, injected into varix and compression dressing applied

4. Valvular reconstruction or valve transplantation infrequent

PORTAL HYPERTENSION

Transjugular intrahepatic portosystemic shunt (TIPSS): Percutaneously created shunt between portal & hepatic veins

- Usual approach: Right IJV to IVC into right hepatic vein
- 'Bridge' created to portal vein
- Tract stented with metallic endoprosthesis



NOTE: Hyperechoic edges of the shunt; in this case the shunt is occluded.

Chapter 30 Quality Assurance

STATISTICS

HINT
$\frac{\text{Numerator}}{\text{Denominator}}$

1. **Sensitivity**

   - Definition - The ability of a test to detect disease
   - Calculation = $\dfrac{\text{\# of true positive tests}}{\text{\# of all positive diagnoses detected by the gold standard}}$

2. **Specificity**

   - Definition - The ability of a test to identify normalcy
   - Calculation = $\dfrac{\text{\# of true negative tests}}{\text{\# of all negative diagnoses detected by the gold standard}}$

3. **Positive predictive value (PPV)**

   - Definition - How often positive study is correct
   - Calculation = $\dfrac{\text{\# of true positive tests}}{\text{\# of all positive noninvasive studies (true positives + false positives)}}$

4. **Negative predictive value (NPV)**

   - Definition - How often negative study is correct
   - Calculation = $\dfrac{\text{\# of true negative tests}}{\text{\# of all negative noninvasive studies (true negatives + false negatives)}}$

5. **Accuracy**

   - Definition - How good a test is
   - Calculation = $\dfrac{\text{Total \# of correct tests}}{\text{Total \# of all studies}}$
   - NOTE: Calculation falls between sensitivity & specificity as well as between PPV & NPV

STATISTICAL CORRELATIONS

The following 2 X 2 factorial table provides a well accepted method of calculating statistics appropriate for the vascular laboratory.

'Gold Standard'

		+	—
TEST	+	A (True Positive)	B (False Positives)
	—	C (False Negatives)	D (True Negatives)

$$\text{Accuracy} = \frac{A + D}{A + B + C + D}$$

$$\text{Sensitivity} = \frac{A}{A + C}$$

$$\text{Specificity} = \frac{D}{B + D}$$

$$\text{Positive Predictive Value} = \frac{A}{A + B}$$

$$\text{Negative Predictive Value} = \frac{D}{C + D}$$

(In an examination situation, the X-axis and Y-axis may be changed, i.e. Gold Standard along vertical axis and noninvasive studies along horizontal axis. In this case, only 'C' and 'B' switch places on the grid. The formulas remain the same)

EXAMPLE OF STATISTICAL CALCULATIONS

140 patients had noninvasive and invasive venous studies. 68 of the noninvasive AND invasive studies showed an acute DVT and are considered to be true positives. 54 showed no evidence of disease (true negatives). 12 were false negative, e.g., the noninvasive study was negative but the venogram was positive. In 6 cases, the noninvasive studies were positive, but the venogram was negative (false positive).

'Gold Standard'

TEST	+	—
+	68 (True Positives)	6 (False Positives)
—	12 (False Negatives)	54 (True Negatives)

$$\text{Accuracy} = \frac{122}{140} = 87\%$$

$$\text{Sensitivity} = \frac{68}{80} = 85\%$$

$$\text{Specificity} = \frac{54}{60} = 90\%$$

$$\text{Positive Predictive Value} = \frac{68}{74} = 92\%$$

$$\text{Negative Predictive Value} = \frac{54}{66} = 82\%$$

PATIENT SAFETY

1. **Infection control**

   a. All personnel follow good hand-washing techniques before/after direct patient contact, after handling bodily fluid/secretions, and after using the bathroom

   b. Standard/universal precautions maintained

      - All bodily fluids considered infectious
      - Personal protection equipment available, e.g. gloves, gowns, masks, and used by personnel when appropriate

   c. Disinfect external ultrasound probes (Always check manufacturers' guidelines as specific recommendations may vary).

      - First, remove visible residue with a soft cloth lightly dampened with mild soap; wipe with water-dampened cloth

      - Isopropyl alcohol solution may be used, but don't let the alcohol air dry

      - Do not use: undiluted bleach, abrasive cleaners, solvents, e.g., thinner or benzene

      - High-level disinfection— commonly used products include:

        - Glutaraldehyde, e.g., Cidex
        - Hydrogen peroxide ($\geq$ 6%)

      - Low-level disinfection—commonly used products include:

        - Quarternary ammonium
        - N-Alkyl, e.g., T-spray II, Sani-cloth
        - Hydrogen peroxide (3%)

      - Sterile saline is not considered a disinfectant agent.

2. **Medical emergencies**

a. All personnel providing direct patient care are trained in basic cardiac life support

b. Emergency response system/s available
- In-house emergency team
- Community emergency number: 911

c. Patients provided with safe environment:

- Escorted to/from examination room
- Assisted on/off exam table as needed
- If patient on stretcher, side rails up before and after study completed

d. For hospitalized patients, make observations with regard to special equipment and care issues, e.g. oxygen tanks, IV medications

e. A test procedure is not carried out if the examiner determines the patient cannot safely tolerate or cooperate with the protocol

f. All electric/electronic equipment and devices meet the institution's safety code requirements

Arterial Self Assessment Test Questions

1. **Which of the following is not true about laminar flow?**
   A. Layers of fluid particles are moving against one another
   B. Fastest flow is in center stream
   C. A stationary layer is at the wall
   D. Flow is considered to be stable flow
   E. Flow patterns have a flattened profile

2. **Which of the following is not true about a hemodynamically significant stenosis?**
   A. There is a notable reduction in volume flow
   B. A notable reduction in pressure is evident
   C. A 75% diameter reduction is equivalent to a 50% area reduction
   D. Distal effects may only be detect able following stress
   E. Flow abnormalities depend on the length of the lesion, degree of narrowing, and other factors

3. **One of the following arteries normally has a lower pulse amplitude than the others.**
   A. Iliac artery
   B. Aorta
   C. Popliteal artery
   D. Femoral artery

4. **All of the following cause vasodilatation of a high resistance bed EXCEPT:**
   A. Body heating
   B. Exercise
   C. Hyperventilation
   D. Stenosis

5. **Mrs. K is scheduled for arterial testing due to pain in her legs. All of the following risk factors suggest Ms. K is a likely candidate for arterial disease EXCEPT:**
   A. Ms. K. has hypertension
   B. She had ligation of her GSV for varicosities one year ago.
   C. She is diabetic
   D. Ms. K. smokes one pack of cigarettes a day
   E. Her mother died from coronary artery disease

6. **All of the following are associated with Buerger's Disease EXCEPT:**
   A. Patients present with history of smoking
   B. Lesions are commonly distal to ankle and/or wrist
   C. Ankle brachial index may be normal
   D. Affects the large arteries of the extremities

7. **Which of the following is not a contraindication to a constant load treadmill?**
   A. Recent history of severe cardiac arrhythmias
   B. 8 year history of smoking.
   C. 10 year history of severe hypertension
   D. Past CVA resulting in unilateral paresis
   E. Recent motor cycle accident and subsequent amputation of left lower extremity

8. **After completing the exercise study, you believe the patient has multi level arterial disease. What is that conclusion based on?**
   A. Recovery to the resting levels takes $\leq$ 2 minutes
   B. The patient takes 2-6 minutes to return to his pre-exercise pressures
   C. Recovery time is 6-12 minutes.
   D. The patient takes < 4 minutes to return to his pre-exercise pressures

9. **Ms. J has a brachial pressure of 124 mmHg on the right and 102 mmHg on the left. What is the significance of this finding?**
   A. The pressures are WNL (within normal limits)
   B. These findings indicate a hemodynamically significant lesion of the left subclavian/axillary arteries
   C. You suspect the patient has a right subclavian steal
   D. Ms. J has a left radial occlusion.
   E. This patient has probably right innominate artery disease

10. **Ms. L's doctor has ordered a lower arterial plethysmographic study because of suspected leg pain. The test has all of the following capabilities EXCEPT:**

A. Combined with pressure measurements, it can help differentiate true claudication from false/pseudo claudication

B. It helps localize the level of an arterial obstruction

C. Plethysmography can discriminate flow from specific vessels

D. The study can be used to follow-up treatment

E. Plethysmography helps define the functional aspect of the disease, i.e. provides hemodynamic or physiologic information

11. **Which of the following is not true about volume plethysmography?**

A. Pneumatic cuffs are wrapped around the extremities.

B. Air is inflated into the cuff bladder to a specific level to provide a good seal

C. Arterial flow beneath the cuff results in a volume change in the limb

D. The increased volume change with diastole results in increased pressure in the cuff bladder

E. Increased pressure in the cuff bladder is shown as a positive deflection on the waveform

12. **All of the following are true about thoracic outlet syndrome (TOS) EXCEPT:**

A. It's the result of a neurovascular bundle compressed by shoulder structure

B. The majority of the symptoms are related to compression of the vasculature

C. It can cause numbness/tingling of the arm

D. It may result in pain/aching of the shoulder /forearm

E. Exercise/upward positions increase the symptoms

13. **All of the following are true about flow related to an AVF EXCEPT:**

A. High resistance flow is expected proximal to it

B. Flow through the communication is very accelerated and turbulent.

C. If close to the heart, it can result in right-sided heart failure

D. Reduced flow distal to it can result in ischemia

E. Increased pulsatility will be evident in the venous side of the AVF

14. **Which is the least likely complication following arteriography?**

A. Hematoma

B. Compartment syndrome

C. Pseudoaneurysm

D. Arterial occlusion

E. Neurologic complications

15. **An ankle/brachial index (ABI) is another example of a physiologic study. Which of the following statements about ABIs is not true?**

A. The ankle pressures are divided by the highest brachial pressure

B. A patient with an ABI of 0.7 is likely to complain of rest pain

C. An ABI of 0.4 means the ankle pressure is 40% of the arm pressure

D. Incompressible vessels can result in falsely elevated pressure measurements

16. **If the ABIs are consistent with incompressible vessels, which of the following tests might provide information about arterial flow?**

A. Segmental pressures

B. Exercise testing

C. Toe pressures

D. Reactive hyperemia

17. **Regardless of the method, what is the correct order of taking the blood pressure measurements?**

A. Brachial, ankle, calf, above the knee, high-thigh (if 4 cuff method)
B. Ankle, calf, thigh (if 3 cuff method), brachial
C. Brachial, high-thigh (if 4 cuff method) above the knee, calf, ankle
D. Thigh (if 3 cuff method), calf, ankle, brachial

18. **A patient presents to your department complaining of pain in the right calf every time he walks 3 blocks. The segmental pressure study is within normal limits and does not document the presence of arterial occlusive disease. What would your next action be?**

A. Assume the patient's symptoms are not related to arterial disease
B. Warn the patient that the doctor will probably order an invasive study next
C. Determine whether the patient can safely walk on a treadmill and complete the exercise test
D. Just discharge the patient from your department after the segmental pressure test is done

19. **All of the following are characteristics of claudication EXCEPT:**
A. Pain in muscle during exercise
B. Pain usually subsides after rest
C. Always vascular in origin
D. Discomfort usually predictable

20. **How could a cold exam room affect pulsatility changes/pulsatile flow patterns on your patients?**
A. It results in higher-resistance flow patterns
B. Cooling decreases the pulsatility seen on an arterial waveform
C. It results in lower-resistance quality
D. Pulsatile flow patterns become less pronounced

21. **Mrs. Darling is scheduled for a lower extremity arterial study. As you go over her previous medical history you know that all of the following are risk factors for peripheral arterial disease (PAD) EXCEPT:**
A. History of heart disease
B. Recent immobility
C. History of collagen disease
D. Recent vascular surgery

22. **In the Doppler equation, the number 2 represents:**
A. The number of times the transducer is multiplied
B. The calculation value for moving reflectors
C. Two times the Doppler angle theta
D. Two Doppler shifts

23. **As you clarify a patient's signs/symptoms, you notice that her right foot is much redder than her left while she is sitting, but when in a supine position for the test, you see that the right leg/foot becomes pale. What is this phenomenon called?**
A. Ischemic rest pain
B. Brawny discoloration
C. Dependent rubor
D. Necrosis

24. **How does secondary Raynaud's differ from primary Raynaud's?**
A. Secondary Raynaud's is only associated with exposure to cold
B. Ischemia is intermittent in secondary Raynaud's
C. In secondary Raynaud's, the patient has fixed arterial disease as well
D. Secondary Raynaud's includes aneurysmal changes

25. **All of the following are true about how Doppler waveforms are processed and displayed EXCEPT:**
A. Waveforms can be displayed with spectral analysis or an analog recorder
B. Analog recorders display true peak systolic and diastolic frequencies
C. A zero crossing frequency meter counts each time the input signal crosses the zero baseline
D. High frequency waves have many oscillations
E. Flow direction varies relative to the cardiac cycle

26. **Which of the following statements about spectral analysis is not true?**
A. It uses the FFT method
B. The displayed frequencies are estimated
C. Time is on the horizontal axis.
D. Frequencies are displayed on the vertical axis
E. This technique has fewer drawbacks than analog recordings

27. **With the three-cuff method, how should the thigh pressure compare to the brachial pressures in the following situations?**
A. You would expect the thigh pressures to be lower than the brachial pressures to exclude proximal disease
B. Thigh pressures are expected to be at least 30 mm Hg higher than the brachial pressures to indicate the absence of in-flow disease
C. The thigh pressures are expected to be similar to the brachial pressures in the absence of AIOD
D. The three-cuff method is not accurate in the evaluation of proximal disease

28. **Which of the following findings reflect an abnormal response to penile imaging?**
A. Increased diameter in cavernous arteries
B. Flow in the dorsal vein increases to >20 cm/sec
C. PSV in the artery increases to at least 30 cm/sec
D. An increase in EDV of the cavernous artery is evident

29. **The toe pressure should be what % of the brachial pressure:**
A. 90%-100%
B. 60%-80%
C. 40%-50%
D. 20%-30%

30. **The most common site of stenosis or occlusion of the hemodialysis access graft is:**
A. Inflow artery stenosis
B. Arterial anastomosis
C. Puncture site
D. Outflow vein

31. **Duplex imaging of the peripheral vessels include all of the following EXCEPT:**
A. Real-time B-mode (gray scale) imaging provides anatomical information
B. Spectral analysis is provided using pulsed Doppler
C. Flow characteristics are displayed based on CW Doppler
D. Color flow information is displayed on the image after it's evaluated for phase and frequency based on the Doppler shift
E. Initially, optimal sample size is 1-1.5 mm

32. **The <u>most important</u> differential diagnostic parameter for pseudoaneurysm is:**
A. Presence of thrombus
B. Swirling blood flow
C. Communication tract between artery and aneurysm
D. Flow where there should not be flow

33. **What is the standard of practice in naming a bypass graft?**
A. The first part of the name refers to the proximal anastomosis; the last part is the location of the blockage
B. The name refers to the location of the incisions
C. The name of the bypass is based on the location of the disease
D. The first part is the proximal anastomosis; the last part is the distal anastomosis
E. The bypass is named after the vessel being bypassed

34. **If the pre-stenotic flow is 50 cm/sec with flows of 100 cm/sec at the narrowing, what is the anticipated DR?**
A. <50% DR
B. ≥50% DR
C. >60% DR
D. ≥75% DR
E. >99% DR.

35. **As you evaluate a bypass graft, all of the following information is abnormal EXCEPT:**
A. ABIs are reduced by < 0.15
B. Flow patterns change from multiphasic to biphasic
C. Wall defect is observed
D. Decreased PSVs are evident at the smallest graft diameter
E. PSVs are >2:1 at any graft segment

36. **As you evaluate Doppler flow patterns in Mr. B, you document a PSV of 104 cm/sec at the distal aorta, 218 cm/sec at the take-off of the left common iliac artery (CIA), and 226 cm/sec at the right proximal CIA. What would your preliminary report include:**
A. This patient has a > 75% DR of the distal aorta
B. These findings are consistent with > 75% DR of the right CIA
C. Your patient has >50% DR of the CIAs bilaterally
D. You have documented a >50% DR of the right CIA and >75% DR of left
E. These flows are WNL

37. **Using a four-cuff setup on the right lower extremity, the examiner notes a high-thigh cuff pressure of 130 mmHg, with the remaining pressures averaging 115 mmHg. The higher brachial pressure is 160 mmHg. All of the following significant conditions could produce these findings EXCEPT:**
A. Aortoiliac arterial obstruction
B. Common femoral arterial obstruction
C. Superficial and deep femoral (profunda femoris) arterial obstruction
D. Deep femoral arterial obstruction

38. **A 64-year-old male presents to the vascular laboratory with a history of left extremity pain, calf claudication after 150 feet, ulceration on the great toe, and ankle Doppler systolic pressures of 300 mmHg (brachial pressures are 140 mmHg bilaterally). What do these findings most strongly suggest?**
A. Small vessel disease
B. Elevated pressure due to the large cuff size
C. Erroneous pressures due to probable arterial calcification
D. Severe hypertension

39. **The Adson maneuver is performed for assessment of:**
A. The palmar arch
B. Dialysis access grafts
C. Thoracic outlet syndrome
D. Carpal tunnel syndrome

40. **A young man presents to the vascular laboratory complaining of impotence. His past medical history is significant for the presence of insulin-dependent diabetes. What would be the most likely penile/brachial index if vasculogenic impotence exists?**
A. 0.8
B. 0.5
C. 1.0
D. 1.3

41. **As you follow the popliteal artery distally you have difficulty imaging the posterior tibial and peroneal arteries. What would your next step be?**
A. State the study is technically limited and the vessels can't be evaluated
B. Go to the ankle level and move proximally
C. Change to a higher frequency transducer
D. Move the transducer to the anterolateral position
E. Get out your CW Doppler

42. **Your next patient is complaining of pain in his left calf muscle every time he gets to the third hole at the golf course. As you scan your patient's proximal superficial femoral artery you get PSVs of 52 cm/sec. Just distal to that you find flows of 258 cm/sec. What would your preliminary report include?**
A. This patient has a >75% DR
B. Velocities are consistent with an occluded SFA
C. Flow changes suggest a >50% DR.
D. Velocity measurement are consistent with a <50% DR
E. The flow patterns are WNL and exclude PAD

43. **As you are imaging the distal anastomosis of a fem-distal bypass graft, you find that you have some reversed flow in the native artery at that site. What conclusion can you draw from that finding?**
A. The reversed flow is the first sign of impending graft failure
B. It is an abnormal finding consistent with a stenosis of the native artery distal to the distal anastomosis
C. This is an expected finding and considered to be WNL
D. This finding suggests increased disease in that native vessel
E. The reversed flow is consistent with a graft fistula

44. **All of the following are points to remember when scanning the abdomen EXCEPT:**
A. Deeper vessels require a lower frequency transducer
B. You may need to use a variety of approaches
C. More pressure on the transducer may be required to see areas of interest
D. You will make large pivoting movements with abdominal imaging
E. Color-flow Doppler assists with vessel identification

45. **The renal artery to aortic ratio is not accurate in which of the following situations?**
A. Occlusion of the contralateral renal artery
B. PSV in the aorta of <35 cm/sec
C. PSV in ipsilateral renal artery is 400 cm/sec
D. Aortic PSV is between 50-90 cm/sec

46. **Patient presents with crampy, dull abdominal pain immediately after meals. If this patient has chronic mesenteric ischemia, which of the following statements is true?**
A. Isolated disease in the celiac artery would account for this problem
B. Ischemia is usually caused by extrinsic compression
C. 2 of 3 mesenteric vessels would be abnormal in most cases
D. Aortic and SMA disease is an expected finding

47. **As you proceed to evaluate the SMA in longitudinal orientation, you remember all of the following is true about that vessel EXCEPT:**
A. It takes off deep to the aorta.
B. The vessel turns shortly after origin.
C. It travels parallel to the aorta.
D. You expect to find the origin distal to the celiac artery

48. **Which vessel(s) will you find just distal to the SMA?**
A. The celiac artery
B. The renal arteries
C. The IMA
D. The common iliac arteries

49. **As you evaluate the distal aorta, all of the following are true EXCEPT:**
A. You will observe for signs of aneurysmal disease
B. The aorta normally enlarges at its bifurcation
C. High resistance flow is expected at the terminal aorta
D. The bifurcation of the aorta is usually at the level of the umbilicus
E. The iliac arteries course posteriorly

50. **All of the following statements are true about Leriche syndrome EXCEPT:**
A. Fatigue in hips and thighs can occur
B. Impotence can occur in males
C. Caused by obstruction of iliac artery(s)
D. Absence of femoral pulses

TEST ANSWERS

1.	E	26.	B
2.	C	27.	C
3.	B	28.	B
4.	C	29.	B
5.	B	30.	D
6.	D	31.	C
7.	B	32.	C
8.	C	33.	D
9.	B	34.	B
10.	C	35.	A
11.	D	36.	C
12.	B	37.	D
13.	A	38.	C
14.	B	39.	C
15.	B	40.	B
16.	C	41.	B
17.	A	42.	A
18.	C	43.	C
19.	C	44.	D
20.	A	45.	B
21.	B	46.	C
22.	D	47.	A
23.	C	48.	B
24.	C	49.	B
25.	B	50.	C

Cerebrovascular Self Assessment Test Questions

1. **Which of the following would be considered a lateralizing symptom?**
   A. Right hemiparesis indicating a lesion of the right hemisphere of the brain
   B. Bilateral loss of vision suggesting poor perfusion to the left hemisphere of the brain
   C. Left arm plegia which may be consistent with decreased blood flow to the right hemisphere of the brain
   D. Right amaurosis fugax related to left internal carotid artery disease

2. **Mr. JP is brought to the emergency room with a sudden onset of aphasia, right hemiparesis (more severely affecting the face and arm than the leg), and behavioral changes. Which artery is the most likely cause of these signs & symptoms?**
   A. Anterior cerebral artery
   B. Middle cerebral artery
   C. Posterior cerebral artery
   D. The vertebrobasilar system

3. **Mrs. HM is having ocular pneumoplethysmography (OPG). Her brachial blood pressure measurements are considered to be within normal limits: 122/76 on the right and 126/74 on the left. As you prepare to perform the study, how much vacuum pressure will you apply?**
   A. 100 mm Hg
   B. 200 mm Hg
   C. 300 mm Hg
   D. 500 mm Hg.

4. **With occlusion of the proximal left internal carotid artery (ICA), perfusion to the left hemisphere would least likely occur from which of the following collateral pathways?**
   A. Left external carotid artery (ECA) to distal ICA via occipital artery
   B. Right anterior cerebral artery (ACA) to left ACA via anterior communicating artery (AcomA).
   C. Basilar artery to posterior cerebral artery (PCA) to posterior communicating artery (Pcom) to distal ICA
   D. Left ECA to facial artery to nasal artery into distal ICA via reversed flow in ophthalmic artery

5. **What percentage of blood in the common carotid artery is normally directed into the internal carotid artery?**
   A. 20%
   B. 30%
   C. 50%
   D. 70%

6. **Which of the following symptoms is most likely related to vertebrobasilar insufficiency?**
   A. Weakness of the right arm
   B. Temporary loss of vision of left eye
   C. Drop attack
   D. Numbness of the left leg
   E. Aphasia

7. **Your next patient has a transcranial Doppler study (TCD) ordered. Which of the following statements about that procedure is not correct?**
   A. You will be using a 2 MHz continuous-wave Doppler
   B. The angle of insonation is assumed to be zero
   C. The middle cerebral, anterior cerebral, posterior cerebral, and terminal ICA are evaluated through the transtemporal approach
   D. TCD is used to assess intracranial flow patterns

8. **As you perform the TCD, the transducer is at the transtemporal window with the sample volume approximately 55-65 mm deep. What is the most likely significance of the bi-directional flow detected (almost identical flow patterns appearing above and below the baseline)?**
   A. This is an example of mirror-imaging or cross-talk because of strong reflectors or increased gain
   B. This is an expected finding
   C. This represents collateral flow likely in response to proximal disease
   D. This flow pattern is consistent with a vasospasm

9. **Your carotid duplex study documents a very rounded , continuous right common carotid artery waveform. What is the least likely reason for that finding?**
   A. Cardiomyopathy
   B. Severe stenosis of the brachiocephalic artery
   C. Occlusion of the right ICA
   D. Critical stenosis at the origin of the right CCA

10. **As you put together the diagnostic information obtained during a carotid duplex study you know that the most significant criteria for an 80-99% diameter reduction is which of the following?**
   A. Peak systolic velocity
   B. End diastolic velocity
   C. ICA/CCA ratio
   D. Spectral broadening
   E. Acceleration time

11. **Which cerebrovascular arteries tend to be asymmetrical?**
   A. ECAs
   B. ICAs
   C. Vertebrals
   D. MCAs
   E. ACAs

12. **Mr. FG is complaining of a 20 minute episode of left vision loss. What does this sign/symptom (S/S) suggest to you?**
   A. The patient had a CVA related to a blockage in the left vertebral artery.
   B. You suspect a lesion in the right ICA resulting in this TIA
   C. This CVA is most likely secondary to a diseased left ICA
   D. You are concerned this CVA is the result of a right ICA occlusion
   E. This TIA is most likely related to a blockage in the left ICA

13. **What is the predominate source of nutrients to the walls of the ICA and ECA?**
   A. Capillary beds in the adjacent tissue
   B. Vasa vasorum
   C. Endothelial arterioles
   D. The flowing blood within the vessel lumen

14. **You document PSV of 54 cm/s with an EDV of 3 cm/s of the right CCA and PSV of 72 cm/s with EDV of 18 cm/s on the left. What do these finding most likely represent?**
   A. An occluded brachiocephalic artery
   B. A hemodynamically significant stenosis of the right subclavian A
   C. An occlusion of the right ICA
   D. A left CCA/IJV fistula

15. **Your carotid duplex study documents an occluded right ICA on a 72 y.o. female. The left ICA has a 50-69% diameter reduction based on flow velocities. What is the most likely explanation for the antegrade flow detected in the right anterior cerebral artery (ACA) with TCD?**
   A. This is normal flow in the ACA and expected because the posterior cerebral artery normally supplies blood to this vessel
   B. The antegrade flow is secondary to cross-over collateralization from the contralateral ACA
   C. This abnormal flow is related to retrograde flow in the right ophthalmic artery secondary to the ICA occlusion
   D. Antegrade flow in the ACA is a normal finding because it's major source of blood flow is from ECA/ICA communications

16. **Which of the following is the least important criteria for determining the presence of a subclavian steal?**
   A. Color-flow Doppler displays blue in common carotid and red in a segment of the ipsilateral vertebral artery
   B. Spectrum analysis displays a negative Doppler shift in the vertebral artery
   C. B-mode imaging is able to visualize the vertebral artery
   D. Flow in the vertebral artery has no flow in end-diastole

17. **CTA documents atherosclerotic plaque in the left ICA with a true lumen of 9 mm and a residual lumen of 3mm. What is the diameter reduction (DR) of this lesion?**
   A. 21%
   B. 33%
   C. 67%
   D. 79%

18. **What is the most likely explanation for the absence of a unilateral carotid bruit in a patient that had one previously documented on a physical exam one year ago? A 50-79% stenosis was also evident on that previous carotid duplex study.**
   A. The lesion causing the turbulent flow has progressed to a >90% diameter reduction (DR)
   B. The previously documented plaque formation has dissolved secondary to the patient's life-style changes
   C. The bruit was probably related to heart disease now under control by medication
   D. Although the 50-79% DR is unchanged from the previous study, the ECA now has a hemodynamically significant stenosis

19. **A cerebral angiogram documents many regularly-shaped areas of narrowing in the distal ICA. What is the most likely explanation for this finding?**
   A. The patient has temporal arteritis
   B. You suspect fibromuscular dysplasia
   C. This is an atheromatous plaque formation
   D. The finding is consistent with neointimal hyperplasia

20. **Your next patient had carotid stenting for recurrent narrowing of the ICA following carotid endartectomy (CEA). How might that stent affect blood flow velocities?**
   A. PSV and EDV within the stent are normally lower then those obtained post-CEA
   B. The same velocity criteria used pre-stenting can be used after stent placement
   C. You may expect some flow acceleration post-stenting
   D. Velocity criteria for peak systole remains the same, but end-diastolic flow is expected to remain reduced in surveillance studies

TEST ANSWERS

1. C	11. C
2. B	12. E
3. C	13. B
4. A	14. C
5. D	15. B
6. C	16. C
7. A	17. C
8. B	18. A
9. C	19. B
10. B	20. C

Venous Self Assessment Test Questions

1. **Which vessel does the axillary vein drain into?**
   A. Brachial
   B. Subclavian
   C. Cephalic
   D. Basilic

2. **Which vessel does the common anterior tibial vein drain into?**
   A. Femoral
   B. Tib-peroneal trunk
   C. Popliteal
   D. Dorsalis pedis

3. **Venous ulcers can be different from arterial ulcers in all of the following ways EXCEPT:**
   A. Venous ulcers are more often located superior to the medial malleolus
   B. They are deep and have a regular shape, e.g., punched-out
   C. The skin surrounding venous ulcers usually have a brawny discoloration
   D. Patients with venous ulcers tend to have less pain than those with arterial ulcers

4. **How is flow to the upper extremity affected when a patient with acute superior vena cava syndrome takes in a breath?**
   A. The flow increases
   B. The flow reverses in direction
   C. The flow remains the same
   D. The flow decreases

5. **How is the left renal vein anatomically related to the aorta?**
   A. The vein is anterior to the artery
   B. The vein is posterior to the artery
   C. The vein runs lateral to the artery
   D. The vein runs medial to the artery

6. **Mrs. JH is a 25 y.o. female who is 32 weeks pregnant and complaining of swelling of her left lower extremity. What might the continuous (non-phasic) venous flow in the common femoral vein signify?**
   A. A more proximal acute deep venous thrombosis (DVT)
   B. Fluid overload
   C. Extrinsic compression
   D. A and C
   E. A and B

7. **During muscle contraction, what is the normal direction of blood flow in the perforating veins of the lower extremity?**
   A. Moves caudal from the superficial veins into the deep
   B. Moves cephalad from the deep veins into the superficial
   C. Moves caudal from the deep veins into the superficial
   D. Moves cephalad from the superficial into the deep

8. **As you continue with your evaluation of the previously described patient, Mrs. JH, what would you do next?**
   A. Reposition the patient into a lateral decubitus position, reevaluate flow patterns in the CFV, and try to image the iliac veins
   B. Call the referring physician and say the test is not reliable on a woman who is 32 weeks pregnant
   C. Place the patient in a Trendelenberg position, reevaluate flow patterns in the CFV, and try to image the iliac veins
   D. Contact the referring physician and provide your preliminary report that the continuous flow in the CFV can only mean there is an acute DVT more proximally

9. **What is the least likely method to evaluate venous insufficiency?**
   A. B-mode imaging
   B. Color-flow Doppler
   C. Spectrum analysis
   D. Plethysmography

10. **What is the most likely reason that a patient might have pulsatile flow in the common femoral veins?**
    A. It may be secondary to an acute DVT proximally
    B. You suspect the patient has significant venous incompetence
    C. The patient's history of congestive heart failure explains this finding
    D. It is consistent with a distal obstruction

11. **Which of the following statements about the portal vein is the least accurate?**
   A. The portal vein carries approximately 30% of the blood flow into the liver
   B. The walls of the portal vein is more echogenic, or brighter, than the hepatic veins
   C. Flow in the portal vein is normally heptopetal
   D. The portal vein is created by the confluence of the superior mesenteric and splenic veins

12. **What is the most likely venous pressure measured at the ankle level of a supine patient?**
   A. 0 mm Hg
   B. 50 mm Hg
   C. 100 mm Hg
   D. 120 mm Hg

13. **You detect a thrombosed femoral vein on your patient, Mrs. GH. Within the vessel lumen, you are able to document areas of cephalad flow and two linear structures that are brightly echoic. What are these findings most likely consistent with?**
   A. Artifact
   B. Collateral flow
   C. Recanalization
   D. Medial wall calcinosis

14. **All of the following are true about post-phlebitic syndrome EXCEPT:**
   A. The patient has had a history of venous thrombosis
   B. A brawny discoloration around the gaiter zone is expected
   C. A palpable cord is detected along the medial aspect of the leg
   D. The patient complains of chronic swelling of the leg

15. **Your next patient is the star pitcher on your alumni's baseball team, a previously healthy 18 y.o. with recent onset of pain & swelling of his right upper extremity. What is the most likely cause of his problem?**
   A. Paget-Schroetter Syndrome
   B. May-Thurner Syndrome
   C. Acute DVT from venous stasis
   D. Superior Vena Cava Syndrome

16. **Your venous duplex study documents spontaneous, phasic flow in the femoral vein which is displayed below the baseline. As you manually compress the external iliac vein just above the inguinal ligament, flow is above the baseline. What is the significance of that finding?**
   A. The flow change is an expected response to a proximal compression
   B. This signifies venous reflux
   C. You are concerned about a proximal obstruction
   D. The flow alteration is secondary to incompetence of Cockett's I, II, and/or III perforators

17. **What is the expected flow pattern in the portal vein?**
   A. Heptofugal (flow into the liver)
   B. Heptopedal (flow away from the liver)
   C. Heptofugal (flow away from the liver)
   D. Heptopedal (flow into the liver)

18. **Your next patient, Mr. LM, is complaining of a 6 month history of swelling in his left lower extremity with increasing discomfort, not exactly pain, but a heaviness. What is the most likely cause of his signs/symptoms (S/S)?**
   A. Incompetent venous valves
   B. Acute venous thrombosis
   C. Chronic arterial occlusion
   D. Decreased capillary pressure

19. **To obtain further information related to the cause of Mr. LM's S/S, which of the following studies would least likely be ordered on that previously described patient?**
   A. Descending venography
   B. Photoplethysmograpy
   C. Ascending phlebography
   D. Air plethysmography

20. **Your next patient has an acute DVT. Unless it is contraindicated, why is heparin routinely administered?**
   A. Heparin decreases thrombus formation
   B. The drug will breakdown/lyse the acute DVT
   C. The medication promotes recanalization
   D. Heparin increases collateral development

TEST ANSWERS

1. B	11. A
2. C	12. A
3. B	13. C
4. C	14. C
5. A	15. A
6. D	16. B
7. D	17. D
8. A	18. A
9. A	19. C
10. C	20. A

Ascending renography : Identifies the presence and Location of DVT.
(Deep vein Thrombosis)

Descending renography : assesses the function of the deep vein valves.
(valvular incompetency)

QA Self Assessment Test Questions

1. **Of the following, which cannot be used to disinfect a transducer?**
   A. Diluted bleach
   B. Cidex
   C. Sterile saline
   D. Hydrogen peroxide

2. **What is the most effective measure to decrease cross-contamination?**
   A. Technologist /sonographer use personal protection equipment/supplies
   B. Technologist /sonographer frequently wash their hands
   C. Patients are given personal protection equipment/supplies
   D. Testing equipment is routinely sterilized

The next seven questions are related to the following scenario: Your vascular laboratory has 38 lower extremity venous duplex exams that had a subsequent venogram for a variety of reasons. Of these, both tests documented venous disease in 22 extremities, and ruled-out venous disease in 10. In 4 cases, the venous imaging study was considered normal while venography documented disease. There were 2 situations where the duplex study was interpreted as abnormal, but no disease was evident on venography.

3. **What is your lab's sensitivity?**
   A. 10/12
   B. 22/26
   C. 10/14
   D. 22/24
   E. 32/38

4. **What is the negative predictive value?**
   A. 10/12
   B. 22/26
   C. 10/14
   D. 22/24
   E. 32/38

5. **What would your overall accuracy be?**
   A. 22/24
   B. 32/38
   C. 22/38
   D. 6/38
   E. 24/26

6. **What is the value of calculating specificity?**
   A. It tells you how often a positive non invasive test is correct
   B. It reflects how good a test is at excluding disease
   C. The calculation indicates how often a negative noninvasive test is correct
   D. It provides you with an idea of how good the test is at picking up disease

7. **What is the significance of a test that is highly accurate?**
   A. The test is good at identifying disease as well as excluding it
   B. The calculation reflects a low sensitivity, but high specificity
   C. The diagnostic information provided by the test should not be used to determine therapeutic intervention
   D. The test is good at excluding disease, but not very good at identifying it

8. **In the previous example, you add 30 more correlations on studies completed on patients being seen in the Coumadin clinic. Of the following calculations, which do you think would be affected the most?**
   A. Specificity and accuracy
   B. Sensitivity and positive predictive value
   C. Specificity and positive predictive value
   D. Sensitivity and negative predictive value

9. **In the previous example where you have 30 additional studies, which cell in the matrix do you anticipate changing the most?**
   A. True negatives
   B. False positives
   C. False negatives
   D. True positives

10. **If accuracy is 96%, what would sensitivity be?**
    A. 0-100%
    B. Less than 96%.
    C. Equal to 96%.
    D. Greater than 96%.

TEST QUESTIONS

1. C

2. B

3. B

4. C

5. B

6. B

7. A

8. B

9. D

10. A